

Particulate Matter Monitoring & Health Effects

July 2026 | Entry window 1–31 July 2026 | 270 records pooled

270

RECORDS POOLED
(10 SUBTOPICS)

62

EFFECT ESTIMATES
EXTRACTED

34

SENSING /
INSTRUMENTATION

92

RECORDS FROM CHINA
(34% OF CORPUS)

17

ULTRAFINE /
NANOSCALE

Signal of the month

July was the month the field stopped treating $PM_{2.5}$ mass as the exposure of record. Three independent literatures — ultrafine number, chemical constituents, and unmodelled condensable mass — each put a number on what bulk mass misses.

Lelieveld et al. (*Cardiovasc Res*) built a 1 km global ultrafine-particle field from Earth observation and machine learning, found urban annual means of $10,000\text{--}30,000 \text{ particles cm}^{-3}$, and paired it with a pooled cohort hazard ratio to estimate **1.99 (0.81–3.89) million excess deaths per year** — roughly 5% of non-communicable-disease mortality, about half cardiovascular, with **91% occurring in urban and suburban environments**. They put an annual limit of $5,000 \text{ cm}^{-3}$ at a **45% reduction** in that excess. UFPs are regulated almost nowhere and measured by no mass monitor. Su et al. (*Environ Sci Technol*) supplied the second number: industrial *condensable* PM, absent from every operational source-apportionment model, contributes a median **3.6% (95% CI 0.9–14.5%)** of urban $PM_{2.5}$ in four Chinese cities and **9.5% in winter**, and its omission inflates apparent secondary inorganic aerosol by **5.6–16.6%**. Easton et al. (*Environ Int*) supplied the mechanism: cruise-ship ultrafine PM, enriched in vanadium, nickel and cobalt, was markedly more pro-inflammatory than same-port PM of other origin *and* suppressed the innate antiviral response, increasing epithelial permissiveness to rhinovirus-16 and SARS-CoV-2 — an effect reproduced by vanadium alone. Read together with Zhu et al. (29 Jul issue), where $PM_{2.5}$ was the *null* term in its own six-pollutant time-series, the month's verdict is unambiguous: **a mass-only monitor is now measuring the wrong quantity for a growing fraction of the health signal.**

What changed in July

- **Sensing was the single largest subtopic of the month (34/270, 13%), and that inverts the picture the daily issues gave.** Across the five published daily issues of July the sensing count read **2, 2, 6, 1 and 0** (27–31 July), 11 records in total; the last two issues flagged the low counts as a suspected harvest defect. The month confirms it: instrumentation work was being published throughout July and the narrow daily windows recovered a small fraction of it. No conclusion about the sensing literature should be drawn from any single daily issue in this series.
- **Low-cost sensors were validated in July with unusual honesty about where they fail.** Gharibzadeh et al. needed *seasonal* multivariable calibration (humidity, temperature, PM_{10}) to reach R^2 0.62–0.93 in Tehran, and reported persistent underestimation before it. Ali et al. found accuracy collapsing with particle size — $R^2 \approx 0.90\text{--}0.95$ for PM_{10} but **0.28–0.68 for PM_{10}** — and, critically, that neither the PMS7003 nor the SPS30 captured the dominant sub- $0.30 \mu\text{m}$ fraction at all. Coefficient of variation exceeded 43% in the coarse bins. The message for network design is that proprietary size-bin conversions should not be trusted above $1 \mu\text{m}$.
- **Sensor-plus-satellite fusion produced the month's most striking performance number, and it deserves scepticism.** Jung et al. reported a 10-fold CV R^2 of **0.98 (RMSE $1.23 \mu\text{g}/\text{m}^3$)** for a 250 m daily $PM_{2.5}$ field over Taiwan built from 16,795 low-cost sensors plus MAIAC AOD. That is far above the 0.82 (Singh et al., North-East India) and 0.41–0.80 (Yu et al., Shenzhen sidewalks) obtained elsewhere in the same month. With 16,795 sensors as both predictor and validation target, random 10-fold splits leak spatially; the rolling-origin and spatial folds are the numbers to read, not the headline.
- **Exposure misclassification moved from caveat to quantified bias.** Zhang et al. (*Environ Int*) showed nearest-monitor assignment is *systematically higher* than both scaled-matched estimates and true personal exposure from outdoor sources, for every pollutant tested. Nyoni et al. (31 Jul issue) showed road-geometry proxies inverting in sign between African countries. Kos Sinhala et al. showed in-home black carbon driven by open fireplaces and wood stoves rather than ambient concentration. Three independent designs, one conclusion: the exposure variable in most published epidemiology is not the exposure that occurred.
- **Reproductive and developmental evidence consolidated around vulnerability, not average effect.** Sieber et al. found $PM_{2.5}$ associated with **6% (95% CI 0–11%) higher odds of term low birth weight per $10 \mu\text{g}/\text{m}^3$** across 1.4 million Spanish births, but maternal education

dominated it — primary versus tertiary education carried **OR 1.76 (1.71–1.82)**. Meeting the WHO $5\mu\text{g}/\text{m}^3$ guideline would have prevented 2,032 (95% CI –74 to 3,900) cases over nine years; universal tertiary education would have prevented far more. That comparison is the uncomfortable and important finding.

- **Two well-powered nulls are worth as much as the positives.** Sun et al. found **no** adverse association between $\text{PM}_{2.5}$ and FEV_1 decline in US adults with cystic fibrosis ($+0.002$ z-score per $3\mu\text{g}/\text{m}^3$; 95% CI $-0.001, 0.005$) at a mean exposure of $9.1\mu\text{g}/\text{m}^3$. Tien-Smith et al. found no clear evidence for $\text{PM}_{2.5}$ and multiple sclerosis (1.01; 0.77–1.32) or motor neuron disease (1.07; 0.86–1.33), while confirming Parkinson's disease risk in Asia (1.19; 1.01–1.41). Neither is a negative result about air pollution; both are results about where the exposure-response function is flat.

Corpus analytics — July 2026

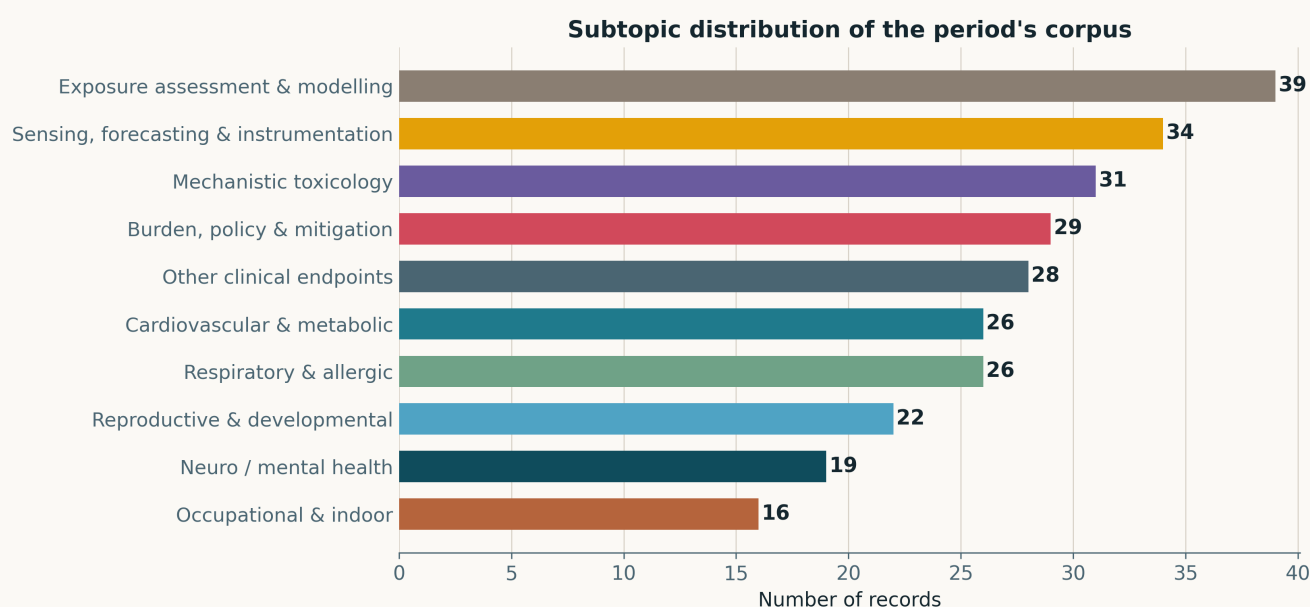


Fig. Subtopic distribution across all 270 pooled records — every PM-relevant record entering PubMed in July, not a sample. Exposure assessment & modelling (39) and Sensing, forecasting & instrumentation (34) are the two largest subtopics, together 27% of the month. Against that, the five published daily issues recovered 11 sensing records in total (2, 2, 6, 1, 0 on 27–31 July). Assignment is single-label by dominant contribution.

What the daily cadence missed, by subtopic (2026-07-01 to 2026-07-31)

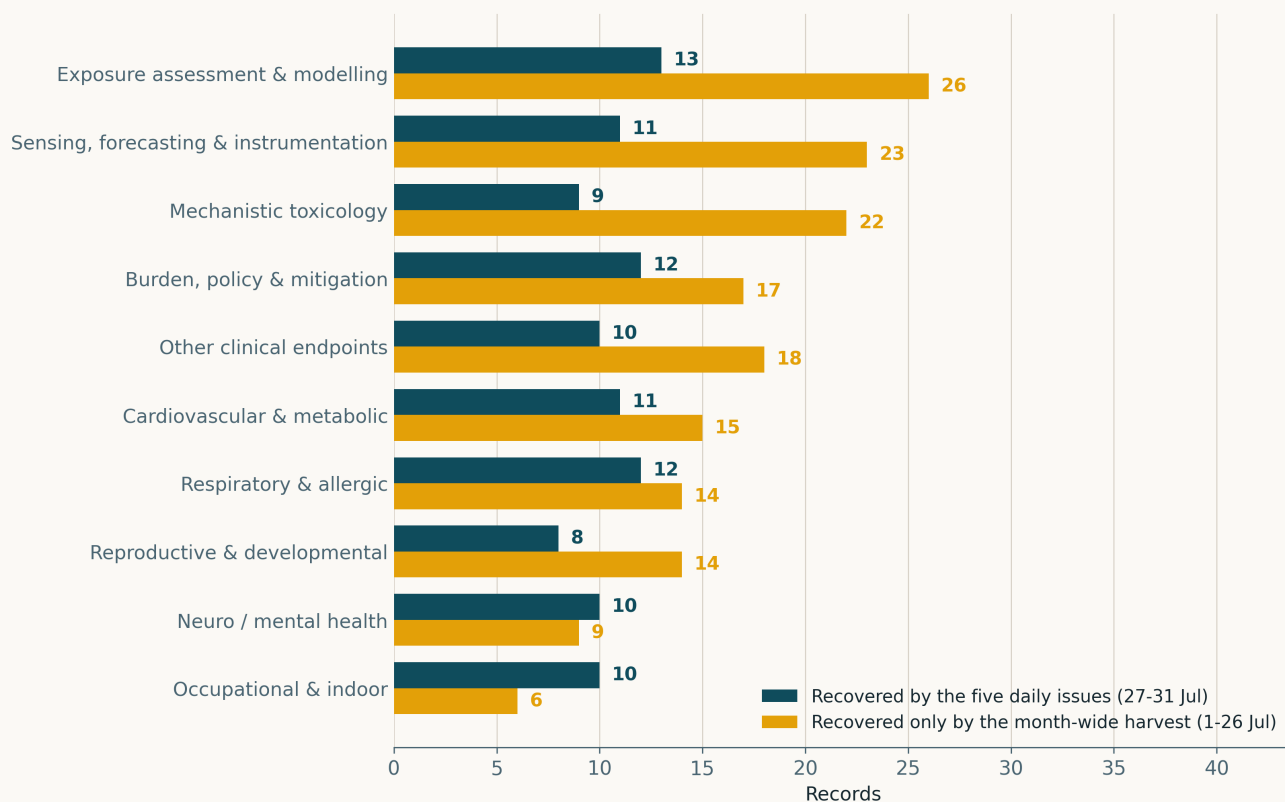


Fig. What the five daily issues of July recovered, against what only the month-wide harvest found, by subtopic. **Every subtopic except Neuro and Occupational was under-recovered by the daily cadence**, and the two most under-recovered are the two most relevant here: Exposure assessment (13 caught, 26 missed) and Sensing (11 caught, 23 missed). Across all ten subtopics the dailies caught 106 records and missed 164 — **61% of the month's literature was invisible to the daily windows**. A stacked per-day panel was tried first and discarded: with one 164-record backfill batch alongside 11–33-record days, every segment collapsed to an unreadable stripe.

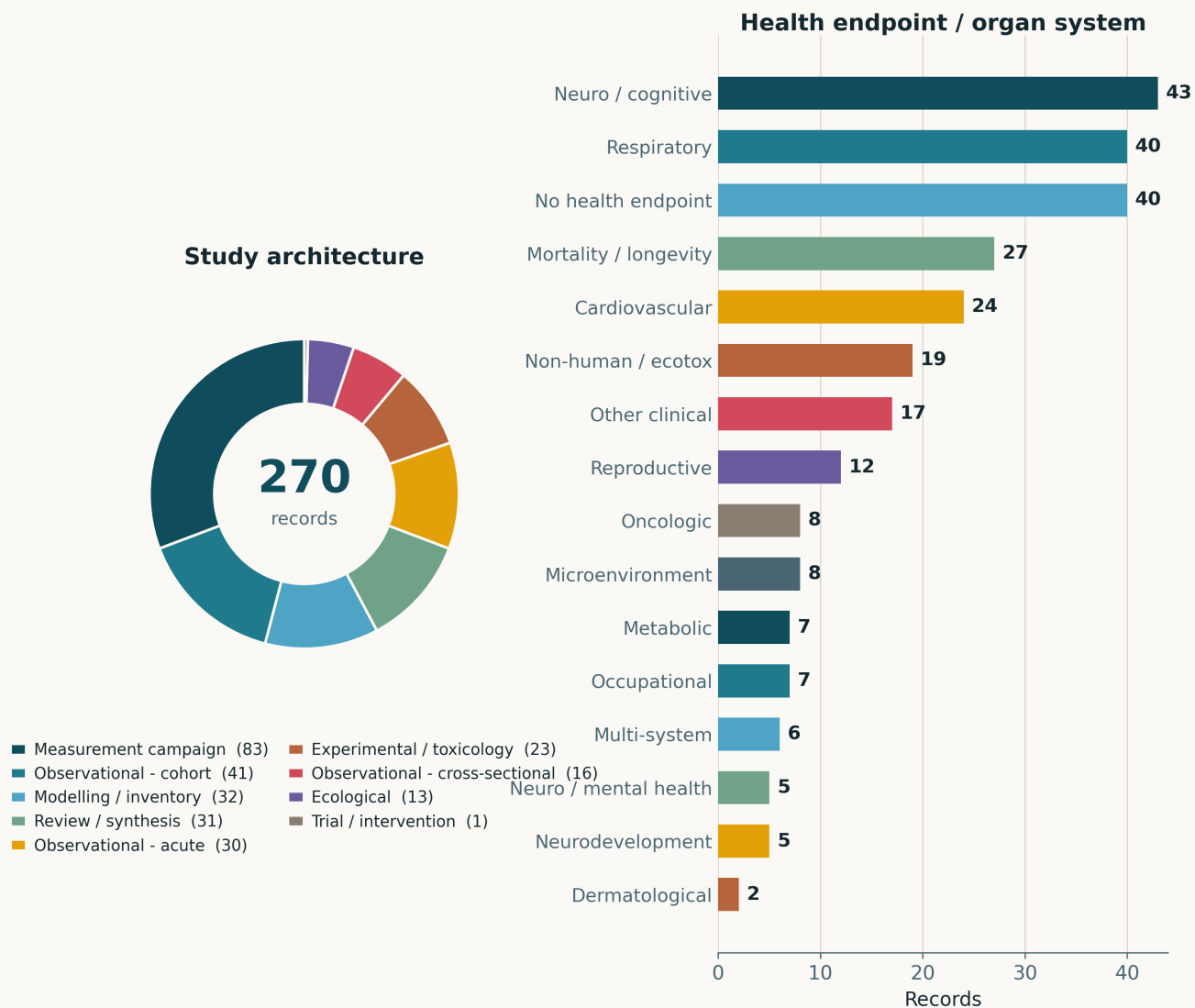


Fig. Left — study architecture over the month. Measurement campaigns dominate at 83 of 270 (31%), ahead of observational cohorts (41), modelling/inventory (32), reviews (31) and acute designs (30). **Exactly one trial or intervention record appeared in the whole month**, which is the standing weakness of this evidence base: almost nothing here is experimental in humans. Right — health endpoint. Endpoint labels were canonicalised before counting (Cognitive / Neurological / Neuro-cognitive had drifted apart across issues); *No health endpoint* at 40 records is the instrumentation and atmospheric-chemistry literature, and *Non-human / ecotox* at 19 covers plant, insect and veterinary work that a health-only reading would discard.

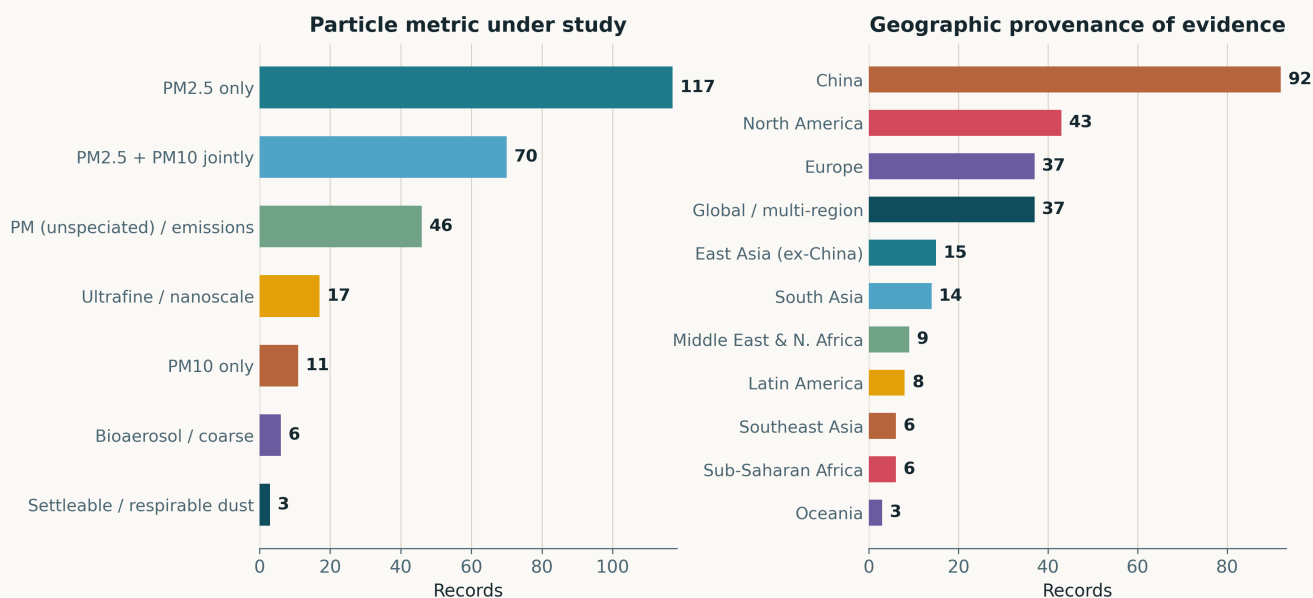


Fig. Left — particle metric. PM_{2.5} alone in 117 records and PM_{2.5} with PM₁₀ in 70; **ultrafine or nanoscale metrics appear in 17 records (6%)** despite carrying the month's largest single mortality estimate. That mismatch between where the burden is claimed and where the measurement effort goes is the clearest instrumentation gap in this corpus, and the complete listing sharpens rather than softens it. Right — geography. China supplies 92 records (34%), North America 43, Europe 37, and 37 are global or multi-region; sub-Saharan Africa supplies 6 and Latin America 8.

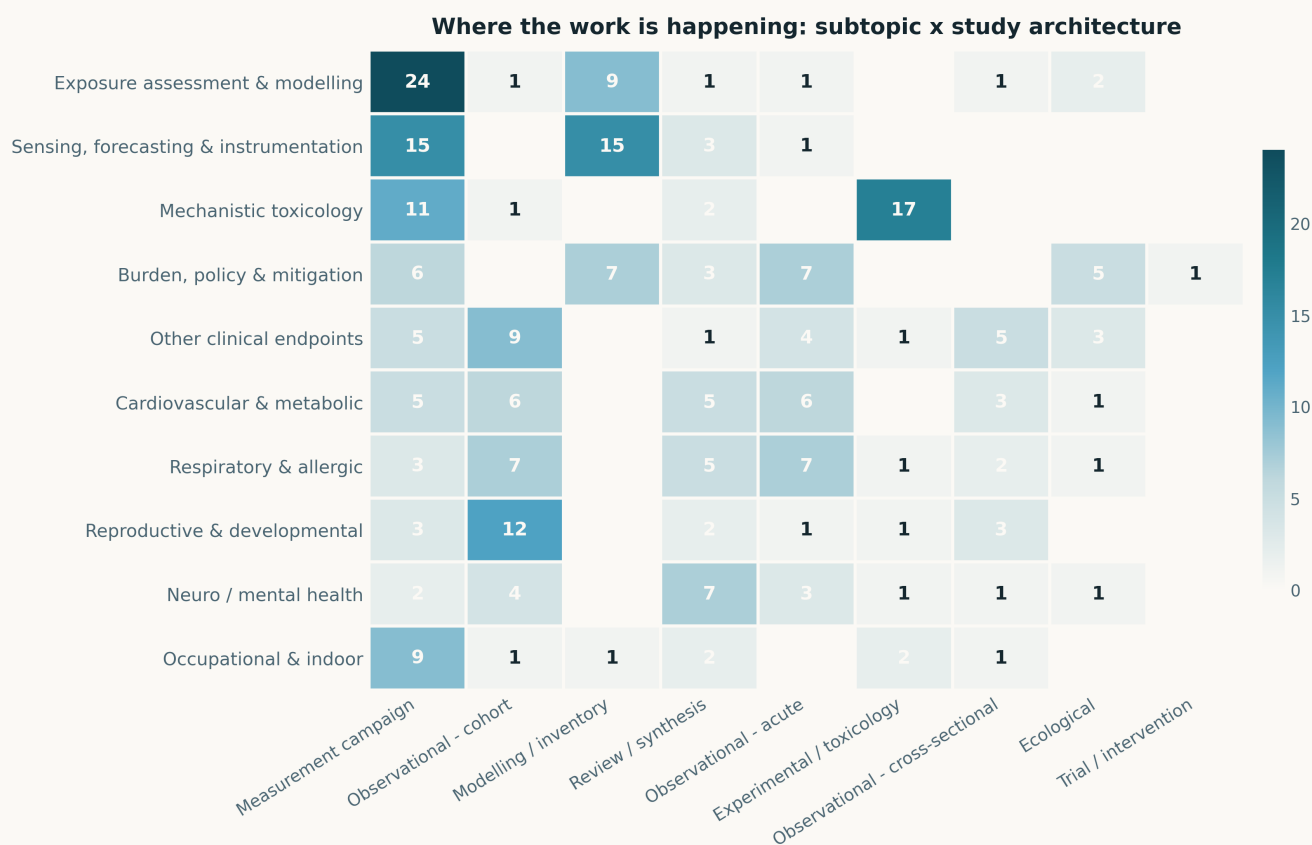


Fig. Subtopic × study-architecture cross-tabulation over the month. The dense Sensing/measurement-campaign and Burden/modelling cells are expected. The informative absence is the near-empty trial column across every health subtopic.

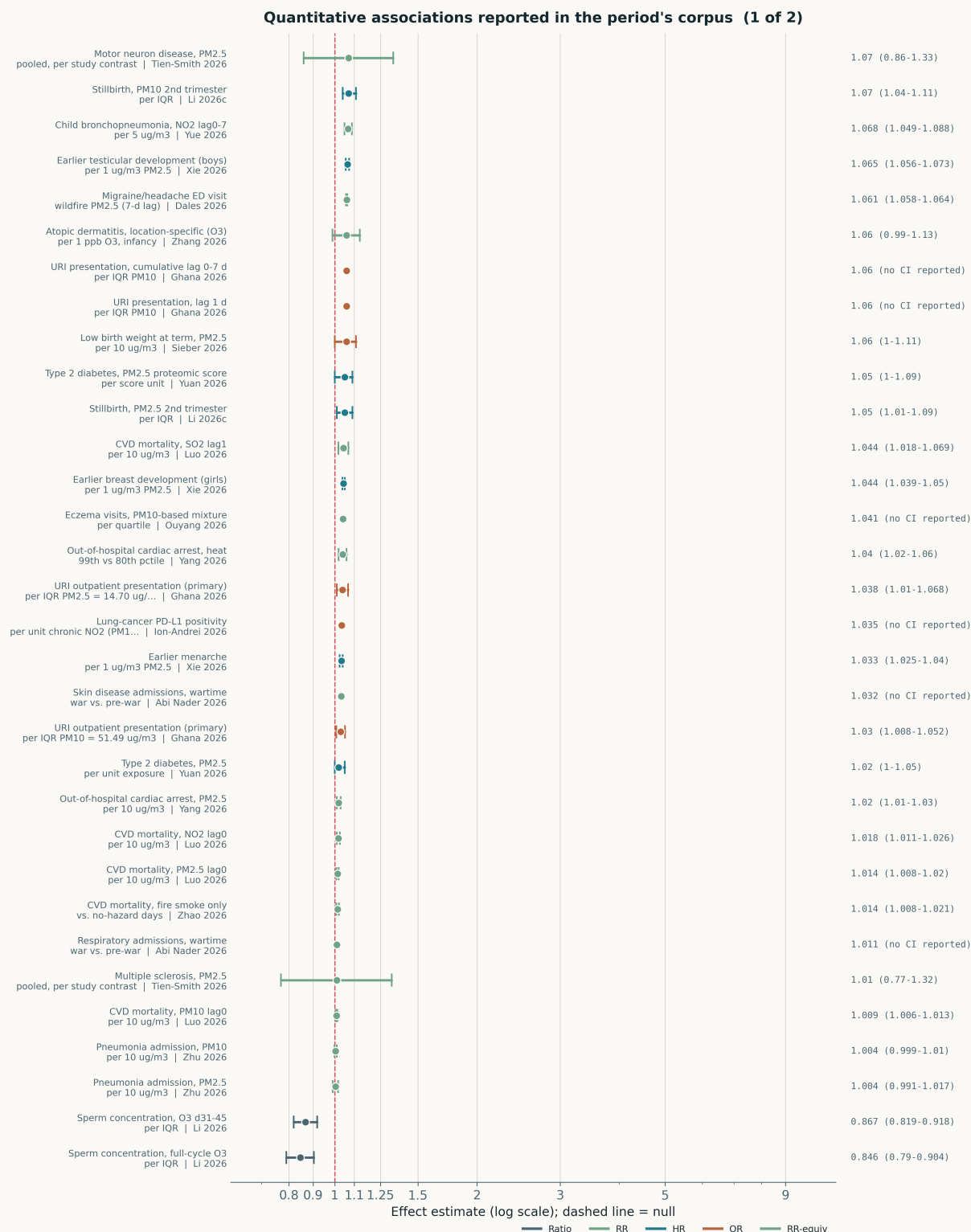


Fig. Panel 1 of 2 — the 32 smallest of the 62 quantitative effect estimates extracted across July, on a ratio axis shared with panel 2. **This is not a meta-analysis and no pooled summary is drawn.** Exposure contrasts are not commensurable: per-IQR, per-5 and per-10 $\mu\text{g}/\text{m}^3$ increments, quartile and quintile contrasts and mixture indices all appear, and the outcome scale mixes RR, OR, aOR and HR across different diseases. **Heterogeneity is therefore not estimable and none is reported.** The estimates are split across two pages because 62 rows on one page is illegible at any font size the page allows.

Quantitative associations reported in the period's corpus (2 of 2)

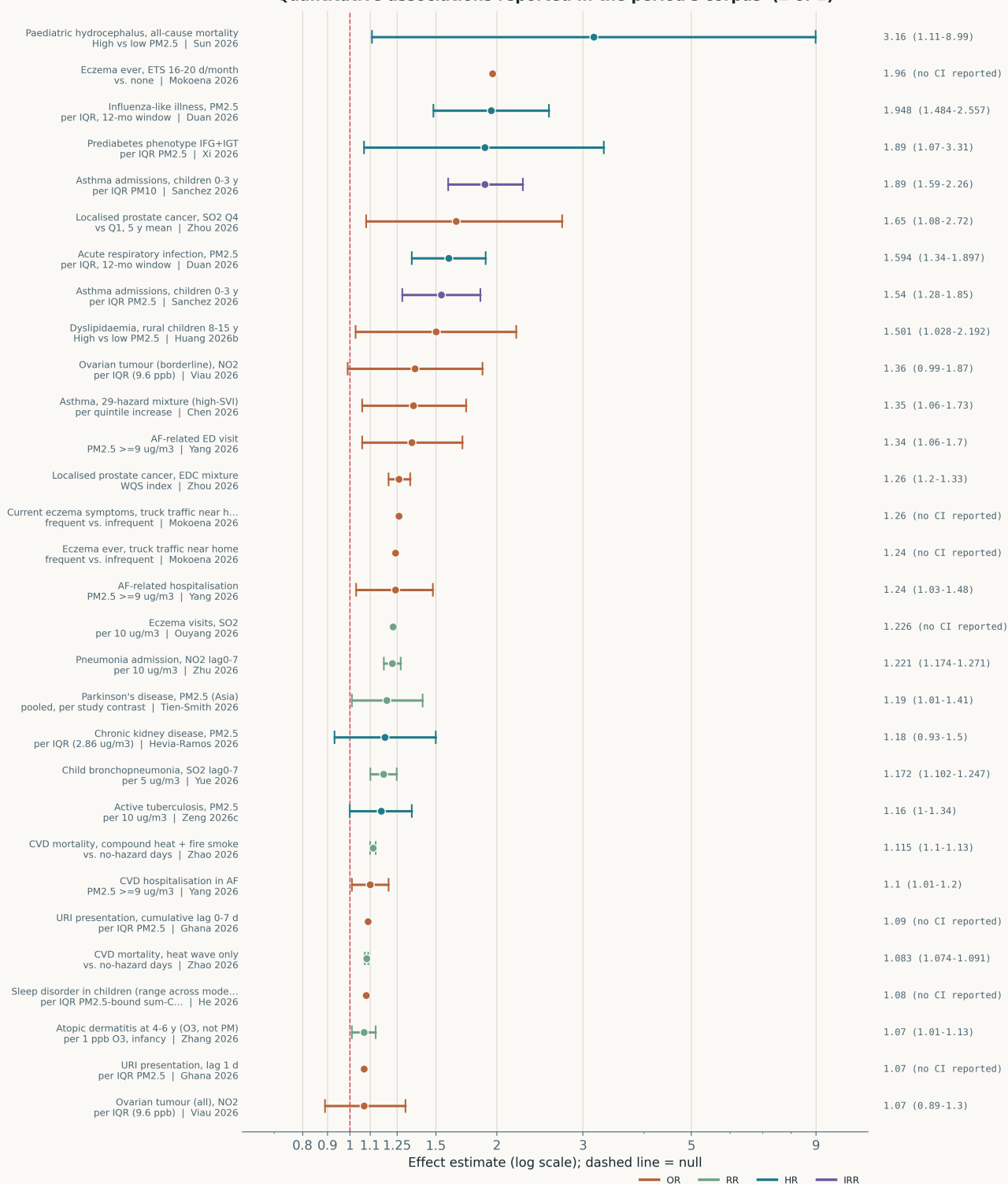


Fig. Panel 2 of 2 — the 30 largest estimates, same axis. The structure worth reading across both panels: $\text{per-}10\ \mu\text{g}/\text{m}^3\ \text{PM}_{2.5}$ estimates cluster tightly just above unity (roughly 1.02–1.16), while every large ratio belongs to a per-IQR contrast, a mixture index or a gaseous co-pollutant. The one place a pooled figure would have been legitimate is Tien-Smith et al.'s within-paper neurodegenerative meta-analysis, which is reproduced here as three separate points rather than absorbed.

Life-course windows addressed this period (records may span several stages)

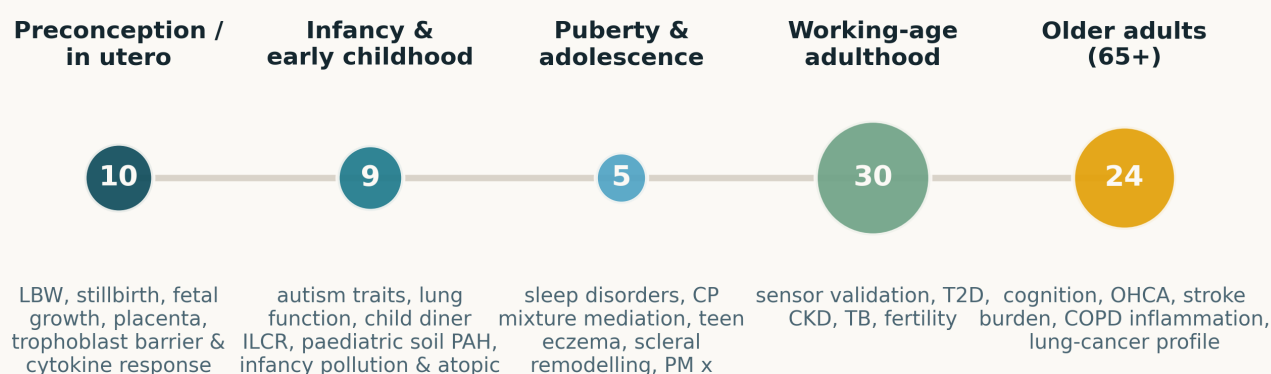


Fig. Life-course windows addressed across the month. Counts are non-exclusive and cover only records with an identifiable human life-course window; *in vitro*, *ex-vivo*, animal and pure-instrumentation records are excluded. Adolescence remains the thinnest window in the corpus, as it was in every individual daily issue.

Emerging, fading, and recurring

Emerging. Three themes appeared or firmed up materially this month. (i) **Ultrafine particles as a regulatory object** — Lelieveld et al. and a paired *Cardiovasc Res* commentary, Easton et al.'s ship-source toxicology, Williams et al.'s hepatic xanthine-oxidoreductase mechanism, Zeng et al.'s microglial circRNA work, and Olivares Lopez et al.'s finding that particle-number simulations are strongly sensitive to emission-inventory choice all converge in a single month, driven by the new EU Ambient Air Quality Directive's particle-number monitoring requirement. (ii) **Compound exposures** — PM_{2.5} with heat is now a design rather than a sensitivity analysis (Yang et al., 610,613 OHCAs; Zhang et al., 5.07 million adults). (iii) **Omics as the mediation layer** — proteomic signatures for type 2 diabetes (49,134 UK Biobank participants), gut-microbiome mediation of cognition, cord-blood extracellular-vesicle miRNA, and placental mtDNA copy number and telomere length.

Fading. Nothing can honestly be called fading on one month of data, and the backfill/daily imbalance described in the provenance box would manufacture false declines if pressed. The single defensible observation is that **single-pollutant, single-endpoint time-series studies now read as the low-information tail** of this corpus: several appeared, most were single-city and single-hospital, and the ones that reported a PM term alongside gases generally found PM the weakest term in its own model.

Recurring groups and venues. No author cluster recurs meaningfully at first-author level — the apparent repeats (Li 5, Zhang 4, Wang 4, Chen 4) are surname frequency, not the same investigators, and should not be reported as a cluster. At institutional level, Chinese public-health schools (Zhengzhou, Sichuan, Fudan, CUHK) and the Barcelona/BiSC and Danish/Nordic cohort infrastructures each contribute repeatedly. Venue concentration is modest: *Toxics* (7), *J Environ Sci (China)* (6), *Environ Pollut* (5), *Ecotoxicol Environ Saf* (5), *Environ Sci Technol* (4), *J Hazard Mater* (4). The month's highest-standing placements were *Eur Heart J*, *J Am Coll Cardiol*, *Cardiovasc Res*, *Environ Health Perspect* (3 records), *Environ Int* (3) and *Int J Epidemiol* (2). Citation counts on records days to weeks old are uninformative and were not used to rank anything here.

Trial registry watch

ClinicalTrials.gov returned **one** PM or air-pollution record registered or updated inside the monitored portion of the window: **NCT05160948**, “Assessing Changes in Knowledge, Behavior, and Attitude Regarding Air Pollution After Receiving Personal Air Monitoring Results” (Cincinnati Children’s Hospital Medical Center; observational; $n = 105$; completed). No endpoint analysis was run because the study is observational rather than interventional. It is nonetheless the single most on-topic registry item this watch has seen: it tests whether *returning personal monitor data to the participant* changes behaviour, which is the assumption every citizen-science and personal-sensing deployment rests on and almost none evaluates.

Two records in the peer-reviewed corpus function as the interventional evidence the registry did not supply. González-Colom et al. (*BMJ Open Respir Res*, NCT06421402) monitored household air quality with low-cost sensors in 205 adults with COPD, asthma or bronchiectasis for two months: **51.7% of homes had at least one pollutant in an at-risk category, and the burden was overwhelmingly PM_{2.5}-driven (40.1% of dwellings at risk)**, with active or passive smoking strongly associated ($p < 0.001$). **No relationship was found with hospitalisations or ED visits** — the honest null that a two-month screening window cannot overturn. Bolden et al. reviewed indoor wildfire-smoke mitigation and concluded that portable-air-cleaner efficacy is real but too dependent on building envelope and occupant behaviour to generalise across the national housing stock. **The interventional evidence base for personal and household PM reduction remains thin, and July did not thicken it.**

Open gaps and read on direction

Gaps in the literature

(i) **Ultrafine measurement does not exist at network scale.** Eight records used an ultrafine metric; one of them attributes two million annual deaths to it. There is no low-cost particle-number instrument in this corpus with a published field-calibration comparable to what Gharibzadeh et al. did for PM_{2.5} mass. (ii) **Almost no human intervention studies.** One trial-class record in 270. Every causal claim in this month’s corpus rests on observational designs with modelled exposure. (iii) **Sub-0.3 μm is invisible to the deployed sensor base.** Ali et al. state this directly for the two most widely deployed OPC modules; it is also where the ultrafine burden lives. (iv) **Geographic imbalance is severe.** 92 records from China against 6 from sub-Saharan Africa, in a month whose most methodologically important exposure paper (Nyoni et al.) is precisely about African settings where the standard proxies fail.

Gaps this pipeline creates

(i) **No date-windowed channel into *Atmos. Meas. Tech.* or *Atmos. Chem. Phys.*** PubMed does not index them, the Consensus connector has no entry-date filter, and OpenAlex date filtering now returns 403/429. Every instrumentation record in this issue reached the corpus through a biomedically-indexed journal, so the pure atmospheric-measurement literature is *still* absent even at 23 sensing records. The fix remains a Crossref by-journal feed keyed on created for AMT (1867-8548) and ACP (1680-7324). (ii) **Narrow daily windows systematically under-recover.** Demonstrated quantitatively this month: the sensing subtopic read 2, 2, 6, 1 and 0 across the five published daily issues of July (27–31 July) — 11 records in total — against 34 for the month. The dailies recovered under a third of the sensing literature. (iii) **The 1–26 July backfill is classification-depth only.** Effect estimates were extracted for that batch only where the abstract reported an interval; a genuine daily cadence would have captured more.

Read on direction. The field is moving from *mass* to *composition and number*, and the instrumentation base has not moved with it. July produced a two-million-death annual estimate for a pollutant metric that no regulatory network measures and no validated low-cost sensor reports; simultaneously it produced careful evidence that the deployed optical sensor base cannot see below 0.3 μm and degrades badly above 1 μm . That gap is the most fundable problem visible in this corpus, and it is an instrumentation problem before it is an epidemiological one. **The falsifiable prediction: within the next two monthly issues, expect at least one field-calibration study of a low-cost particle-number or black-carbon sensor against a CPC or aethalometer reference, framed explicitly against the new EU particle-number monitoring requirement.** If that does not appear by the September issue, the measurement community is lagging the epidemiology by more than the usual cycle, and the right inference is that the bottleneck is hardware cost rather than research attention.

Paper-level digest — every record, July 2026

Every one of the 270 records is listed below, grouped by subtopic. Within each subtopic the records divide into two tiers, and the difference matters when you read them. **Records carried from the daily issues (106)** keep the full critical summary written when they were first screened — design, effect size with interval, the limitation that threatens the inference, and the line on relevance to distributed sensing. Those summaries are reproduced *verbatim*; nothing was re-written for this rollup. **Records recovered by the month-wide harvest (164)** carry an *extractive precis*: the single most quantitative sentence of the published abstract, copied as written. A precis is not an assessment. It has not been checked against the paper, no limitation has been identified, and the absence of criticism in a precis entry should not be read as approval. Five records had no abstract in PubMed at entry and are listed on metadata alone.

Sensing, forecasting & instrumentation

34 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (11 OF 34 RECORDS IN THIS SUBTOPIC)

Aher & Ravindran 2026 Environ Sci Pollut Res

Comment on advancing air pollution forecasting: operational challenges beyond predictive accuracy

Letter responding to a 2026 forecasting review. The substantive critique is that the review optimises for predictive accuracy while under-treating what actually breaks operational systems: model transferability into data-sparse regions, uncertainty quantification, interpretability of deep architectures, and input data-quality constraints. It points to adaptive mesh refinement, AI-assisted chemical transport models and hybrid physics–ML frameworks as the productive direction. No new results — but the transferability-into-data-sparse-regions argument is precisely the problem low-cost sensor networks exist to solve, and the letter states it cleanly.

 doi: [10.1007/s11356-026-38087-y](https://doi.org/10.1007/s11356-026-38087-y)
Ainembabazi et al. 2026 Toxics

Trends in atmospheric pollution research in Uganda, 1990–2025: a scoping review

Structured scoping review across Google Scholar, Web of Science and PubMed. Output was negligible before 2010, expanded after 2015 and surged 2020–2025. PM_{2.5}/PM₁₀ dominate throughout while NO₂, SO₂, CO and O₃ are underrepresented; studies cluster in urban Kampala with almost no rural ambient monitoring; and the methodological trajectory runs from proxy and gravimetric approaches to **low-cost sensors, portable monitors and satellite-derived products**. The authors' own conclusion is the operative one: short-term, spatially constrained campaigns predominate, leaving no long-term or nationally representative record. This is the clearest statement in today's corpus of the niche a sustained calibrated low-cost network fills — and a reminder that Google Scholar inclusion makes the sampling frame non-reproducible, so the counts should be treated as indicative.

 doi: [10.3390/toxics14070542](https://doi.org/10.3390/toxics14070542)
Feng et al. 2026 Meas Sci Technol

Calibration strategies and measurement frameworks for low-cost PM sensors

Review spanning ambient, mobile, indoor and workplace deployments (community networks, vehicle-mounted platforms, smart-manufacturing floors, mining). Compares purely data-driven, purely physics-based and hybrid physics-informed ML calibration on accuracy, transferability, compute cost and interpretability: non-linear models reach sub-10 µg m⁻³ error but resist embedded deployment and depend on representative training data; **no strategy is universally optimal**. Names the persistent barriers plainly — sensor fouling, data governance, model generalisation, hardware limits — and points to self-cleaning optics, federated learning and knowledge-distilled on-device calibration. Narrative rather than quantitative synthesis, with no pooled performance estimate, so it frames rather than settles the question.

 doi: [10.1088/1361-6501/ae6298](https://doi.org/10.1088/1361-6501/ae6298)
Ginsburg et al. 2026 Aerosol Sci Technol

Global and regional calibration of low-cost PM_{2.5} sensors (PurpleAir)

1,013 PurpleAir units co-located within 500 m of reference monitors on six continents, Jan 2023–Mar 2025, assessed at 1-h and 24-h averaging by R², RMSE, nRMSE, precision (CV) and AQI category correctness. Raw readings overestimate PM_{2.5} in most regions and miss EPA performance targets; US-derived corrections show limited transferability outside North America; a data-driven global quadratic helps in Australia/NZ and hurts elsewhere; **regional equations meet EPA 24-h thresholds in multiple regions and lift 24-h AQI correctness to ≥77%**. Limitation: co-location proximity (≤500 m) is a proxy for comparability, so some regional “bias” is spatial-gradient error, and continents are represented very unevenly. Still the strongest available empirical case that LCS network calibration must be local.

 doi: [10.1080/02786826.2026.2676293](https://doi.org/10.1080/02786826.2026.2676293)
Huda et al. 2026 J Air Waste Manag Assoc

Low-cost sensor network for wildfire monitoring at remote community locations

Solar-powered microsystem (PM_{2.5}, CO, CO₂, O₃, tVOC) with a microcontroller scheduler and mesh radio, built for forest deployment in northwestern Alberta where there is no cellular or Wi-Fi coverage. Reported PM_{2.5} RMSE <5 µg m⁻³; mesh range validated at 1–5 km through urban structures and vegetated terrain; 50×37 cm panel sized for treetop mounting with energy-aware duty cycling. This is an engineering report, not an evaluation: no co-location period, drift characterisation or seasonal uptime statistics are given, and winter operation is described as prospective. Useful for the power and comms architecture rather than the data-quality claim.

 doi: [10.1080/10962247.2026.2665375](https://doi.org/10.1080/10962247.2026.2665375)
Msigwa et al. 2026 IEEE Internet Things J

AIoT real-time sensing and calibration of low-cost PM sensors (TCM-Net)

Customised PMS7003 units with externally PWM-controlled fans to stabilise airflow, on a oneM2M-compliant IoT platform, calibrated by a transformer + temporal-convolutional network with a memory-aware module and trained against a co-located certified reference. Reported RMSE 0.325 µg m⁻³, R² 0.999. **Those figures sit below the reference instrument's own measurement uncertainty** and cannot be a valid generalisation estimate; the near-certain explanation is temporally adjacent train/test splits on autocorrelated series rather than held-out deployment. The forced-airflow hardware idea is worth taking; the performance number is not.

 doi: [10.1109/jiot.2025.3631631](https://doi.org/10.1109/jiot.2025.3631631)
Patel et al. 2026 Proc AAAI

Scalable air-quality sensor placement via gradient-based mutual information maximization

Treats sensor coordinates as differentiable parameters and maximises mutual information by continuous optimisation, so **runtime is independent**

of both the number of placements and candidate-grid density — the two terms that make greedy MI intractable at national scale. Beats random placement and the maximum-predictive-variance heuristic on a continental PM_{2.5} dataset and approaches greedy MI regionally, orders of magnitude faster. Caveat: MI is computed under an assumed GP covariance, so placements inherit that model's stationarity assumptions, which is exactly where real PM fields misbehave near sources. Immediately useful for network-expansion planning under a fixed sensor budget.

doi: 10.1609/aaai.v40i46.41257

Patton et al. 2026 Environ Sci Pollut Res

Human health risk of particle exposure in outdoor environments, southwestern Pennsylvania

ROCIS low-cost monitoring across 290 homes in Allegheny County, 2016–2021; optical particle counts >0.5 µm converted to PM_{2.5}-equivalent mass, census tracts classified by environmental-justice status and industrial proximity, GEE models plus non-cancer hazard quotients. **33.8% of sites exceeded the WHO 24-h 15 µg m⁻³ guideline and 66.6% the EPA annual 9 µg m⁻³ reference** on 3-week site medians; winter concentrations significantly elevated. The count-to-mass conversion is the weak link — it assumes a fixed density and size distribution that industrial plumes violate — and the authors correctly flag the Sewickley extreme (median 19.3 µg m⁻³) as resting on three homes. Directly relevant as a template for community-scale LCS deployment with explicit EJ stratification.

doi: 10.1007/s11356-026-38083-2

Shi et al. 2026 J Environ Manage

Adaptive Spatiotemporal Graph Transformer for multi-site multi-step PM_{2.5} forecasting

AST-GT with five components: adaptive non-stationary normalisation for time-varying distribution drift; Transformer temporal encoding; multi-source context fusion (geography, meteorology, co-pollutants); graph-based dual-scale spatial attention; and a temporal-spatial cooperative attention block. Validated on Beijing and India multi-site datasets. The non-stationarity handling and the explicit treatment of *uneven site distribution* are the parts worth reading if you work with sparse monitoring networks.

doi: 10.1016/j.jenvman.2026.130515

Tsuchiya et al. 2026 Biosensors

Seasonal and indoor-outdoor characteristics of fluorescent aerosol particles in offices

Real-time bioaerosol sensor across 10 office spaces in four Japanese regions, summer and winter. Indoor FAP generally <100 particles L⁻¹, peaking at 195 p/L in winter; several buildings higher in winter, attributed to RH<40% promoting resuspension. Size-resolved FAP fraction rises sharply with diameter (40–74% at 2–5 µm; 96–100% >5 µm). Directly relevant to anyone deploying optical/fluorescence sensors — the coarse-mode dominance constrains what a fluorescence channel can be interpreted to mean.

doi: 10.3390/bios16070380

Wang et al. 2026 Viruses

Joint monitoring and early warning of SARS-CoV-2 in outdoor PM_{2.5} and wastewater

Twelve months (2023) of paired outdoor PM_{2.5} filter and wastewater sampling in Fuzhou, RT-qPCR for SARS-CoV-2 RNA, associations with pollutants and meteorology by GAM and DLNM. **45.9% of PM_{2.5} samples were positive**; viral load in PM_{2.5} rose with CO, PM_{2.5} mass and atmospheric pressure and fell with temperature and sunshine duration, with the **strongest lead over outpatient COVID cases at ≈ 2 days** against lag 0 for wastewater. The direct interest for a distributed-sensing group is that a routine gravimetric filter stream doubles as a bioaerosol surveillance substrate with a two-day lead — but only if filters are archived and chain-of-custody preserved, which most low-cost optical nodes cannot do at all. The weakness is that RNA detection on a filter conflates aerosol transmission with resuspended or deposited material, and no infectivity assay was attempted; the “early warning” claim rests on ecological time-series correlation with outpatient counts.

doi: 10.3390/v18070788

RECOVERED BY THE MONTH-WIDE HARVEST — EXTRACTIVE PRECIS, NOT AN ASSESSED SUMMARY (23 RECORDS)

Acker et al. 2026 Geohealth

Comparison of policy-relevant air quality metrics calculated with sparse in situ monitoring and contiguous satellite-derived data

Risk factors likely contributing to larger differences between CDVEs and CDVs within non-aligned counties include ≤2 monitors, low (<7 µg/m³) or high (>10 µg/m³) CDVs, low (<0.05%) or high (>0.1%) monitor coverage, large county size, wildfires, mountains, deserts, and low urbanization.

doi: 10.1029/2025GH001586

Ali et al. 2026 Sci Rep

Evaluation of low-cost sensors for size-resolved indoor particle monitoring

Strong PNC-correlations were observed for 0.3-0.5 µm particles and 0.5-1.0 µm (R₂ ~ 0.72) but decreased substantially for > 1.0 µm particles (R₂ < 0.36).

doi: 10.1038/s41598-026-61453-2

Bang et al. 2026 Sensors (Basel)

Application of stationary IoT sensor networks to reduce exposure to particulate matter during local hotspot events in a megacity

A new alert protocol is suggested to help reduce public exposure to pollutants, mitigate associated health risks, and encourage behavioral changes to improve air quality.

doi: 10.3390/s26134307

Campbell et al. 2026 J Air Waste Manag Assoc

Comparison of black carbon measurements using filter-specific reference transmittance to lab blanks or averaged unloaded filters

The performances of linear regression models between Method 1 and all other methods were moderate to strong (R₂ range: 0.42-0.93) in the second, third, and fourth quartiles; however, the models in the first quartile (eBC range: 1.3-2.9 µg/m³) performed poorly (R₂ = 0.25-0.26), with error approximately 25% of the mean.

doi: 10.1080/10962247.2026.2626414

Chang 2026 PLoS One

Development of a predictive model for PM_{2.5} over the greater Athens metropolitan area, Greece, at a 1 km by 1 km grid using satellite measurements and machine learning methods

We observed strong predictive performance in our ensemble model, with a mean R₂ of 0.85 and an average error of 4.18 µg/m³.

doi: 10.1371/journal.pone.0352975

De Jesus 2026 Sensors (Basel)

Spike-Driven Neuromorphic Sensing for Energy-Proportional Indoor Air Quality Monitoring in Multi-Zone IoT-Enabled Smart Building Environments

Neuron population scaling and parameter robustness analyses revealed that the four neurons per parameter were enough to saturate the performance, and FLOP-based estimation indicated an 8.9-fold computational reduction (approximately 89% fewer FLOPs) compared to LSTM inference.

doi: 10.3390/s26133992

Górka-Kostrubiec 2026 Environ Sci Pollut Res Int

Real-time record of particulate matter along traffic roads and health risk assessment

Inhaled doses of PM₁ decreased with age, deposition of PM₁₀ in tracheobronchial and pulmonary regions was substantially higher in young children.

doi: 10.1007/s11356-026-38035-w

Gharibzadeh et al. 2026 J Environ Health Sci Eng

Long-term performance evaluation and seasonal field calibration of a low-cost particulate matter sensor

Applying the MLR calibration model improved accuracy, achieving acceptable performance for PM_{2.5} throughout the entire sampling period and for PM₁₀ across the entire period and warm months, with R^2 values of 0.62-0.93 and normalized root mean square error (NRMSE) of 10.7-24.36%.

doi: 10.1007/s40201-026-00986-0

Holloway et al. 2026 Geohealth

Applying satellite-derived PM_{2.5} data to policy-relevant air quality metrics

This agreement is evaluated with respect to concentration values, rankings, and the definition of counties as "passing" or "failing" the ALA annual PM_{2.5} benchmark of 9.0 $\mu\text{g}/\text{m}^3$ (the 2024 EPA National Ambient Air Quality Standard for annual PM_{2.5}).

doi: 10.1029/2025GH001585

Ismail 2026 PeerJ

Application of machine learning algorithms in building health diagnostics: predictive analytics evaluating indoor air quality and sick building syndrome in educational settings

Validation also showed that multiple IAQ parameters were strongly associated with health complaints, especially in mechanically ventilated environments.

doi: 10.7717/peerj.20964

Jung et al. 2026 J Expo Sci Environ Epidemiol

Development of a high-resolution PM_{2.5} model in Taiwan using integrated low-cost air quality sensors and satellite observations

The model demonstrated excellent performance, with a 10-fold CV R^2 (RMSE) of 0.98 (1.23 $\mu\text{g}/\text{m}^3$).

doi: 10.1038/s41370-026-00953-9

Kos Sinhala et al. 2026 Int J Environ Health Res

Determinants of in-home personal exposure to black carbon: the MobiliSense sensor-based study

Our study suggests that in-home pollution can be reduced by improving home features, such as heating systems, or by reducing the infiltration of outdoor air pollutants into indoor microenvironment either by using air filters or redesigning the ventilation system.

doi: 10.1080/09603123.2026.2702527

Majeed 2026 PLoS One

Air quality index prediction using machine learning regression models: A comparative analysis

The results indicate that the models effectively predicted AQI using evaluation metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the Coefficient of Determination (R^2).

doi: 10.1371/journal.pone.0349858

Mamani-Sucilla 2026 Sensors (Basel)

PM_{2.5}/PM₁₀ Forecasting System with Benchmarking of 44 Machine Learning Algorithms and Ensemble Learning Approaches

This unified comparative analysis-applying consistent metrics (MAE, RMSE, R^2 , and MAPE), the same prediction horizon, and a shared dataset-provides replicable empirical evidence that had not previously been reported for the urban context of Arequipa.

doi: 10.3390/s26134315

Olivares Lopez et al. 2026 Environ Pollut

Sensitivity of ultrafine particle and black carbon simulations to model resolution and emission data in urban environments

At 5 km resolution, HERMESv3 reduces the systematic underestimation of pollutant concentrations compared to EMEP.

doi: 10.1016/j.envpol.2026.128618

Rocha 2026 An Acad Bras Cienc

Non-destructive characterization of atmospheric particulate matter using SEM-EDS: Elemental mapping as a tool for chemical composition assessment

Blank filter characterization revealed that fiberglass filters contained Na, Ba, and Zn (~6%), while quartz filters is cleaner.

doi: 10.1590/0001-3765202620250649

Ryu et al. 2026 Environ Pollut

Spatial variability and pollutant-specific mitigation effects of an urban vegetated park observed with a high-density air quality sensor network

These results suggest that the limited PM_{2.5} reduction observed in this study is partly attributable to the small spatial extent of the park, highlighting the importance of vegetation area, density, and continuity for effective PM_{2.5} exposure mitigation.

doi: 10.1016/j.envpol.2026.128696

Salari 2026 Sci Rep

Hybrid multilayer perceptron models optimized by evolutionary algorithms for urban air quality forecasting: a case study of Shiraz, Iran

The aim is to enhance the prediction of key air quality indicators, carbon monoxide, ozone, nitrogen dioxide, sulfur dioxide, and the Air Quality Index, using long-term air quality measurements collected from Shiraz Province, Iran, from April 2019 to November 2024.

doi: 10.1038/s41598-026-52228-w

Shishegaran 2026 Environ Sci Pollut Res Int

Ensemble model based on time series and regression to predict daily air quality in Tehran

The results show that the proposed second ensemble model (Model 4) achieved the highest prediction accuracy across all seasons, with coefficients of determination (R^2) ranging from 0.942 to 0.969 and RMSE values between 3.861 and 12.783 in the validation dataset.

doi: 10.1007/s11356-026-37989-1

Singh et al. 2026 Environ Sci Pollut Res Int

A spatiotemporal machine learning framework for high-resolution PM_{2.5} estimation using reconstructed satellite aerosol observations

Multiple ML models are evaluated, and a multi-layer perceptron is identified as the best-performing estimator, achieving a coefficient of determination (R^2) of 0.82 with substantially reduced prediction errors compared to models driven by raw satellite inputs.

doi: 10.1007/s11356-026-37976-6

Sribudda 2026 Stud Health Technol Inform

Development of a Hospital Air Quality Monitoring Application Utilizing IoT Technology and Google Workspace: A Case Study of a Hospital in Maharakham, Thailand

By integrating IoT technology with low-code platforms, the system reduces administrative workload and promotes sustainable risk management practices.

doi: 10.3233/SHIT260873

Yang 2026 Anal Chem

A Chemical and Biological Portable Mass Spectrometer (CB-pMS) with Automated Sampling and Hybrid Ionization for Online Detection of Bioaerosols and VOCs

The CB-pMS achieves detection limits of 1 ng/mL for liquid-phase hyacinthine and 10 ppbv for gas-phase dimethyl methyl phosphonate (DMMP), exhibiting excellent linearity ($R^2 > 0.99$) and MS/MS confirmation capability for analyte identification.

doi: 10.1021/acs.analchem.6c03165

Yu et al. 2026 Environ Sci Technol

Integrating audiovisual data for short-term particulate matter concentration prediction on urban sidewalks

LightGBM achieved the best performance, yielding R^2 values of 0.64-0.65 for 10 s predictions and 0.80 for 1 min predictions under random cross-validation.

doi: 10.1021/acs.est.6c03195

Exposure assessment & modelling

39 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (13 OF 39 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026 J Hazard Mater*Dual-isotope evidence for shipping contributions to atmospheric nitrate in coastal Chinese cities*

615 daily PM_{2.5} samples across Tianjin, Qingdao, Shanghai, Xiamen and Guangzhou during the cold-season transition, with nitrate $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ fed to a Bayesian isotope mixing model against an emissions-inventory prior. Shipping-derived nitrate ranged $17.6 \pm 4.7\%$ (**Tianjin**) to $37.9 \pm 11.4\%$ (**Xiamen**). Northern cities show shipping contribution rising with pollution severity and marine air-mass association; southern cities show it falling as inland transport intensifies. The inference threat is the mixing-model prior — source endmember $\delta^{15}\text{N}$ ranges for shipping overlap substantially with those for coal combustion, and no endmember sensitivity analysis is reported. Relevant to sensing because it quantifies how much of coastal PM_{2.5} mass a mass-only monitor cannot attribute.

doi: 10.1016/j.jhazmat.2026.143129

Chen et al. 2026 J Hazard Mater*Geospatial-based neighborhood hazards and asthma in adults: interaction with social vulnerability*

Exposome-wide association study over 29 residential hazards — air pollutants, industrial VOC emissions, highway density and proximity to 13 pollution-source classes — in 7,092 North Carolina adults (2013–2022), stratified at the median census-tract Social Vulnerability Index, with quantile g-computation for the mixture. In the overall population only proximity to dry-cleaning-solvent-contaminated sites was associated with asthma. In high-SVI tracts, **eight hazards — predominantly PM_{2.5} constituents and VOCs — were positively associated at FDR < 0.05**, and each one-quintile increase in the hazard mixture carried **35% higher asthma odds (95% CI 6–73%)**. Toluene, dry-cleaning proximity, PM organic matter and carbon monoxide each contributed over 10% of the joint association. Hazard correlations were themselves stronger in high-SVI areas, which is both the substantive finding and the analytical problem: with collinearity differing by stratum, the g-computation weights are not directly comparable across strata. Cross-sectional, self-reported physician-diagnosed asthma, residence-only exposure.

doi: 10.1016/j.jhazmat.2026.143083

Chengdu supersite 2026 J Environ Sci (China)*Gas-phase reactions accelerate nitrate formation in spring and early summer, Sichuan Basin*

Urban supersite speciation of PM_{2.5} over April–June 2018–2021 in Chengdu, targeting the under-studied warm-season nitrate problem. During complex PM_{2.5}/O₃ episodes nitrate rose **47.8%, from 5.52 to 8.16 $\mu\text{g m}^{-3}$** , with its share of PM_{2.5} climbing from 13.4% to 22.2%. The NO₃[−]/PM_{2.5} ratio scales with aerosol liquid water, and rapid early-morning ALW growth drives hygroscopic growth and heterogeneous production on top of the gas-phase channel. Directly consequential for optical sensing: a nitrate- and water-dominated aerosol has a very different hygroscopic growth factor and refractive index from the winter carbonaceous mixture most low-cost sensor calibrations are fitted to, so a humidity correction trained in winter will bias warm-season estimates. Observational and single-city, so the mechanism is inferred from covariation rather than tested.

doi: 10.1016/j.jes.2026.02.002

Ding et al. 2026 Geo-spat Inf Sci*Urban–rural balanced estimation framework for PM_{2.5} exposure across China*

Targets a structural bias rather than a model-fit problem: because ground stations cluster in cities, AOD-to-PM_{2.5} retrievals systematically overestimate rural concentrations. Integrates AOD, reanalysis fields and ground measurements under data-assimilation principles, adaptively constraining non-urban estimates with low-weighted reanalysis samples, then ensemble-averages across dynamically weighted background/observation settings. China 2015–2021: **clean-site RMSE 18.18 → 8.31 $\mu\text{g m}^{-3}$, national mean PM_{2.5} down ~55%**, with corresponding correction of PM-attributable mortality in low-density areas. The 55% figure is the paper's own counterfactual, validated at clean sites that are still station locations — true rural accuracy remains unvalidated by construction. If even partly right, it implies an inherited upward exposure bias across the satellite-based exposure–response literature.

doi: 10.1080/10095020.2026.2624335

Hu et al. 2026 Geo-spat Inf Sci*Fusion of ground monitoring, satellite retrieval and CTM for seamless hourly 1-km PM_{2.5}*

Fills satellite AOD gaps with continuous WRF-Chem AOD, injects physics-informed features (PM_{2.5} temporal components, AOD vertical structure) from the CTM, then predicts with a geospatially aware LightGBM that exploits monitoring-site spatial structure. Site-based cross-validation R^2 0.87, RMSE 13.42 $\mu\text{g m}^{-3}$, MAE 8.68 $\mu\text{g m}^{-3}$, with day-and-night coverage demonstrated on one pollution episode. Site-based CV overstates performance away from stations — the standard weakness — and an RMSE of 13.4 is large relative to the WHO annual guideline, so this is useful for episode mapping and weak for long-term exposure assignment. The physics-feature injection is the transferable step.

doi: 10.1080/10095020.2026.2680855

Nguyen et al. 2026 Eng Res Express*Satellite-driven PM_{2.5} prediction: machine learning and explainable AI*

Systematic review of 2015–2025 work integrating satellite products (LEO and geostationary AOD, top-of-atmosphere reflectance, Ångström exponent) with ML/DL. Finds Random Forest still a robust benchmark, hybrid and transformer architectures better at spatiotemporal dependence, and AOD gap-filling plus normalisation the dominant determinants of model stability; positions XAI as the route to physical plausibility. No pooled quantitative comparison and no protocol registration, so the ranking claims are narrative. Adds little beyond Hu and Ding above; retained as a citation anchor for the physics-guided-learning argument.

doi: 10.1088/2631-8695/ae7028

Nyoni et al. 2026 Air Qual Atmos Health*Performance of road-traffic-based exposure proxies against personal PM_{2.5} measurements in three sub-Saharan African countries*

One year of filter-based outdoor personal PM_{2.5} from 343 postpartum participants in The Gambia, Kenya and Mozambique, used to validate population-weighted road network density (wRND), Euclidean distance to highway (EH) and to main road (EM). Spearman correlations were weak and **sign-inconsistent across countries**: wRND significant only in Mozambique ($\rho = 0.351$), EH positive in The Gambia (0.335) and Mozambique (0.292) but **negative in Kenya** (-0.224 , $p = 0.018$). Single-proxy random forests gave $R^2 < 0.45$ everywhere; combined models reached $R^2 = 0.52$ (RMSE 31.7 $\mu\text{g m}^{-3}$) in Kenya and 0.60 (8.9 $\mu\text{g m}^{-3}$) in Mozambique. The sampled population — postpartum women — is not representative of general mobility patterns, and only outdoor-filtered personal exposure was measured, so the household cooking contribution that most plausibly drives the proxy failure is inferred rather than partitioned. Still the strongest available empirical case that road-geometry proxies require local validation before use, and a direct argument for node deployment over GIS surrogacy.

doi: 10.1007/s11869-026-02060-y

Pająk et al. 2026 Toxics

Toxicological legacy of PAHs from an urban tire fire: soil contamination and cancer risk

Source-to-receptor assessment of a large landfill tire fire coupling HYSPLIT dispersion modelling against air-quality monitor data with soil sampling on the fireground and in surrounding allotment gardens, screened for Zn as a tire tracer and 16 priority PAHs. Emissions **PM₁₀ 1.34 t and PAHs 17.7 kg**; fireground soil ΣPAH 148.9 mg kg⁻¹; cumulative child ILCR 9.7×10^{-4} , an order of magnitude above the 10^{-4} regulatory benchmark, with dermal contact rather than inhalation the dominant pathway and elevated risk (10^{-5} – 10^{-4}) persisting at distal residential sites. Ecological risk quotients exceeded unity for PAHs everywhere and for Zn/Cu near the fire. Limitation: no pre-fire soil baseline, so urban background PAH — itself substantial in a coal-heating region — is attributed to the fire; and HYSPLIT at this scale is validated against a sparse monitor network. The transferable point is that acute source events deposit a multi-year soil legacy that airborne monitoring alone will not detect.

doi: 10.3390/toxics14070543

Patwary 2026 Res Sq (preprint)*Spatio-temporal assessment of PM₁₀ and meteorological correlates in Cumilla, Bangladesh*

Single Department of Environment monitoring station, January–April 2026, Pearson correlation and trend analysis against temperature, RH and rainfall. Dry-season peak **210.7 µg/m³ monthly mean PM₁₀ in February**, roughly 4.7× the WHO 24-h guideline. No instrument model, calibration history or data-completeness statistics are reported, and one station cannot support the “spatio-temporal” framing in the title. Logged because South Asian monitoring records are structurally scarce in this corpus, not because the analysis is sound.

doi: 10.21203/rs.3.rs-10519411/v1

Raj & Dhada 2026 Res Sq (preprint)*Spatiotemporal ambient air quality across urban, tourist and industrial Himachal Pradesh*

Five Himalayan towns (Solan, Baddi, Shimla, Kullu, Manali) contrasted by land use, 25 June–24 July 2026, 150 site-days, using AQI plus PM_{2.5}, PM₁₀, CO, SO₂, NO₂ and O₃ taken from the public AQI.in platform. The land-use contrast (industrial Baddi vs. tourist hill towns) is the design's one genuine idea. The fatal weakness is provenance: AQI.in aggregates heterogeneous and partly low-cost sources with no published calibration chain, so between-town differences cannot be separated from between-instrument differences. Preprint, not peer reviewed.

doi: 10.21203/rs.3.rs-10492149/v1

Sanchez et al. 2026 Indian J Occup Environ Med*Morbidity related to the level-two eruption of Popocatepetl*

Ecological correlational study in communities 15 km from the crater, May 2023, with daily PM₁₀/PM_{2.5} and Kriging used to interpolate ash dispersion because the monitoring network is too sparse to resolve it. 14% of presentations were ash-related, predominantly non-infectious respiratory; Spearman correlations between ash exposure and morbidity were not statistically significant. The authors read this as possible biological adaptation, which the design cannot support — a null in a small ecological study with interpolated exposure is far more likely to be measurement error and low power. Its real value here is as a candid worked example of exposure assignment where no formal network exists.

doi: 10.4103/ijocem.ijocem_146_24

Solórzano-Araque et al. 2026 GeoHealth*Multiscale exposure modelling with LOTOS-EUROS, mobility surveys and particle toxicity*

Medellín / Aburrá Valley, 2019. LOTOS-EUROS CTM driven by WRF at 1 km×1 km; mobility from an origin–destination survey of >180,000 trips across 240 traffic analysis zones; morpho-chemical characterisation of metals and carbonaceous fractions; and a weighted exposure model combining exposure time, concentration and cytogenotoxic indicators. Cluster analysis identified central and southern zones and highway corridors — ~60% of the metropolitan population — exceeding 40 µg/m³ at peak hours. The toxicity-weighting step is the transferable idea.

doi: 10.1029/2025GH001621

Zhang et al. 2026 npj Clim Atmos Sci*High-resolution aerosol liquid water content across the contiguous US using machine learning*

ML surrogates trained on GEOS-Chem output to learn ALWC–predictor thermodynamics, then applied to bias-corrected high-resolution inputs, giving **1 km daily ALWC for 2000–2019**. Outperforms the parent CTM on daily variability and spatial heterogeneity. Regional ordering Midwest > East > West is driven by PM_{2.5} mass, composition, temperature and RH; the secular decline tracks sulfate. Demonstrated as a constraint on water-soluble iron variability. The limitation is structural: the ML model inherits every bias of GEOS-Chem's thermodynamic module, so this is a downscaling of a simulation, not an observational product — there is no independent ALWC validation set. Still the most directly useful record here for hygroscopic-growth correction of optical sensors.

doi: 10.1038/s41612-026-01371-2

RECOVERED BY THE MONTH-WIDE HARVEST — EXTRACTIVE PRECIS, NOT AN ASSESSED SUMMARY (26 RECORDS)

Abinash 2026 Environ Monit Assess*Modulation of fine aerosol ionic profiles and chemical characteristics by crop residue burning over the North-Western Indo-Gangetic Plain*

PM_{2.5} concentrations increased significantly from 48.5 ± 22.6 µg m⁻³ (pre-CRB) to 77.6 ± 24.3 µg m⁻³ during CRB, showing a 60% increase during the CRB phase.

doi: 10.1007/s10661-026-15612-7

Al-Saleh 2026 Chemosphere*Long-term trends and exposure implications of ambient air pollution in Riyadh, Saudi Arabia: A nine-year multi-site assessment*

Temporal analyses showed declines in PM_{2.5} (-40.7%), PM₁₀ (-32.5%), and CO (-22.1%), alongside modest increases in NO₂ (+7.8%) and O₃ (+9.9%).

doi: 10.1016/j.chemosphere.2026.145018

Christoforou et al. 2026 Int J Popul Data Sci*Ascertaining household-level exposure to air pollution in people with schizophrenia and other psychotic disorders*

Area-level deprivation was associated with higher prevalence of schizophrenia/OPD, whereas no clear differences were observed by urbanicity or air pollution exposure.

doi: 10.23889/ijpds.v11i1.3360

Collarin 2026 Environ Pollut*Semi-parametric methods for long-term trend detection in air quality: The North-Eastern Italian Alps case*

Model performance varied by pollutant, with higher accuracy for gases (R₂ up to 0.86 for O₃) than for particulate matter (R₂ of approximately 0.66).

doi: 10.1016/j.envpol.2026.128671

Devan 2026 Environ Toxicol Chem*Integrated weighted geospatial analysis and spatiotemporal assessment of key criteria air pollutants for hotspot identification and air quality risk assessment*

Monsoon rains significantly reduced pollutant levels.

doi: 10.1093/etojnl/vgag151

Firooz et al. 2026 Environ Pollut*Characterization of traffic-related particulate matter in an urban tunnel: source apportionment of exhaust and non-exhaust emissions*

Results indicated that the EF for PM_{2.5} and PM₁₀ were significantly higher during the warm days (26 ± 7.7 for PM_{2.5}, 40.4 ± 9.7 for PM₁₀) than during the cold days (12.9 ± 1.3 for PM_{2.5} and 16 ± 1.14 for PM₁₀).

doi: 10.1016/j.envpol.2026.128766

Goel 2026 Environ Monit Assess*Ambient black carbon and its dependence on in situ stubble burning*

Absorption Ångström exponent (AAE) was also calculated and observed to be 1.7 ± 0.3 , with hourly variations ranging from 1.1 to 2.4, suggesting a significant influence of CRB.

doi: 10.1007/s10661-026-15679-2

Hoppa 2026 Sci Rep*Roadside trees along school routes enhance foliar particulate-matter retention but increase children's pedestrian-level exposure*

Higher horizontal crown density (LAI), by contrast, improved foliar retention - a 1% increase in LAI was associated with 0.52% and 0.56% higher foliar PM₁ and PM_{2.5}, respectively.

doi: 10.1038/s41598-026-59046-0

Lee 2026b J Environ Manage*Spatiotemporal response of urban bike-sharing ridership to weather, air quality, and future climate change in major U.S. cities*

Projecting these sensitivities forward, we show that future warming will enhance annual cycling suitability, particularly in seasonally cold cities, by reducing prohibitive winter days.

doi: 10.1016/j.jenvman.2026.130431

Liu 2026 J Environ Sci (China)*Synergistic control of greenhouse gases and air pollutants in the Greater Bay Area, China: Regional, sectoral and supply chain pathways*

Regionally, these emissions originated mainly from resource-intensive provinces such as Shandong, Hebei, and Henan, while at the city level, flows from Foshan and Guangzhou to Hong Kong dominated, contributing 64 % of intercity embodied emissions.

doi: 10.1016/j.jes.2026.01.062

Lu 2026 Bull Environ Contam Toxicol*Combustion Sources Dominated Ammonia Concentrations During Severe Air Pollution Episodes During "Two Sessions" 2025 in Beijing*

An unexpected heavy pollution episode occurred during this period, with NH₃ concentrations reaching 59.7 ppb, significantly higher than those observed on clean days (17.8 ± 4.5 ppb).

doi: 10.1007/s00128-026-04287-y

Mărmureanu 2026 Environ Pollut*Spatial clustering of air pollutant trends and exposure burden across Romania (2013-2024)*

The Mann-Kendall test, Sen's slope estimator, and Local Indicators of Spatial Association (LISA) are used to quantify trends and detect spatially coherent clusters.

doi: 10.1016/j.envpol.2026.128759

Nam 2026 Bull Environ Contam Toxicol*Assessment of Air Pollution in Hai Phong City (Vietnam) Using Lichens as Biomonitors*

Analysis data showed that mean content of all elements, except Hg and S, was higher in exposed lichens.

doi: 10.1007/s00128-026-04301-3

Sablan 2026 Geohealth*Wildfire Smoke and Cardiorespiratory Emergency Visits in New Mexico 2022: Sensitivity to Exposure Estimates and Referent Periods*

Our study compared a range of exposure methods: (a) PM_{2.5} from the Environmental Protection Agency (EPA) regulatory-grade monitors, (b) PM_{2.5} from both the EPA regulatory-grade monitors and low-cost PurpleAir observations, (c) modeled 24-hr average wildfire smoke PM_{2.5} from the Community Multiscale Air Quality Modeling System (CMAQ), and (d) CMAQ daily 1-hr maximum ...

doi: 10.1029/2025GH001492

Su et al. 2026 Environ Sci Technol*Industrial condensable particulate matter: a missing source in PM_{2.5} source apportionment driving winter pollution in China*

Based on receptor modeling, CPM contributes a median of 3.6% (95% CI: 0.9%-14.5%) to urban PM_{2.5} across all four seasons in 2023, in the industrialized cities examined (Shanghai, Xi'an, Hefei, Shijiazhuang).

doi: 10.1021/acs.est.6c02727

Tang 2026 PLoS One

Global and local impacts of multi-dimensional urban form on PM_{2.5} concentrations: A case study of the downstream urban agglomeration in the Yangtze River Delta
The regional annual average PM_{2.5} concentration shows a continuous downward trend (decreasing from 60.77 $\mu\text{g}/\text{m}^3$ in 2014 to 29.52 $\mu\text{g}/\text{m}^3$ in 2022), with reductions ranging from 29.55% to 40.61% across individual cities.

doi: 10.1371/journal.pone.0354380

Teng 2026 Environ Sci Technol*Greenland Emissions Contribute to Black Carbon Aerosols in the Arctic Marine Boundary Layer*

rBC concentrations were highest in the Greenland coastal region (2.4 ng m⁻³), compared with the open ocean (1.7 ng m⁻³) and the marginal ice zone (0.6 ng m⁻³).

doi: 10.1021/acs.est.6c03343

Wang 2026c Environ Sci Technol*Nighttime Organic Nitrates Drive New Particle Formation and Aerosol Growth in Urban Beijing*

This study presents observational evidence from summer 2017 in urban Beijing, supported by model simulations and a case study from summer 2023, demonstrating the crucial role of nighttime organic nitrates (ONs) production in subsequent daytime SOA formation.

doi: 10.1021/acs.est.6c01200

Xiang 2026 Environ Sci Technol*Identifying Tracing Species from Source Regions of PM_{2.5} on the Tibetan Plateau by Integrating Measurement and Modeling*

However, increased contribution from the northwestern China to PM_{2.5} was associated with increased concentrations of low-molecular-weight polycyclic aromatic hydrocarbons, Cl-, and several anthropogenic metals, reflecting impacts from mixed industrial and residential sources.

doi: 10.1021/acs.est.6c04954

Xie 2026 Proc Natl Acad Sci U S A*Matrix effects reshape organic aerosol volatility and atmospheric persistence*

The results show that species in simplified proxies and reference mixtures with limited components follow a higher-volatility trend that approaches their intrinsic values, whereas species present in ambient and biomass-burning organic aerosols exhibit the opposite behavior, with systematically reduced apparent volatility attributable to matrix effects.

doi: 10.1073/pnas.2614944123

Yan 2026 Sci Rep

Evaluating the impact of different green infrastructure strategies on particulate pollutant concentrations in street canyons: a numerical simulation study in Qingdao
In contrast, street trees and multi-layer GI configurations tended to increase PM10 concentrations in the leeward zone, with the maximum leeward ARDC indicating an approximately 9% increase relative to the non-vegetated reference scenario.

doi: [10.1038/s41598-026-59675-5](https://doi.org/10.1038/s41598-026-59675-5)

Zerin 2026 Environ Sci Pollut Res Int

Integrated spatial analysis and Gaussian dispersion modelling of airborne particulate matter from rice processing mill hubs in rural Bangladesh and associated public health risks

AirQ⁺ health risk analysis revealed that PM_{2.5} exposure contributed to 39.78% of ALRI-related mortality in children under five, while in adults (≥ 30 years), it accounted for 48.01% of Chronic Obstructive Pulmonary Disease (COPD), 56.25% of Ischemic Heart Disease (IHD), and 65.05% of stroke deaths.

doi: [10.1007/s11356-026-38036-9](https://doi.org/10.1007/s11356-026-38036-9)

Zhang 2026b Environ Sci Technol

Investigating Ultrafine Aerosol Turbulent Fluxes during Atmospheric New Particle Formation Events

These findings suggest that NPF events associated with entrainment from the overlying residual/stable layer during planetary boundary layer growth represent a significant and potentially underestimated source of boundary layer aerosols.

doi: [10.1021/acs.est.6c06924](https://doi.org/10.1021/acs.est.6c06924)

Zhang et al. 2026b Environ Int

Extrapolating personal exposure measurements to air pollution from outdoor sources into annual mean estimates

We found that ambient measurements assigned by the nearest monitor approach were consistently higher than those estimated by the matched-scaled approach and higher than personal exposure to air pollution from outdoor sources for all pollutants, indicating potential bias in epidemiological associations, under specific conditions concerning variability and random error.

doi: [10.1016/j.envint.2026.110414](https://doi.org/10.1016/j.envint.2026.110414)

Zhou 2026 Environ Pollut

Identifying sources of black carbon associated with high mortality risk in Beijing, China based on long-term hourly continuous measurements

The findings indicated that the health risk of BC could vary by source, with traffic BC posing the highest non-accidental mortality risk (RR: 1.014, 95% CI: 1.003, 1.025), followed by biomass burning and atmospherically aged BC.

doi: [10.1016/j.envpol.2026.128791](https://doi.org/10.1016/j.envpol.2026.128791)

Zhu 2026 J Environ Sci (China)

Significant contributions of sea spray aerosol to vertical CINO₂ formation over coastal South China

Moreover, SSA not only enhances CINO₂ production but also increases the mixing ratios of chlorine and hydroxyl radicals at higher altitudes the following day (400 m), significantly boosting AOC, with an increase in AOC of up to 10 %.

doi: [10.1016/j.jes.2026.01.008](https://doi.org/10.1016/j.jes.2026.01.008)

Cardiovascular & metabolic**26 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (11 OF 26 RECORDS IN THIS SUBTOPIC)

Belay et al. 2026 Eur J Prev Cardiol

Long-term bushfire-related PM_{2.5} and coronary plaque burden

365 asymptomatic adults aged 40–70 with a family history of CAD, serial coronary CTA at baseline and 3 years, with daily total and bushfire-attributed PM_{2.5} from validated 5 km models assigned by residential postcode. Per 1 µg m⁻³ cumulative bushfire PM_{2.5}: **total plaque volume +10.14 mm³ (1.45–18.82)** at baseline and +4.92 mm³ (1.33–8.52) at follow-up; total PM_{2.5} associations were weaker. Note the exposure setting — 10-year means of 5.88 and 5.93 µg m⁻³, i.e. near the WHO guideline — which makes a plaque-volume signal notable but also means the bushfire and total metrics are highly collinear at these levels. Postcode-level assignment in a 365-person sample is the binding limitation.

doi: [10.1093/eurjpc/zwag399](https://doi.org/10.1093/eurjpc/zwag399)

Chen et al. 2026 Environ Pollut

Long-term PM exposure and coronary inflammation by pericoronary fat attenuation index

5,528 CCTA patients with a 656-person serial subcohort; FAI extracted for LAD, LCx and RCA by a deep-learning pipeline; individual PM_{2.5}, PM₁₀ and black carbon from a validated 1 km spatiotemporal model; GAM and linear mixed-effects models. Per IQR of 1-year exposure, **RCA FAI rose 7.57 (PM_{2.5}), 8.72 (PM₁₀) and 2.35 HU (BC)**, with accelerated annual FAI progression longitudinally. FAI is technically sensitive to tube voltage, contrast timing and epicardial adiposity, and a retrospective clinical population undergoing CCTA is selected on suspected disease — so absolute HU shifts should not be read as population effects. The local-inflammation endpoint nonetheless complements Belay's structural one.

doi: [10.1016/j.envpol.2026.128840](https://doi.org/10.1016/j.envpol.2026.128840)

Chrysohoou et al. 2026 Eur J Prev Cardiol

Preventing cardiovascular disease within the built and living environment: an expert opinion

Expert opinion placing air and noise pollution alongside food environment, green space, physical activity and psychosocial stress in a single built/living/social framework, with inflammation, oxidative stress, neuroendocrine dysregulation and chronic stress as the shared mechanistic substrate. No new data and no quantitative synthesis; its value is as a citable framing document for grant and introduction text, nothing more.

doi: [10.1093/eurjpc/zwag387](https://doi.org/10.1093/eurjpc/zwag387)

Huang et al. 2026 Zhonghua Liu Xing Bing Xue Za Zhi

PM_{2.5} and dyslipidaemia in rural Chinese children; dietary effect modification

10,788 children aged 8–15 across 103 surveillance counties; 3-year mean PM_{2.5} 30.36±10.62 µg/m³; dyslipidaemia prevalence 21.84%. **OR 1.501 (1.028–2.192)**, with no significant association among children with greater vegetable intake diversity. Effect-modification-by-diet claims of this type are fragile to residual confounding, but the sample size is substantial.

doi: [10.3760/cma.j.cn112338-20251120-00843](https://doi.org/10.3760/cma.j.cn112338-20251120-00843)

Kaźmierska et al. 2026 Przegl Epidemiol

Air pollution and cardiovascular disease: review of current data

Standard mechanistic review (oxidative stress, inflammation, endothelial dysfunction, autonomic imbalance) anchored to the 2021 WHO 5 µg/m³ annual guideline. Adds little new; useful only as an introductory citation.

doi: [10.32394/pe/218966](https://doi.org/10.32394/pe/218966)

Luo et al. 2026 Front Public Health

Short-term ambient air pollution and cardiovascular mortality: an 11-year time-series in Nanning

87,913 CVD deaths, 2014–2024; generalized additive quasi-Poisson adjusted for long-term trend, temperature, RH and day of week, across single-day lags 0–7 and cumulative windows 0–1 to 0–7. At lag 0, **RR 1.014 (1.008–1.020) for PM_{2.5}**, 1.009 (1.006–1.013) for PM₁₀ and 1.018 (1.011–1.026) for NO₂; SO₂ peaks at lag 1 with **1.044 (1.018–1.069)**, the largest coefficient in the set. Subtropical southwestern China is genuinely under-represented in this literature. Two-pollutant models were run, but the paper does not resolve the SO₂/PM_{2.5} collinearity its own results imply, so the apparent SO₂ dominance should not be read as source-specific.

doi: [10.3389/fpubh.2026.1878398](https://doi.org/10.3389/fpubh.2026.1878398)

Rusconi et al. 2026 GeoHealth*Long-term ultrafine particle exposure 2010–2019 and cardiovascular burden in high-income countries*

Applies a new global UFP dataset to compute population-weighted exposure for high-income countries and correlates it with country-level CVD DALYs, YLLs and YLDs, reporting “consistent patterns” between long-term UFP exposure and CVD burden. This is a country-level ecological correlation with no individual data, no lag structure and no control for the PM_{2.5} with which UFP is strongly collinear — the authors say so, restrict to high-income countries deliberately, and warn against generalisation. It cannot support causal or even population-attributable inference. Included only because it is the corpus’s sole ultrafine-focused record, which is itself the point: UFP epidemiology still lacks the exposure infrastructure PM_{2.5} has.

doi: [10.1029/2025GH001636](https://doi.org/10.1029/2025GH001636)

Xi et al. 2026 Diabetologia*Air pollution, built environment and temperature vs. prediabetes phenotypes (KORA)*

KORA cohort, Southern Germany 2006–2022, 1,618 participants, complementary log-log models plus quantile g-computation for mixtures. Per-IQR PM_{2.5} gave **HR 1.89 (1.07–3.31) for combined IFG+IGT**; lower NDVI HR 1.59 (1.03–2.46). Crucially, associations were with *incidence and non-remission* of impaired glucose tolerance rather than progression to type 2 diabetes — a phenotype-resolution result that most air-pollution/diabetes studies cannot deliver.

doi: [10.1007/s00125-026-06814-2](https://doi.org/10.1007/s00125-026-06814-2)

Yang et al. 2026 J Am Heart Assoc*Short-term PM_{2.5} and hospitalisation/ED use in atrial fibrillation*

2021 Medicare data, 26,158 western-US patients with established AF; ZIP-code daily PM_{2.5} from a 1 km geospatial model, threshold $\geq 9 \mu\text{g}/\text{m}^3$, lags 0/1/3 d. **OR 1.10 (1.01–1.20) for CVD hospitalisation; 1.24 (1.03–1.48) for AF hospitalisation; 1.34 (1.06–1.70) for AF ED visits**. No significant PM $\times$ extreme-heat interaction — a negative finding worth noting given how often synergy is assumed.

doi: [10.1161/JAHA.125.046917](https://doi.org/10.1161/JAHA.125.046917)

Zhao et al. 2026 J Am Coll Cardiol*Compound extreme heat and fire-sourced air pollution vs. cardiovascular mortality*

Multi-country time-series across six countries, 5.9 million CVD deaths, with each day classified into mutually exclusive states (no hazard / heat-wave only / substantial fire-sourced air pollution only / compound) and quasi-Poisson RRs plus formal additive interaction. **Compound RR 1.115 (1.100–1.130)** vs. heat-only 1.083 (1.074–1.091) and SFAP-only 1.014 (1.008–1.021); RERI 0.018 (0.002–0.035), attributable proportion 1.598% (0.181–3.129). The interaction is statistically present but small — most compound risk is the heat term — and the mutually-exclusive day classification discards dose information within each state. Fire-sourced attribution rests on modelled smoke fields, so exposure misclassification is non-differential but real. Sets the design standard for compound-hazard work.

doi: [10.1016/j.jacc.2026.06.005](https://doi.org/10.1016/j.jacc.2026.06.005)

Zhou et al. 2026 iScience*Air pollution and hypertension: an umbrella review of systematic reviews and meta-analyses*

Umbrella review over three databases to September 2025, AMSTAR-2-appraised, retaining eleven meta-analyses (eight on essential hypertension, three on gestational hypertension). Conclusion: cumulative evidence supports PM_{2.5} and PM₁₀ raising essential hypertension risk, while the gestational-hypertension association remains inconclusive on three studies. The value of this record is as a citation anchor and a map of between-review heterogeneity; the abstract reports **no pooled estimate, no I^2 , and no AMSTAR-2 distribution**, so the strength of the “plausible role” claim cannot be checked without the full text. Umbrella reviews also inherit the primary literature’s exposure-model heterogeneity — land-use regression, satellite-derived and nearest-monitor assignments are pooled as though interchangeable, which is precisely the assumption a co-located sensor network is positioned to test.

doi: [10.1016/j.isci.2026.116811](https://doi.org/10.1016/j.isci.2026.116811)

RECOVERED BY THE MONTH-WIDE HARVEST — EXTRACTIVE PRECIS, NOT AN ASSESSED SUMMARY (15 RECORDS)

Huang 2026b Front Endocrinol (Lausanne)*Association between air pollution and incident cardiovascular diseases among a population with cardiovascular-kidney-metabolic syndrome stages 0-3: the first evidence from the China Health and Retirement Longitudinal Study*

Per 10- $\mu\text{g}/\text{m}^3$ increase in PM₁, PM_{2.5}, PM₁₀, O₃, and NO₂ corresponded to a 14.8% (hazard ratio [HR], 1.148 [95% CI, 1.086–1.214]), 9.6% (HR, 1.096 [95% CI, 1.064–1.128]), 6.8% (HR, 1.068 [95% CI, 1.052–1.084]), 2.5% (HR, 1.025 [95% CI, 0.962–1.091]), and 12.4% (HR, 1.124 [95% CI 1.063–1.189]) higher incident risk of CVD, respectively, ...

doi: [10.3389/fendo.2026.1852623](https://doi.org/10.3389/fendo.2026.1852623)

Kim 2026 Int Arch Occup Environ Health*Associations of short-, mid-, and long-term particulate matter exposure with elevated LDL-C levels in Korean adults: analysis of KNHANES 2016–2022*

We examined associations of short- (1–7 days), mid- (1–6 months), and long-term (1–5 years) PM₁₀ and PM_{2.5} exposures with elevated LDL-C ($\geq 160 \text{ mg}/\text{dL}$ or medication use) in Korean adults. Methods In this cross-sectional analysis of 14,867 Korean adults from the 2016–2022 the Korea National Health and Nutrition Examination Survey (KNHANES), a nationally representative survey.

doi: [10.1007/s00420-026-02220-8](https://doi.org/10.1007/s00420-026-02220-8)

Kuntic 2026 Herz*[Effects of air pollution on cardiovascular health]*

Long-term and short-term exposure to fine particulate matter and traffic-related pollutants is consistently associated with an increased risk of ischemic heart disease, stroke, heart failure, arrhythmia and cardiovascular death.

doi: [10.1007/s00059-026-05388-4](https://doi.org/10.1007/s00059-026-05388-4)

Lelieveld et al. 2026 Cardiovasc Res*Air quality and health implications of exposure to ultrafine particle pollution*

We estimate a mortality density of 35.7 (15.8–65.5) per 100,000 people annually in Europe, and 27.4 (12.9–47.4) per 100,000 in North America, the latter being close to the global mean.

doi: [10.1093/cvr/cvag136](https://doi.org/10.1093/cvr/cvag136)

Li 2026 Environ Pollut

Associations of green space and air pollution with insulin resistance: Analysis of repeated-measurements health examination data

Each standard deviation increase in NDVI was associated with a decrease in METS-IR ($\beta = -0.030$; 95% CI: -0.039 to -0.021).

doi: 10.1016/j.envpol.2026.128782

Li et al. 2026b Ecotoxicol Environ Saf

Sequential mediating roles of MMP-12 and DNA methylation in air pollution-induced atherosclerotic cardiovascular disease

The results indicated that an IQR increment of PM_{2.5}, PM₁₀, PM₁, and their components (BC, NH₄⁺, NO₃⁻, SO₄²⁻, OM) was significantly associated with elevated serum MMP-7 (percentage change: 3.8%-12.4%) and MMP-12 (7.7%-19.9%).

doi: 10.1016/j.ecoenv.2026.120473

Liang 2026 Environ Pollut

Synergistic effects of daytime, nighttime, and compound extreme temperature events and ambient air pollution on cardiovascular mortality

For each 10 $\mu\text{g}/\text{m}^3$ increment in pollutant concentration, the odds ratios (ORs) were 1.019 for PM_{2.5}, 1.006 for PM₁₀, and 1.005 for ozone.

doi: 10.1016/j.envpol.2026.128730

Liu 2026b Blood Press

Association of air pollutant levels and blood pressure variability in patients with essential hypertension: a distributed lag analysis study

For every 10- $\mu\text{g}/\text{m}^3$ increase in PM₁₀, the maximum lag effect on nDBPSD (night time diastolic blood pressure standard deviation) and nDBPCV (night time diastolic blood pressure coefficient of variation) was observed at lag 5, with increments of 0.57 (95% CI: 0.16, 0.99) and 1.00 (95% CI: 0.19, 1.8) respectively.

doi: 10.1080/08037051.2026.2705619

Luo 2026 J Am Coll Cardiol

Beyond PM_{2.5} Mass: Constituent-Specific Cardiovascular Risks and Implications for Air Pollution Control

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1016/j.jacc.2026.05.045

Paolisso et al. 2026 Eur Heart J

Micro- and nano-plastics in the coronary circulation and air pollution exposure in ischaemic heart disease presentation

STEMI patients showed higher levels of interleukin-6 and tumour necrosis factor- α ($P \leq .006$) and were exposed to higher levels of PM_{2.5} ($P \leq .012$).

doi: 10.1093/eurheartj/ehag447

Rajzer 2026 Cardiovasc Res

Ultrafine particles - rethinking air pollution risk

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1093/cvr/cvag161

Wang 2026 Int J Environ Health Res

Size-specific health effects of particulate matter components on cardiovascular mortality: a weekly time-series study in Tianjin, China

Generalized linear models with a quasi-Poisson distribution were used to estimate the excess risk (ER) associated with an interquartile range increase in exposure, with further analyses by cardiovascular disease (CVD) subtype, sex, and age, and additional adjustment for ozone (O₃) and nitrogen dioxide (NO₂).

doi: 10.1080/09603123.2026.2706019

Xu 2026 Ecotoxicol Environ Saf

Independent and interactive associations of green space and air pollution with blood lipid biomarkers: A longitudinal study

Each one-standard deviation (SD) increase in NDVI was associated with lower NHHR ($\beta = -0.103$; 95% CI: -0.114, -0.091) and AIP ($\beta = -0.009$; 95% CI: -0.012, -0.006).

doi: 10.1016/j.ecoenv.2026.120475

Yang et al. 2026 J Am Heart Assoc

Joint effects of heat and fine particulate matter on out-of-hospital cardiac arrest in Japan

Heat (relative risk, 1.04 [95% CI, 1.02-1.06]; 99th versus 80th percentile); and PM_{2.5} (relative risk, 1.02 per 10 $\mu\text{g}/\text{m}^3$ [95% CI, 1.01-1.03]) were independently associated with increased OHCA risk.

doi: 10.1161/JAHA.125.047917

Zhang et al. 2026c iScience

Association of joint exposure to PM_{2.5} and extreme heat with cardiovascular disease: a population-based cohort study

Each 1% of AREHD and AREHH increase reduced onset time by 0.446% (95% CI: 0.304-0.587) and 1.379% (95% CI: 1.043-1.714), stronger in the elderly, males, and sedentary individuals.

doi: 10.1016/j.isci.2026.116526

Neuro / mental health**19 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (10 OF 19 RECORDS IN THIS SUBTOPIC)

Calderón-Garcidueñas et al. 2026 J Alzheimers Dis

Abnormal eye movements reflect early cortical and volumetric brain changes in PM_{2.5}-exposed urban youth

Video-based eye tracking (EyeLink 1000-plus) in 80 volunteers aged 33±11 from Metropolitan Mexico City and lower-exposure Cuernavaca, with structural MRI in 45 MMC subjects. **Oculomotor variables did not differ significantly between cities** — the headline null. Abnormal antisaccades, low pursuit gain and square-wave jerks were merely *more frequent* in MMC, and the significant findings are correlations between oculomotor measures and frontal-temporal-parietal cortical, hippocampal, thalamic, caudate, amygdalar and cerebellar structure *within* the exposed group. With $n = 45$ and that many regions the multiple-comparison exposure is severe, and no correction is described. The group's long-running MMC series is valuable, but this instalment does not establish an exposure contrast.

doi: 10.1177/13872877261471048

Cattarinussi et al. 2026 medRxiv

Independent and joint effects of outdoor air pollution and genetic risk on adolescent mental health trajectories

ABCD Study, $n = 10,620$, ages 9–16, sex-stratified trajectories of internalising symptoms and psychotic-like experiences against residential PM_{2.5} and polygenic scores for schizophrenia and major depression. PM_{2.5} was associated with **smaller decreases in PLEs over time in females** ($p\text{-FDR} = 0.005$), with no effect on internalising trajectories in either sex, and a **PM_{2.5} × PRS-MDD interaction on PLE trajectories in females** ($p\text{-FDR} = 0.048$) — a borderline interaction that will not survive replication unless the effect is real. PM_{2.5} is residential-address-assigned, so within-person exposure misclassification is large and non-differential, which biases toward the null for main effects but has unpredictable sign for interactions. Preprint.

doi: 10.64898/2026.07.12.26357864

Dales et al. 2026 JAMA Netw Open*Wildfire-sourced PM_{2.5} and ED visits for migraine and other primary headache*

Case-crossover over 997,701 ED visits, 2010–2023 Alberta/Ontario, using the Canadian Optimized Statistical Smoke Exposure Model to separate wildfire-sourced from background PM_{2.5}. **+6.07% (5.75–6.39) per wildfire exposure**; background PM_{2.5} null. Effect modification by material/social deprivation is large (3.61% → 10.42% across quintiles). The methodological contribution — explicit source attribution within a case-crossover frame — is more transferable than the headache endpoint itself.

doi: [10.1001/jamanetworkopen.2026.25015](https://doi.org/10.1001/jamanetworkopen.2026.25015)

Glover et al. 2026 Age Ageing*Umbrella meta-review of urban environmental exposures and cognitive health*

54 systematic reviews, 66 meta-analysed outcomes, 270 exposure–outcome associations, graded by sample size, effect strength and bias. PM_{2.5} is the strongest-graded exposure (10/11 significant outcomes, “highly probable” detrimental), ahead of NO₂/NO_x (5/10). Social relationships were probably protective; green/blue space associated with lower dementia risk (3/6). Useful as a citation anchor for exposure hierarchy claims.

doi: [10.1093/ageing/afag215](https://doi.org/10.1093/ageing/afag215)

Goossens et al. 2026 Neurosci Appl*In utero air pollution exposure and pre-/postnatal brain development*

Imaging-focused review (fetal ultrasound + childhood/adolescent MRI) stratified by pollutant. Prenatal ultrasound studies in high-exposure regions show reduced biparietal diameter and head circumference; childhood MRI in low-exposure regions shows no global volume effect but regional cortical, subcortical grey and white matter differences. The exposure-contrast dependence of the discrepancy is the point worth carrying forward.

doi: [10.1016/j.nsa.2026.107019](https://doi.org/10.1016/j.nsa.2026.107019)

Kabir et al. 2026 Health Place*Environmental attributes as moderators of neighbourhood deprivation and mental health*

Systematic review across seven databases plus grey literature, 19 studies included, asking specifically whether objectively measured environmental attributes moderate the deprivation–mental-health association. Moderation findings were mixed: greenness and blue space generally protective in disadvantaged neighbourhoods, air pollution exacerbated poor mental health, parks and walkability inconsistent. Noise and elder-support centres were each examined in exactly one study. With 19 heterogeneous studies and no meta-analysis, the moderation claim is directional at best, and the authors’ call for longitudinal designs is the honest conclusion. Relevant if you intend to use LCS network output as an effect modifier rather than a main exposure.

doi: [10.1016/j.healthplace.2026.103719](https://doi.org/10.1016/j.healthplace.2026.103719)

Lu et al. 2026b Environ Res*Subchronic PM_{2.5} exposure exacerbates chronic stress-induced depression-like pathology in mice*

Factorial murine design crossing subchronic PM_{2.5} inhalation with chronic unpredictable mild stress (CUMS), read out by behavioural batteries, hippocampal histopathology, transcriptomics, electrophysiology and synaptic-plasticity markers. The result that matters is the **interaction, not the main effect**: PM_{2.5} alone produced no depression- or anxiety-like behaviour, only modest neurobiological change, whereas PM_{2.5} + CUMS worsened anhedonia, behavioural despair and anxiety relative to CUMS alone, with greater hippocampal neuroinflammation, redox disruption and impaired synaptic plasticity. This supports an effect-modifier rather than a direct-cause model. The abstract reports no exposure concentration, no particle characterisation and no effect sizes, so dose–response is unassessable and the inhalation regime cannot be compared with ambient exposure; “real-ambient” status is not claimed. For sensing, the implication is that a monitoring design aimed at mental-health endpoints needs to resolve *co-occurrence* of exposure and psychosocial stressor, which is a personal-monitoring problem, not a fixed-site one.

doi: [10.1016/j.envres.2026.125327](https://doi.org/10.1016/j.envres.2026.125327)

Ma & Yau 2026 iScience*PM_{2.5}-induced neurotoxicity and depression: exposure risks and mechanisms*

Narrative integration of epidemiological, animal and mechanistic evidence. Component-focused: metals, black carbon and organic compounds are implicated via oxidative stress, neuroinflammation, blood–brain barrier disruption, HPA-axis dysregulation and kynurenine-pathway alteration. Explicitly calls for component-resolved exposure assessment — relevant if your instrumentation work can deliver speciated output.

doi: [10.1016/j.isci.2026.116617](https://doi.org/10.1016/j.isci.2026.116617)

Pan et al. 2026 Zhonghua Yu Fang Yi Xue Za Zhi*Maternal PM exposure and offspring neurodevelopment: effects and mechanisms*

Chinese-language review covering placental barrier disruption and particle translocation, oxidative stress, epigenetic alteration, endocrine dysfunction and gut microbiota imbalance. Largely confirmatory of the above; included for completeness of the mechanistic map.

doi: [10.3760/cma.j.cn112150-20250804-00752](https://doi.org/10.3760/cma.j.cn112150-20250804-00752)

Sarkar et al. 2026 J Appl Toxicol*PM-induced neurotoxicity: neuroinflammation and organelle dysfunction*

Narrative review mapping CNS entry routes (olfactory-bulb translocation, blood–brain barrier disruption, peripheral inflammatory signalling) onto subcellular failure modes: unfolded protein response with CHOP and caspase-12 induction, impaired oxidative phosphorylation, ROS accumulation, mitochondrial permeability transition, and compromised autophagic flux and lysosomal integrity. Coherent as a mechanistic map but adds no new data, and — as with most of this literature — draws heavily on instillation and high-concentration in vitro models whose dosimetry does not correspond to ambient exposure. Use for the organelle-level framing, not for dose–response.

doi: [10.1002/jat.70360](https://doi.org/10.1002/jat.70360)

RECOVERED BY THE MONTH-WIDE HARVEST — EXTRACTIVE PRECIS, NOT AN ASSESSED SUMMARY (9 RECORDS)

Cao 2026 BMC Public Health*Short-term effects of ambient air pollutants on neurological hospital admissions: a time-series analysis in a northern Chinese city*

Exposure to PM_{2.5} (RR = 1.23, 95% CI: 1.14–1.32), NO₂ (RR = 1.18, 95% CI: 1.10–1.27), PM₁₀ (RR = 1.12, 95% CI: 1.06–1.19), and CO (RR = 1.10, 95% CI: 1.03–1.19) was associated with increased hospital admissions for overall neurological disorders.

doi: [10.1186/s12889-026-28245-9](https://doi.org/10.1186/s12889-026-28245-9)

Ghassabian et al. 2026 Environ Health Perspect*Prenatal air pollution exposure and autism spectrum disorder in the ECHO Consortium*

Higher O₃ was associated with higher SRS scores at the median (β per IQR increment = 0.83, 95% CI: 0.05, 1.61) and highest quantile (β = 2.19, 95% CI: 0.06, 4.32) in the meta-analysis.

doi: [10.1021/EHP.6c00106](https://doi.org/10.1021/EHP.6c00106)

Kwon 2026b NPJ Parkinsons Dis*Untargeted serum metabolomics and air pollution in Parkinson's disease*

Air pollution was associated with inflammation-related lipid metabolism (reduced arachidonate; elevated leukotriene), membrane phospholipid remodeling (reduced choline phosphate and LysoPC), amino acid metabolism (reduced aspartate and serine), and dopamine biosynthesis (reduced tyrosine) in PD patients.

doi: [10.1038/s41531-026-01451-3](https://doi.org/10.1038/s41531-026-01451-3)

Lichtiger 2026 Environ Res*Air pollution and autism-like traits: Sensitive periods and joint effects of exposure to sources and constituents of fine particulate matter*

Exposures to elemental carbon (EC) PM_{2.5} during gestation and to EC and traffic PM_{2.5} during the first year of life were associated with higher odds of autism-like traits (prenatal EC adjusted odds ratio (aOR): 1.72, 95% CI: 1.04-2.83; first year EC aOR: 1.72, 95% CI: 1.21-2.46; traffic aOR: 1.31, 95% CI: 1.01-1.71).

doi: [10.1016/j.envres.2026.125213](https://doi.org/10.1016/j.envres.2026.125213)

Ma et al. 2026 J Health Econ*Impacts of contemporaneous air pollution exposure on cognitive performance in Kenya*

Higher PM_{2.5} exposure during testing is associated with lower cognitive performance, particularly in executive function, memory, and visuospatial tasks.

doi: [10.1016/j.jhealeco.2026.103166](https://doi.org/10.1016/j.jhealeco.2026.103166)

Qi et al. 2026 Environ Health Perspect*The role of gut microbiota in the association between air pollution and cognitive function in older adults*

Higher air pollution levels were associated with increased circulating levels of 2-oxoglutarate and l-glutamine (key metabolites in neurodegenerative progression), which were further linked to higher odds of concurrent mild cognitive impairment (OR: 1.39-1.56) and an increased 2-year risk of cognitive decline (OR: 1.26-1.37).

doi: [10.1021/EHP.6c00026](https://doi.org/10.1021/EHP.6c00026)

Qiao 2026 iScience*Beyond air pollution: Green space, water minerals, and genetic predisposition in migraine onset*

Higher residential green space was associated with a reduced migraine risk (HR 0.90 [0.82-0.99]), whereas greater coastal distance increased risk (HR 1.30 [1.18-1.43]).

doi: [10.1016/j.isci.2026.116646](https://doi.org/10.1016/j.isci.2026.116646)

Tien-Smith et al. 2026 Environ Int*Long-term outdoor air pollution and incidence of Parkinson's disease, multiple sclerosis and motor neuron disease: a systematic review and meta-analysis*

There was no clear evidence that PM_{2.5} (1.01; 0.77-1.32) or nitrogen dioxide (NO₂, 0.98, 95% CI: 0.95-1.01) were associated with MS risk or PM_{2.5} with MND risk (1.07, 95% CI: 0.86-1.33).

doi: [10.1016/j.envint.2026.110377](https://doi.org/10.1016/j.envint.2026.110377)

Wang et al. 2026 Neurol Sci*The evolving clinical landscape of PM_{2.5}-attributable hemorrhagic stroke in China, 1990-2050*

Despite severe demographic headwinds, China achieved substantial reductions in PM_{2.5}-attributable hemorrhagic stroke mortality.

doi: [10.1007/s10072-026-09261-4](https://doi.org/10.1007/s10072-026-09261-4)

Respiratory & allergic

26 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (12 OF 26 RECORDS IN THIS SUBTOPIC)

Bayar Muluk 2026 Int J Biometeorol*Climate change and otolaryngology: airway disease and surgical emission reduction*

Dual-domain narrative review. The relevant claim is a proposed *synergy* between elevated CO₂ and PM_{2.5} increasing pollen allergenicity and mucosal penetration. The second half (operating-room carbon footprint) is outside this watch's scope.

doi: [10.1007/s00484-026-03289-z](https://doi.org/10.1007/s00484-026-03289-z)

Chaiwong et al. 2026 Int J Mol Sci*PM_{2.5} and systemic inflammatory biomarkers in elderly COPD versus healthy elderly*

Pilot panel of 38 COPD patients (73.6 ± 7.1 y) and 20 age-matched controls, sampled in a high- (79.1 ± 23.3 µg/m³) and a low-pollution (13.9 ± 3.6 µg/m³) Chiang Mai season, measuring hsCRP, IL-6, IL-8 and TNF-α. IL-8 was higher in COPD across both periods, but **no significant within-group increase accompanied the 5.7-fold seasonal exposure contrast** — a null the authors attribute to inhaled corticosteroids (65.8% of COPD) and statins (42% COPD, 65% controls). That explanation is untestable at *n* = 58; equally consistent are ceiling effects on chronic inflammation, ambient-monitor exposure misclassification, and simple lack of power. Reported as a pilot, and should be cited as one. Relevance: a seasonal contrast this large failing to move systemic biomarkers argues for personal rather than municipal exposure metrics in susceptible-population panels.

doi: [10.3390/ijms27146283](https://doi.org/10.3390/ijms27146283)

AIRCLARE-Ghana 2026 (preprint) Research Square*Short-term ambient PM_{2.5}/PM₁₀ and acute respiratory morbidity in Ghana*

See *Signal of the day*. Time-stratified case-crossover, Jan 2023–Dec 2024, 9,511 acute respiratory presentations from 213,241 outpatient records, conditional logistic regression adjusted for temperature and relative humidity. **URI: OR 1.038 (1.010–1.068) per IQR PM_{2.5}; OR 1.030 (1.008–1.052) per IQR PM₁₀**, peaking at lag 1 and persisting to cumulative lag 0–7 d. Pneumonia null, asthma null. Threat to inference: one monitoring site stands in for a whole catchment, so Berkson-type and classical exposure error coexist and the true effect is plausibly larger; the case definition is bespoke and unvalidated; and it is a preprint. The 96.03% post-QC completeness figure is the number a network operator should take away.

doi: [10.21203/rs.3.rs-10441248/v1](https://doi.org/10.21203/rs.3.rs-10441248/v1)

Li et al. 2026 mSystems*Host genetic variation, airway microbiome and environmental exposure vs. respiratory health*

Sputum metagenomes from 1,651 individuals, with Mendelian randomisation implicating SO₂, CO and PM_{2.5} effects on lung function potentially mediated by *Actinomyces*, *Haemophilus influenzae* and *Veillonella* spp., and genotype-dependent exposure-microbiome interactions (MEOX1 with lower *Filifactor alocis* under pollutant exposure; FAM110D and USP36 with higher *Ralstonia pickettii*/*Neisseria* and lower *Oribacterium*, tracking COPD risk). Fungal loci also reported. This is many hypotheses from one cohort: MR on aggregate exposures with microbial mediators is fragile to pleiotropy and the variant identifiers are rare or novel, so the loci need replication before they mean anything. Mechanistically suggestive for why identical exposure yields divergent lung-function trajectories.

doi: [10.1128/msystems.00442-26](https://doi.org/10.1128/msystems.00442-26)

Liao et al. 2026 Med Gas Res*Short-term air pollution, respiratory symptoms and lung function in asthma*

Small panel study (32 patients, 266 visits, Beijing 2015–2016) with mixed-effects models. Per-IQR $PM_{2.5}$ ($74.5 \mu g/m^3$, 6-day) reduced Asthma Control Test scores 3.6%; PM_{10} 7-day moving average reduced FEV_1 predicted by 3.6% and $FEF_{25\%}$ by 6.0%. Exposure levels are far above current WHO guidance — limits extrapolation to low-exposure settings.

doi: [10.4103/mgr.MEDGASRES-D-25-00164](https://doi.org/10.4103/mgr.MEDGASRES-D-25-00164)

Mokoena et al. 2026 Public Health Chall*Socio-economic conditions, air pollution sources and eczema symptoms in Soshanguve teenagers*

Cross-sectional modified ISAAC III survey of 2,885 schoolchildren aged 13–14 across 13 schools in Tshwane, South Africa; multiple logistic regression. Current eczema symptoms 26.8%, eczema ever 23.3%. Frequent truck traffic near the home: AOR 1.26 (current) and 1.24 (ever); ETS 16–20 days/month: AOR 1.96; female sex, monthly paracetamol use (AOR 2.37) and pet ownership also associated. No confidence intervals are reported for most estimates, exposure is entirely self-reported with no measurement, and the paracetamol association is the textbook reverse-causation trap. Value is geographic coverage — one of very few sub-Saharan records in this corpus — not effect estimation.

doi: [10.1002/puh2.70325](https://doi.org/10.1002/puh2.70325)

Sánchez de Prada et al. 2026 Epidemiologia*Ambient PM and asthma-coded admissions in children aged 0–3, Spain*

Nationwide municipality-year ecological panel, 2020–2022, 20,746 municipality-years, 3,231 admissions — but 93.7% of municipality-years are zero, which is the dominant statistical feature. Poisson IRRs per IQR: $PM_{2.5}$ **1.54 (1.28–1.85)**, PM_{10} **1.89 (1.59–2.26)**; attenuated but persistent under negative-binomial and province fixed effects. Multipollutant models were collinear and unstable. Ecological design — do not read as individual-level risk.

doi: [10.3390/epidemiologia7040097](https://doi.org/10.3390/epidemiologia7040097)

Wang et al. 2026 Front Med*Environmental and socioeconomic influences on childhood asthma and pneumonia burden*

Retrospective cohort from paediatric electronic medical records at one Wuhu hospital, Anhui, January 2022–December 2025: 936 children aged 1–14 (47.4% asthma, 36.5% pneumonia, 16.0% both), with regional NO_2 , $PM_{2.5}$, temperature and humidity and census-based socioeconomic status. Higher NO_2 and $PM_{2.5}$ were associated with asthma; biomass fuel use and poor household ventilation with pneumonia; lower SES strongly with pneumonia but not asthma. Pneumonia peaked in winter, asthma in spring. Hospital-based case series with area-level exposure and no comparison population — the associations describe who among the diagnosed had which condition, not incidence.

doi: [10.3389/fmed.2026.1872723](https://doi.org/10.3389/fmed.2026.1872723)

Yang et al. 2026 J Hazard Mater*Allergenic vs. toxic risk of airborne seed hairs compared with ambient $PM_{2.5}$*

Simultaneous urban/suburban sampling of poplar-willow seed hairs and $PM_{2.5}$ in Hohhot, May 2025, with sIgE-based allergenic quantification and FVOC/heavy-metal toxicity. Seed hairs carry $13.1 \pm 12.1 \times$ the pollen and $92.9 \pm 55.2 \times$ the bacterial load per unit mass, giving sIgE levels $14.9 \times$ and $124 \times$ higher than concurrent $PM_{2.5}$; conversely $PM_{2.5}$ FVOC toxicity exceeds seed hairs by $\sim 4 \times 10^4$. A clean demonstration that mass-based monitoring misallocates seasonal allergic risk entirely.

doi: [10.1016/j.jhazmat.2026.143076](https://doi.org/10.1016/j.jhazmat.2026.143076)

Yue et al. 2026 Front Public Health*Short-term traffic-related gaseous pollutant exposure and childhood bronchopneumonia hospitalization in Guangzhou*

20,623 paediatric bronchopneumonia admissions (ICD-10 J18, ages 0–14) at a single tertiary hospital in Huadu District, Guangzhou, 2014–2023, against daily NO_2 and SO_2 in a quasi-Poisson generalised additive model. Per $5 \mu g/m^3$, peak single-day excess risk at lag 4 was **2.74% (95% CI 1.73–3.75)** for NO_2 and **8.81% (4.94–12.82)** for SO_2 ; over the 0–7-day moving average these rose to **6.83% (4.93–8.77)** and **17.21% (10.18–24.70)**. Children aged 6–14 and boys were consistently more susceptible. Two problems: no particulate term is reported at all, so the gaseous estimates absorb whatever correlated PM signal exists and cannot be read as pollutant-specific; and a single-hospital catchment in a ten-million-person megacity leaves the denominator undefined and exposed to referral-pattern drift over a decade. Included because the lag structure and the age–sex susceptibility gradient are informative, not because the point estimates are attributable.

doi: [10.3389/fpubh.2026.1885590](https://doi.org/10.3389/fpubh.2026.1885590)

Zhang et al. 2026 Epidemiology*Infancy air pollution exposure and atopic dermatitis at 4–6 years*

Pooled longitudinal analysis of 1,587 children across six US cities in the ECHO-PATHWAYS Consortium (2017–2023); first-year-of-life $PM_{2.5}$, NO_2 and O_3 from validated national spatiotemporal models; three caregiver-reported outcomes; Poisson models with prespecified modification by child sex and maternal education. $PM_{2.5}$ and NO_2 null; O_3 associated with current AD (aRR 1.07 per 1 ppb, 1.01–1.13) and location-specific AD (1.06, 0.99–1.13), evident only in children of lower-educated mothers. Outcome ascertainment is caregiver questionnaire, not clinical examination, and the education-stratified finding is a subgroup result on a modest sample. A well-powered $PM_{2.5}$ null against a validated exposure model is worth as much as a positive.

doi: [10.1097/EDE.0000000000002033](https://doi.org/10.1097/EDE.0000000000002033)

Zhu et al. 2026 Medicine (Baltimore)*Air pollution and pneumonia hospitalization in Qingyang, China: a time-series analysis*

Five years of pneumonia admissions (2015–2019) in an inland Chinese energy city against six pollutants, generalised additive plus distributed-lag nonlinear models. Per $10 \mu g/m^3$: $PM_{2.5}$ **RR 1.004 (0.991–1.017)** and PM_{10} **1.004 (0.999–1.010)** — both null — against NO_2 **1.221 (1.174–1.271)**, SO_2 **1.035 (1.022–1.048)**, and CO 1.382 (1.230–1.552) per $1 mg/m^3$. Effects peaked at lag 1 for $PM_{2.5}$ and lag 0–7 for the gases; men were more vulnerable to SO_2 , women to CO, and children ≤ 14 formed the high-risk group. The single-pollutant framing is the limitation — in a heating-dominated city $PM_{2.5}$, SO_2 , NO_2 and CO share a common combustion source and a common seasonal cycle, and no multi-pollutant model is presented, so the apparent NO_2 dominance is a statement about correlation structure rather than about which pollutant acts. Useful mainly as a demonstration that PM mass can be the null term in its own time-series.

doi: [10.1097/MD.00000000000049962](https://doi.org/10.1097/MD.00000000000049962)

RECOVERED BY THE MONTH-WIDE HARVEST — EXTRACTIVE PRECIS, NOT AN ASSESSED SUMMARY (14 RECORDS)

Bbuye 2026 BMC Public Health*Ambient air pollution and risk of respiratory tract infections in low- and middle-income countries: a systematic review and meta-analysis*

Subgroup analyses suggested stronger associations in studies conducted in Africa (OR: 1.94, 95% CI: 1.46–2.58) compared to Asia (OR: 1.16, 95% CI: 0.92–1.45), and for pneumonia (OR: 1.74, 95% CI: 1.38–2.18) compared to other respiratory tract infections; though based on few studies.

doi: [10.1186/s12889-026-28257-5](https://doi.org/10.1186/s12889-026-28257-5)

Duan et al. 2026 JMIR Public Health Surveill*Long-term exposure to PM2.5 and ozone and the risk of acute respiratory infections: community-based prospective cohort*

In the fully adjusted model using the 12-month exposure window, each IQR increase in PM2.5 was associated with higher risks of ARIs (hazard ratio [HR] 1.594, 95% CI 1.340-1.897), with stronger associations observed for ILI (HR 1.948, 95% CI 1.484-2.557).

doi: 10.2196/88045

He 2026 Eur Respir Rev*Air pollution-induced airway epithelial barrier dysfunction in COPD: a narrative review of mechanisms and therapeutic implications*

Some metals or organic chemicals with redox properties can initiate oxidation reactions and induce pathologic responses.

doi: 10.1183/16000617.0171-2025

Jackson 2026 Inhal Toxicol*Respiratory health outcomes following co-exposure to subchronic high temperature, episodic wildfire smoke exposure, and high-cholesterol diet*

To determine effects of ongoing exposure, 4-5-week-old rats were subchronically housed at RT or high temperature (HT; 30-31 °C) and fed ND or high-cholesterol (2%) diet (HCD).

doi: 10.1080/08958378.2026.2697433

Kaewsri 2026 Environ Res*Does diet modify the effects of air pollution on lung function? A large cross-sectional study*

Exposure to PM2.5 per 5 µg/m3 increment was associated with reduced FEV1 in the low of -14.4 mL(95%CI: -26.8, -2.2) but not in medium and high fruit intake groups (+2.9 mL(95%CI: -13.8,19.7) and +9.7 mL(95%CI: -4.1,23.6), respectively).

doi: 10.1016/j.envres.2026.125252

Krecl 2026 Environ Health (Wash)*Wildfire Smoke and Human Health: A Global Perspective on Respiratory Risks and Emerging Interventions*

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1021/envhealth.6c00112

López Domínguez 2026 Expert Rev Respir Med*Short-term exposure to urban air pollution and asthma-related morbidity in children under five: a PRISMA-guided systematic review with narrative synthesis (2021-2025)*

Across settings, short-term increases in particulate matter (PM_{2.5}/PM₁₀) and nitrogen dioxide (NO₂) were consistently associated with increased asthma-related hospitalizations or urgent visits.

doi: 10.1080/17476348.2026.2699458

Matheny 2026 J Burn Care Res*Residential Air Pollution, Inhalation Injury, and Burn Recovery: A Burn Model System Study*

These associations were attenuated after adjustment; though PM2.5 exposure was associated with inhalation injury, which was strongly associated with prolonged ventilation, tracheostomy, lower Global Health scores, and lower employment at follow-up.

doi: 10.1093/jbcr/irag120

Meng 2026 Eur Respir Rev*Genetic susceptibility to respiratory health effects from outdoor air pollution: a structured narrative review*

rs1800629 (TNF), rs3804099 (TLR2) and rs848 (IL13)), consistently modifying pollution-related lung function decline and airway inflammation.

doi: 10.1183/16000617.0326-2025

Milibari 2026 J Air Waste Manag Assoc*The impact of air pollution on paediatric hospital admissions for acute lower respiratory infections (ALRI) in Makkah, Saudi Arabia, during Hajj events*

Exposure to NO2 was significantly associated with an increased risk of same-day ALRI admissions in children during the Hajj events (RR = 1.033; 95% CI: 1.003-1.063 per IQR rise) but not outside the Hajj events.

doi: 10.1080/10962247.2026.2694481

Odimba et al. 2026 Inhal Toxicol*Sex-specific associations between ambient air pollution exposure thresholds and lung function in middle-aged and older Canadians*

low NO2 exposure was associated with greater reductions in FEV1 among males (-5.00%; 95% CI: -7.16, -2.84) than among females (-1.99%; 95% CI: -2.79, -1.19), and significantly reduced FVC among males (-3.58%; 95% CI: -5.15, -2.00) but not among females (-1.64%; 95% CI: -3.39, 0.11).

doi: 10.1080/08958378.2026.2700549

Sinha 2026 Clin Infect Dis*Incorporating ambient air pollution as a risk factor for tuberculosis mortality in Delhi, India*

Among 61,925 adults initiating tuberculosis treatment in Delhi, India, higher ambient PM2.5 exposure during the 30 days before treatment was independently associated with increased mortality, most pronounced at the highest exposures.

doi: 10.1093/cid/ciag424

Sun et al. 2026 Ann Am Thorac Soc*Fine particulate matter exposure and long-term lung-function trajectory in adults with cystic fibrosis*

After covariate adjustment, we did not find evidence of an adverse association between PM2.5 and annual change in FEV1 z-score in the overall cohort [+0.002 per 3 µg/m3 change in PM2.5 (95% CI -0.001, 0.005)] or across birth cohorts.

doi: 10.1093/annalsats/aaag188

Tangsangwornthamma 2026 BMC Public Health*Ambient PM2.5 and respiratory tract disorder-related outpatient visits among healthcare workers: an ecological time-series study*

Among 12,458 healthcare workers, short-term increases in weekly ambient PM2.5 concentrations were associated with higher outpatient visit counts for overall RTDs (IRR 1.139, 95% CI 1.059-1.225), common cold (1.164, 1.082-1.252), allergic rhinitis (1.240, 1.126-1.365), sinusitis (1.168, 1.017-1.342), and asthma (1.429, 1.003-2.037).

doi: 10.1186/s12889-026-28580-x

Reproductive & developmental

22 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (8 OF 22 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026 J Environ Sci (China)*Prenatal air pollution, fetal growth trajectories and child neurodevelopment*

Time-series *k*-means clustering to derive fetal growth trajectory phenotypes, then mediation analysis. Sensitive windows identified at 0–20 and 27–41 weeks; higher exposure raised odds of high-risk trajectories by 9–32%, and first-trimester crown-rump length plus mid-pregnancy estimated fetal weight mediated effects on the 2-year psychomotor development index. The trajectory-clustering approach is the methodologically interesting part.

doi: 10.1016/j.jes.2025.10.043

Huang et al. 2026 Ecotoxical Environ Saf*Joint prenatal bisphenol + air-pollutant mixture exposure and preschool IQ*

Guangxi Zhuang Birth Cohort, 231 mother–child pairs, WISC indices, analysed by generalized linear models, restricted cubic splines, PCA, quantile g-computation and BKMR. Prenatal PM_{2.5} and PM₁₀ associated with lower verbal comprehension and full-scale IQ; effects stronger in boys; BKMR shows an inverted-U for fluid reasoning against the 10-pollutant mixture. Small *n* — treat the mixture results as hypothesis-generating.

doi: 10.1016/j.ecoenv.2026.120566

Li et al. 2026 Ecotoxical Environ Saf*Ambient air pollutant exposure and semen quality: a refined multi-window panel study*

One-year prospective panel of 32 male undergraduates aged 18–22 with four quarterly follow-ups spanning the 90-day spermatogenic cycle, using both daily-average and cumulative-dose exposure metrics and — the methodological contribution — stratifying the 15–69-day meiotic window into three biologically distinct sub-stages. Linear mixed-effects, two-pollutant and Bayesian weighted quantile sum models. Per IQR of full-cycle O₃, **sperm concentration fell 15.45% (95% CI –20.95 to –9.58)**, with the sub-stage peak at days 31–45 (–13.30%, –18.09 to –8.24); NO₂ acted earliest (15–30 d), CO later (31–69 d). With *n* = 32 the sub-stage resolution outruns the power, and homogeneous undergraduates limit generalisability by design. The window-stratification framework is worth borrowing regardless.

doi: 10.1016/j.ecoenv.2026.120581

Lin et al. 2026 Environ Pollut*Maternal PM exposure during pregnancy and offspring liver function*

Fujian Birth Cohort, 1,046 mother–offspring pairs, with gestational and trimester-specific PM₁, PM_{2.5} and PM₁₀, serum ALT and AST at ages 1–2 years, restricted cubic splines for shape and distributed lag models for timing. **Second-trimester IQR increments gave ALT +1.42 (0.82–2.03), +1.41 (0.77–2.05) and +0.91 (0.35–1.46) U/L** for PM₁/PM_{2.5}/PM₁₀, with AST increases of similar magnitude. The effect sizes are small relative to the assay's biological variability in toddlers and clinical significance is unestablished; a single post-natal measurement precludes trajectory inference, and PM₁/PM_{2.5}/PM₁₀ are too collinear to separate as cleanly as the paper implies. The distributed-lag identification of a second-trimester window is the durable part.

doi: 10.1016/j.envpol.2026.128835

Liu et al. 2026 Int J Mol Sci*Deoxynivalenol, cytokines and particulate matter in human trophoblast cell models*

Head-to-head sensitivity comparison of HTR-8/SVneo and BeWo trophoblast lines against three maternal-exposome stimuli — the mycotoxin deoxynivalenol, a TNF- α /IL-1 β /IFN- γ cocktail, and PM_{2.5} — at two penicillin/streptomycin concentrations, 24 h exposure. **The PM arm was null:** no change in IL-6, IL-8 or hCG in either line, while DON and the cytokine cocktail moved IL-6/IL-8, hCG and junctional markers (ZO-1, OCLD, CLDN-3/4, E-CAD). A negative result in a 24 h submerged monoculture at an unreported dose is weak evidence of placental safety and strong evidence that these two lines are the wrong model for particle toxicity — there is no air–liquid interface, no deposition physics and no particle characterisation. Worth logging precisely because prenatal-PM epidemiology keeps citing trophoblast inflammation as its mechanism.

doi: 10.3390/ijms27146434

Peterson 2026 Curr Environ Health Rep*Ambient air pollution, PAHs and pubertal development: critical review*

Component-level synthesis. Organic matter and sulfate fractions of PM_{2.5} are associated with earlier menarche and precocious puberty; OM is enriched in combustion-derived PAHs. Notably, high *chronic* exposure in heavily polluted settings has also been associated with *delayed* menarche — a non-monotonic pattern that any dose–response modelling should accommodate. Proposed mechanism: disruption of hypothalamic kisspeptin–GnRH signalling.

doi: 10.1007/s40572-026-00552-8

Sezer et al. 2026 Birth Defects Res*Preconception and early pregnancy exposure to ambient air pollution and risk of cleft lip and/or palate*

Hospital-based case-control study, 127 nonsyndromic CLP cases against 271 controls, with PM_{2.5}, PM₁₀, O₃, NO_x, SO₂ and CO assigned for the three months before conception and the first trimester. Higher O₃, PM_{2.5} and PM₁₀ were each associated with elevated CLP risk in single-pollutant models; **O₃ was the dominant contributor in the WQS mixture** and the only exposure stable across windows and model specifications, while traffic-related pollutants did not survive multi-pollutant adjustment. The study is underpowered for a mixture analysis — 127 cases against a six-pollutant WQS index invites weight instability, and no bootstrap weight distribution is reported — and hospital-based control selection risks referral bias. Read as hypothesis-generating. The methodological point that carries: with correlated pollutants and a small case series, the pollutant that wins the mixture is often the one with the widest exposure contrast, not the one with the strongest biology.

doi: 10.1002/bdr2.70102

Xie et al. 2026 Environ Res*Long-term PM_{2.5} and timing of pubertal development in boys and girls*

Chongqing Child Health Cohort 2014–2024, 3,833 children, time-varying cumulative average residential PM_{2.5} from the prenatal period, Cox models across nine milestones. **Boys: testicular HR 1.065 (1.056–1.073), penile 1.063 (1.055–1.071). Girls: breast 1.044 (1.039–1.050), menarche 1.033 (1.025–1.040)**, all per 1 $\mu\text{g}/\text{m}^3$. Associations were stronger in overweight/obese girls and in areas with higher long-term temperature and lower humidity — i.e. meteorological effect modification, not just confounding. The male-inclusive design is the field's main gap being closed here.

doi: 10.1016/j.envres.2026.125313

RECOVERED BY THE MONTH-WIDE HARVEST — EXTRACTIVE PRECIS, NOT AN ASSESSED SUMMARY (14 RECORDS)

Gao 2026b J Hazard Mater*Association of atmospheric oxidants and fine particulates with polycystic ovary syndrome and the reproductive trajectory: Protective role of residential greenness*

Each 1 $\mu\text{g}/\text{m}^3$ increase in oxidants was linked to higher odds of PCOS [odds ratio (OR): 1.07, 95% confidence interval (95% CI): 1.05–1.08], exceeding any individual component.

doi: 10.1016/j.jhazmat.2026.142881

Gomez-Herrera et al. 2026 Int J Hyg Environ Health*Prenatal air pollution exposure and placental biomarkers of oxidative stress and ageing: the BiSC cohort*

Increasing levels of NO₂ and PM_{2.5} during the 1st and 3rd trimesters and across the entire pregnancy, as well as black carbon during the 3rd trimester, were significantly associated with an increase in placental mtDNAcn.

doi: 10.1016/j.ijheh.2026.114865

Hu et al. 2026 Environ Health Perspect*Prenatal exposure to ambient air pollution and fetal growth trajectories in Wuhan, China*

Compared with the reference trajectory group, exposure to higher PM_{2.5} during weeks 1-16 was significantly associated with lower odds of being in the "fast growth group" for AC (OR = 0.68, 95% CI: 0.55, 0.84) and EFW (OR = 0.71, 95% CI: 0.56, 0.91).

doi: 10.1021/EHP.6c00033

Jiang 2026 Sci Rep*Association of surrounding greenness with preterm birth and the potential mediating role of ambient air pollution*

Greenness exposure was associated with lower odds of PTB, and this association was partially mediated by reduced ambient PM_{2.5} and NO₂ pollution.

doi: 10.1038/s41598-026-63002-3

Kahroba et al. 2026 Environ Res*Cord-blood black carbon particle burden and a C19MC extracellular-vesicle miRNA signature enriched for neurodevelopmental pathways*

Five of the ten candidate miRNAs were associated with BC at BH-FDR ≤ 0.10 : hsa-miR-25-3p and hsa-miR-433-3p (positive associations) and hsa-miR-518a-3p, hsa-miR-519a-3p/519b-3p, and hsa-miR-520g-3p/520h (negative associations).

doi: 10.1016/j.envres.2026.125159

Lee 2026c Environ Res*Prenatal exposure to particulate matter and risk of autism spectrum disorder in children: a population-based cohort study in South Korea*

Prenatal PM_{2.5} exposure was not associated with ASD risk during the total pregnancy period (HR, 0.99; 95% CI, 0.95-1.03) or during the first (1.02, 0.99-1.05), second (0.99, 0.97-1.02), or third trimester (0.98, 0.95-1.00).

doi: 10.1016/j.envres.2026.125161

Li et al. 2026c Environ Epidemiol*Maternal ambient air pollution exposure and risk of stillbirth in Georgia, USA*

Particulate matter showed strong associations in the second trimester exposure window for all stillbirths (PM₁₀: HR = 1.07; 95% confidence interval [CI] = 1.04, 1.11; PM_{2.5}: HR = 1.05; 95% CI = 1.01, 1.09).

doi: 10.1097/EE9.0000000000000505

Liu 2026e Environ Res*Impacts of extreme air pollution events on pregnancy outcomes: A systematic review*

Studies were selected using the PECOS (Pregnant women, Exposure to extreme air pollution events, Comparison with lower or no exposure, Pregnancy outcomes, Observational study designs) framework and synthesised following Synthesis Without Meta-Analysis (SWiM) guidelines.

doi: 10.1016/j.envres.2026.125242

McCracken 2026 Respiriology*Prenatal Air Pollution Exposure and Child Lungs: Innovations in Measurement and Causal Inference*

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1002/resp.70280

Sieber et al. 2026 Int J Epidemiol*The impact of improved air quality on low birth weight and health inequalities in Spain, 2010-2018*

Relative to tertiary education, secondary and primary education were associated with increased LBW odds [1.32 (95% CI: 1.30-1.35); 1.76 (95% CI: 1.71-1.82)].

doi: 10.1093/ije/dyag130

Smith 2026 Curr Dev Nutr*Air Pollution Mixture and Folate Status Indicators among Pregnant Women in Canada: Maternal-Infant Research on Environmental Chemicals (MIREC) Study, 2008 to 2011*

In the third but not the first trimester, the mixture was positively associated with plasma total folate [12.0 nmol/L per one-quartile increase, 95% confidence interval (CI): 1.5, 22.7 nmol/L], negatively associated with %5MTHF (-2.8, 95% CI: -5.1, -0.70), and positively associated with %UMFA (13.4, 95% CI: 4.4, 23.0).

doi: 10.1016/j.cdnut.2025.107617

Tang 2026b Environ Pollut*Window-specific effects of ultraviolet radiation and air pollution on ovarian reserve in infertile women: A retrospective cohort study*

Higher UV exposure during the earliest window was associated with lower odds of diminished ovarian reserve (AMH ≤ 1.1 ng/mL; odds ratio 0.73, 95% CI: 0.56-0.94), whereas positive associations during intermediate windows were less consistent across adjustment models.

doi: 10.1016/j.envpol.2026.128740

Yin et al. 2026 Ecotoxicol Environ Saf*PM_{2.5}-dominant disruption of male reproductive hormone homeostasis: fine versus coarse particulate matter*

In the 0-360 day exposure window, each 10 $\mu\text{g}/\text{m}^3$ increase in PM_{2.5} was associated with a 0.0675 increase in log-transformed FSH (95% CI: 0.0231, 0.1120) and a 0.1136 decrease in log-transformed T (95% CI: -0.1604, -0.0667).

doi: 10.1016/j.ecoenv.2026.120446

Zhang 2026 Front Public Health*Maternal exposure to fine particulate matter and autism spectrum disorder in children: population based case-control study*

Positive associations were observed between PM_{2.5} and ASD (OR = 5.197, 95% CI: 3.294-8.200), and between SO₄ 4⁻², NO₃ 4⁻, NH₄ 4⁺, BC, and OM and ASD.

doi: 10.3389/fpubh.2026.1848913

Mechanistic toxicology**31 records**

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (9 OF 31 RECORDS IN THIS SUBTOPIC)

Balu et al. 2026 J Biochem Mol Toxicol*SRM 1650b administration to isolated rat heart aggravates ischemia-reperfusion injury via mitochondrial dysfunction*

Langendorff-perfused male Wistar hearts exposed to NIST SRM 1650b heavy-duty diesel particulate at 10, 100 and 300 $\mu\text{g}/\text{mL}$ before 30 min ischemia and 60 min reperfusion. Dose-dependent deterioration of haemodynamics and elevated tissue injury, with increased oxidative stress in tissue and isolated mitochondria, reduced bioenergetic enzyme activity and respiratory efficiency, downregulated *Pgc-1 α* , *Tfam*, *Polg*, *Fis1*, *Mfn1* and *Pink1*, lower mtDNA copy number, and suppressed PI3K/Akt signalling that fell further after the ischemic challenge. The design isolates a direct cardiomyocyte effect cleanly — no autonomic, pulmonary or systemic inflammatory route is available in an ex-vivo preparation — which is both its strength and its ceiling. The concentrations are far above any inhaled dose delivered to myocardium in vivo, and direct coronary perfusion bypasses the alveolar barrier entirely, so this establishes a mechanism that *can* operate, not one that does at ambient exposure.

doi: 10.1002/jbt.71038

Ding et al. 2026 J Environ Sci (China)

PM_{2.5} promotes vascular calcification via miR-27b-5p/HES1 signalling

Mouse aortic smooth-muscle cells, hiPSC-derived 3D blood-vessel organoids, and a murine model. Proposed axis: PM_{2.5} suppresses miR-27b-5p transcription in an XBP1-dependent manner → HES1–Runx2 interaction → osteogenic transformation of VSMCs. miR-27b-5p mimics rescued the organoid phenotype. The organoid arm is the credible translational bridge; dose and duration dependence shown in vivo.

doi: 10.1016/j.jes.2025.12.074

Fakfum et al. 2026 Toxics

Seasonal PM_{2.5} exposure and plasma metabolome changes related to metabolic syndrome

Prospective within-subject comparison of plasma ¹H-NMR metabolomes in 25 healthy adults in Samoeng District, Chiang Mai, between haze and non-haze seasons. Twenty-six metabolites separated the seasons by PLS-DA (VIP > 1.5): eleven up (maleylacetoacetic acid, deoxyribose 5-phosphate, betaine, 3-hydroxyanthranilic acid, 1-methyladenosine) and fifteen down (deoxyguanosine, D-arabitol, glycerophosphocholine, ophthalmic acid, oxaloacetic acid), implicating inflammation, ROS generation, NO inhibition and disrupted energy metabolism across 56 pathways. The design cannot separate PM_{2.5} from everything else that changes with the Thai dry season — temperature, humidity, diet, activity, and co-emitted gases — and PLS-DA at *n* = 25 over hundreds of features is prone to overfitting without the permutation testing the abstract does not report. Hypothesis-generating.

doi: 10.3390/toxics14070544

Liu et al. 2026 Environ Toxicol Pharmacol

Resveratrol ameliorates PM_{2.5}-aggravated IBS-D via microbiota and metabolite restoration

Three-arm murine design (IBS-D, PM_{2.5}+IBS-D, resveratrol+PM_{2.5}+IBS-D) with untargeted metabolomics, 16S microbiota profiling, colonic histopathology, oxidative-stress and barrier-function assays and visceral-hypersensitivity testing. PM_{2.5} worsened IBS-D with dysbiosis and disrupted hypoxanthine and linoleic-acid metabolism; resveratrol restored both. No exposure concentrations, particle source or dose–response are reported in the abstract, group sizes are unstated, and the resveratrol arm is a single dose — so this is proof-of-concept, not a therapeutic result. The gut-axis endpoint remains underexplored, which is why it is retained.

doi: 10.1016/j.etap.2026.105120

Lu et al. 2026 Toxics

Toxicity of PM_{2.5} from incomplete solid-fuel burning in Caenorhabditis elegans

Synchronised L4 *C. elegans* exposed to aqueous suspensions of PM_{2.5} collected from incomplete combustion of six solid fuels (rice, wheat, peanut and rapeseed straw; poplar and paulownia branch), scored for body length, fertilised-egg count, lipofuscin accumulation and ROS, alongside OC/EC, water-soluble ion and PAH characterisation. **Oilseed-crop straw (peanut, rapeseed) was the most toxic**, and the effect tracked lipid-soluble PAH content via oxidative stress. The source-specificity result is the useful part — it says fuel type, not mass concentration, drives toxicity, which is an argument for speciation-capable monitoring. The model is the limitation: an aqueous suspension delivered to a nematode cuticle and gut has no relationship to pulmonary deposition, and no mammalian or human-cell confirmation is offered.

doi: 10.3390/toxics14070597

Maraschi et al. 2026 J Comp Physiol B

Osmoregulatory disruption from settleable atmospheric PM in estuarine fish

96 h exposure of *Centropomus parallelus* to 1 g L^{−1} settleable APM. Al, Fe, V and Cr accumulated in gill and kidney; branchial Na⁺/K⁺-ATPase activity and NKA⁺ ionocyte density rose while renal ATPase activity fell. Ecotoxicological rather than human-health relevant, but a useful reminder that deposition, not inhalation, is the dominant PM pathway in aquatic receptors.

doi: 10.1007/s00360-026-01698-5

Pan et al. 2026 Front Pharmacol

Effects of air pollution exposure on nasal mucosal immune-inflammatory markers in animal models of allergic rhinitis

PRISMA systematic review and random-effects meta-analysis of 18 rodent studies, standardised mean differences for nasal mucosal markers. Air pollutant exposure was associated with increased eosinophils and elevated IL-4, IL-5, IL-13 and OVA-specific IgE — a coherent enhancement of type 2 inflammation — with scattered evidence of epithelial barrier loss (reduced ZO-1) and innate activation (IL-1β, NLRP3). IFN-γ showed no consistent direction. Heterogeneity was substantial and the authors decline to claim a causal pathway. The pooling is the weak point: 18 studies spanning diesel exhaust particulate, PM_{2.5}, PM₁₀ and mixed urban aerosol at uncontrolled doses and durations, in an ovalbumin-sensitisation model whose relationship to human allergic rhinitis is itself contested, cannot yield a transferable effect size. No dose–response meta-regression is attempted, which is the analysis that would have justified the synthesis. Value here is as a marker map for the type 2 axis, not as an effect estimate.

doi: 10.3389/fphar.2026.1870023

Wang et al. 2026 Exp Eye Res

PM exposure, mitochondrial stress and ECM remodelling in scleral fibroblasts

Combined in vivo and in vitro: topically applied PM produced particle-surrogate distribution to ocular tissues with scleral thinning and axial elongation, while human scleral fibroblasts showed raised oxidative stress, mitochondrial ultrastructural abnormality, phenotypic change and altered collagen metabolism; transcriptomics implicated ECM organisation, mitochondrial metabolism and stress-response pathways. The framing as a myopia mechanism is plausible but the exposure route is the problem — topical application is not inhalation, and axial elongation in a rodent eye is a weak proxy for human myopia progression. Notable mainly as one of very few records addressing an ocular endpoint at all.

doi: 10.1016/j.exer.2026.111187

Wu et al. 2026 J Environ Sci (China)

Macrophage NLRP3 inflammasome in real-ambient PM-induced hepatic glucose dysregulation

Individually-ventilated-cage real-ambient exposure system, 15 weeks, plus transwell macrophage–hepatocyte co-culture and *Nlrp3* knockdown. Lysosomal damage and cathepsin B release → NLRP3 activation → IL-1β → suppressed hepatocyte insulin signalling. Both NLRP3 inhibition and IL-1β blockade rescued. Real-ambient exposure (rather than instillation) makes the dosimetry considerably more defensible than most of this literature.

doi: 10.1016/j.jes.2025.11.018

RECOVERED BY THE MONTH-WIDE HARVEST — EXTRACTIVE PRECIS, NOT AN ASSESSED SUMMARY (22 RECORDS)

Cai 2026 Mol Biol Rep

Network toxicology and multi-omics identify potential interactions between air pollutants and interferon-related signaling in tuberculosis

We integrated human genes associated with seven air pollutants with TB-associated genes from public databases.

doi: 10.1007/s11033-026-12289-6

Chen 2026 J Cardiovasc Pharmacol*Asiatic acid mitigates PM2.5-elicited cardiomyocyte pyroptosis via suppression of mtDNA-driven cGAS-STING-NLRP3 signalling*Oral administration of AA (12.5 or 25 mg kg⁻¹ from day 7) dose-dependently restored TFAM expression, reduced cytosolic mtDNA, blunted cGAS-STING-NLRP3 axis activation, attenuated pyroptosis and preserved cardiac architecture and function.

doi: 10.1097/FJC.0000000000001858

Easton et al. 2026 Environ Int*Ultrafine particulate matter emitted from ships drives inflammation and susceptibility to viral infection*

In cell models of airway and alveolar epithelium, this cruise ship-associated ultrafine PM was markedly more pro-inflammatory than PM from other sites, while simultaneously reducing expression of a range of genes associated with the innate antiviral response.

doi: 10.1016/j.envint.2026.110381

Ediagbonya 2026 J Appl Toxicol*Dosimetry of Inhaled Aerosols: From Exposure to Target Tissue Dose in Risk Assessment and Drug Delivery*

Inhaled aerosol dosimetry is a critical discipline bridging exposure to biological effect in both therapeutic and toxicological contexts.

doi: 10.1002/jat.70327

Ei Mais 2026 Environ Int*In vitro bioanalytical assessment of aryl hydrocarbon and steroid receptor-mediated activities of primary and aged particulate emissions from residential pellet and logwood stoves*

After aging, biological activities generally decreased or remained comparable, reflecting the degradation or inactivation of active primary ligands and the formation of new bioactive compounds.

doi: 10.1016/j.envint.2026.110412

Gao 2026 J Appl Toxicol*Epigenetically Regulated NOX4/NRF2 Axis Mediates PM2.5-Induced Ferroptosis and Inflammatory Response in Membranous Nephropathy*

Notably, pretreatment with GKT137831 or ferrostatin-1 attenuated the PM2.5-mediated reduction in cell viability and prevented the activation of ferroptosis and pro-inflammatory phenotypes in PAN-stimulated MPC5 cells.

doi: 10.1002/jat.70354

Giorgione 2026 Environ Sci Pollut Res Int*Exploring oxidative stress and epigenetic biomarkers in bees: insights into health effects of air pollution and diet - a pilot study*

amara or probiotics tended to exhibit lower levels of oxidative and epigenetic biomarkers.

doi: 10.1007/s11356-026-38040-z

Karuna 2026 Cardiovasc Toxicol*Nanoparticle-Encapsulated 5-Azacytidine Attenuates Air Pollution-Induced Human Cardiac Fibroblast Activation*

HCFs were exposed to PM0.49 (low and high concentrations), with or without pre-treatment using free 5-Aza or 5-Aza-loaded hyaluronic acid nanoparticles (5-Aza-NP).

doi: 10.1007/s12012-026-10161-4

Kern-Allely 2026 Am J Vet Res*Dairy calves exposed to ambient fine particulate matter have increased bovine respiratory disease risk*A 5 µg/m³ increase in daily PM2.5 was associated with 66% higher odds of incident BRD (OR, 1.66; 95% CI, 1.08 to 2.53).

doi: 10.2460/ajvr.26.04.0158

Killur 2026 Environ Monit Assess*Air pollution risks to wheat and mustard in Delhi agricultural airshed during the winter cropping season (2022-2025)*

Potential yield-loss risks for rabi wheat in Bahadurgarh, Rohat and Sonipat could be up to 15, 14.25 and 11.4% while that of mustard could be potentially reduced by 12, 11.4 and 10.54%, respectively.

doi: 10.1007/s10661-026-15716-0

Kováts 2026 Bull Environ Contam Toxicol*Combined Effects of Air Pollution and Drought Stress in Tomato Landraces*

3 commercial varieties as well as 2 Hungarian landraces were selected, effects of the combined stressors were evaluated based on biomass reduction and biochemical end-points such as photosynthetic pigments concentrations and peroxidase (POD) activity.

doi: 10.1007/s00128-026-04305-z

Kwon 2026 Cutan Ocul Toxicol*Ambient fine particulate matter (PM2.5) induces AhR-dependent proinflammatory responses and reactive oxygen species production in human conjunctival epithelial cells*

PM2.5 induced dose-dependent reductions in cell viability and significant increases in intracellular ROS at 25 and 50 µg/mL.

doi: 10.1080/15569527.2026.2695393

Lee 2026 Int J Mol Sci*Heat-Killed Lactocaseibacillus paracasei ATG-E1 Improves Particulate Matter 10 Plus Diesel Exhaust Particles (PM10D)-Induced Airway Inflammation*

Furthermore, it decreased various inflammatory mediators, such as C-X-C Motif chemokine ligand (CXCL)-1, macrophage inflammatory protein (MIP)-2, interleukin (IL)-1α, and tumor necrosis factor (TNF)-α, in the BALF and lung tissue, as well as serum symmetric dimethylarginine (SDMA) levels in the PM10D-induced airway inflammation model.

doi: 10.3390/ijms27135940

Massarsky 2026 Comp Biochem Physiol C Toxicol Pharmacol*Developmental exposure to cigarette smoke particulate matter induces cross-generational DNA methylation changes in zebrafish*

Embryos were exposed to TPM (0.1, 0.2, or 0.3 µg/mL equi-nicotine units) from 6 to 96 h post fertilization (hpf).

doi: 10.1016/j.cbpc.2026.110628

Noor 2026 Environ Sci Pollut Res Int*Wildfire smoke exposure impacts wing development of two pest outbreaking moths in boreal forests*

Smoke exposure significantly increased malformation rates, particularly folded and crumpled wings, while undeveloped wings decreased.

doi: 10.1007/s11356-026-38023-0

Peng 2026 Environ Health (Wash)*From Size-Resolved PAC Emissions to Age-Specific Respiratory Deposition: Toxicity Potential of Particulate PACs from Residential Coal and Biomass Combustion*

Respiratory deposition modeling showed clear age dependence: under the same exposure scenario, children had 40% higher total respiratory deposition of particulate PACs than adults.

doi: 10.1021/envhealth.6c00050

Ribble et al. 2026 Toxicol Sci*Exposure timing is a determinant of fine particulate matter pulmonary, vascular, and metabolic toxicity in male mice*

Exposure to fine particulate matter (PM2.5) air pollution is associated with an increased cardiometabolic disease risk.

doi: 10.1093/toxsci/kfag080

Shi 2026 Immun Inflamm Dis*Mometasone Furoate Alleviates PM_{2.5}-Induced Pyroptosis of Nasal Mucosa Cells via Blocking JAK2/STAT3/AIM2 Inflammasome*

Luciferase and chromatin immunoprecipitation assays were used to verify the interaction between signal transducer and activator of transcription 3 (STAT3) and absent in melanoma 2 (AIM2).

doi: 10.1002/iid3.70476

Strader 2026 Environ Sci Pollut Res Int*Near-road particulate matter from Washington D.C. has variable levels of 6PPD and 6PPD-quinone: acute and developmental impacts in zebrafish*

PM_{2.5} exposure markedly inhibited E-cadherin expression and aggravated epithelial ferroptosis, while CC16 intervention efficiently restored E-cadherin levels, elevated NRF2/GPX4/SLC7A11 expression, suppressed ACSL4, and reduced lipid peroxidation.

doi: 10.1007/s11356-026-38048-5

Wang 2026d Curr Res Toxicol*CC16 alleviates PM_{2.5}-induced airway inflammation in asthmatic mice by targeting the E-cadherin-mediated ferroptosis regulatory axis*

PM_{2.5} exposure markedly inhibited E-cadherin expression and aggravated epithelial ferroptosis, while CC16 intervention efficiently restored E-cadherin levels, elevated NRF2/GPX4/SLC7A11 expression, suppressed ACSL4, and reduced lipid peroxidation.

doi: 10.1016/j.crttox.2026.100310

Williams et al. 2026 Free Radic Biol Med*Liver-derived, circulating xanthine oxidoreductase drives vascular impairment associated with inhalation of ultrafine particulates*

Drinking water supplemented with the XOR inhibitor febuxostat or nitrite (NaNO₂) partially prevented nTiO₂-induced impairment of vascular reactivity.

doi: 10.1016/j.freeradbiomed.2026.07.023

Zeng et al. 2026b Environ Health (Wash)*Circular RNA circDNAJC5 regulates ultrafine particle-disrupted microglial lipid metabolism via sphingolipid signalling*

We identified circDNAJC5, a lipid metabolism-associated circRNA, as significantly downregulated during PM_{0.1}-induced lipid metabolic disruption.

doi: 10.1021/envhealth.5c00701

Occupational & indoor

16 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (10 OF 16 RECORDS IN THIS SUBTOPIC)

Chen et al. 2026 Front Med Technol*Nested suction instrument to reduce intraoperative surgical-smoke PM_{2.5}*

Bench study on porcine dorsal tissue with electrosurgical cutting, PM_{2.5} monitored by laser dust counters. Three orifice diameters tested (1.5/2.0/2.5 mm) with local smoke evacuation; reduction rates 24.2%, 66.9% and 43.0% respectively — the 2.0 mm “all-in-one” design (simultaneous smoke and blood suction) was optimal. Small, single-setting, but a clean engineering-control result.

doi: 10.3389/fmedt.2026.1796828

Cole & Driscoll 2026 Ann Work Expo Health*Determinants of respirable crystalline silica exposure in Australian tunnelling*

3,288 personal exposure measurements across 12 tunnel projects (2016–2024), obtained via freedom-of-information requests — the largest such dataset in Australian tunnelling. Linear mixed models explained 53–68% of variance. Underground workers had **2.3× the exposure of above-ground workers** (GM 0.026 vs. 0.012 mg/m³); each 1% increase in silica fraction of respirable dust raised RCS exposure 5%; exposures fell after the July 2020 standard reduction (GM 0.023 → 0.013 mg/m³). A model for what transparent exposure-data access enables.

doi: 10.1093/annweh/wxag055

Ding et al. 2026 J Hazard Mater*Stage-resolved hair Zn-isotope dynamics as a firefighting exposure tracer*

53 hair samples from 15 Beijing firefighters across five stage-defined periods. $\delta^{66}\text{Zn}$ ranged -0.79 to -0.13‰ , with non-linear incident–recovery trajectory: heavy-isotope enrichment after repeated missions, partial return toward baseline in late recovery. Linear mixed effects with stage-dependent Ni/Pb/Co associations; ICC 0.82 (i.e. dominated by between-subject variability). A genuinely novel time-integrated biomarker concept for combustion-derived PM.

doi: 10.1016/j.jhazmat.2026.143081

Ghosh et al. 2026 Indian J Occup Environ Med*Respiratory quality of life and pulmonary function in traffic police, eastern India*

Cross-sectional comparison of 62 traffic police against 120 unexposed controls (Oct 2023–Jan 2024) using the St George’s Respiratory Questionnaire and digital spirometry. Exposed personnel had significantly higher SGRQ scores and lower FVC and FEV₁ (both $p < 0.001$), with SGRQ correlating positively with pulmonary function measures. No exposure measurement whatsoever — occupation is the exposure proxy — and no adjustment for smoking, duration of service or socioeconomic position is reported, while simple random sampling from unequal groups invites selection differences. Reads as a hazard-surveillance signal for a large occupational group, not as an exposure–response result.

doi: 10.4103/ijoom.ijoom_150_25

Gwak et al. 2026 Risk Anal*Size-resolved modelling of airborne viral RNA copies in indoor environments*

Size-resolved framework over 1–2000 μm coupling saliva-specific evaporation and residue formation with gravitational settling, across temperature, RH and two ambient pressures (101.325 and 75.3 kPa), combined with saliva-mass uncertainty, viral load, half-life, exhalation mode and ventilation. **Talking dominates airborne RNA persistence** because coughing and sneezing generate larger droplets that settle quickly. Predictions matched hospital air-sampling data within roughly one order of magnitude at the 50th saliva-mass percentile. The framework’s own caveat is the binding one: RNA decay stands in for infectivity decay, and if viral RNA preferentially concentrates in long-suspended fine aerosol the size-resolved conclusions shift. Directly relevant to size-resolved optical sensing of bioaerosol in occupied spaces.

doi: 10.1111/risa.70318

Komonnithiphong et al. 2026 Toxics*PM_{2.5}-bound PAHs and carcinogenic risk in charcoal barbecue (Moo Kratha) restaurants*

Ten area samples taken at customer dining tables in ten Chiang Mai charcoal-barbecue restaurants during peak evening hours in February 2026 (early dry season), 2 L min⁻¹ for 180 min onto quartz fibre filters, 16 US-EPA priority PAHs by GC-MS. Mean PM_{2.5} **87.64 $\mu\text{g m}^{-3}$** , total PAHs 132.74 ng m⁻³, carcinogenic fraction 52.3%, TEQ_{BaP} 15.34 ng m⁻³. Inhalation ILCR for a frequent adult diner **2.82×10^{-5}** (child 2.12×10^{-5}), rising to 6.16 and 4.62×10^{-5} at the worst site; even monthly diners exceed 10^{-6} . Two hard limitations: $n = 10$ single visits with no repeat sampling gives no within-site variance, and February in Chiang Mai means the regional haze background is confounded with the cooking source

— no outdoor or upwind reference was reported. A paired indoor/outdoor low-cost node would have resolved that for a fraction of the cost.

doi: [10.3390/toxics14070643](https://doi.org/10.3390/toxics14070643)

Park et al. 2026 Sci Total Environ

Environmental evaluation of wood-based remodeling in aging buildings: indoor air quality and life-cycle carbon

Mock-up chamber characterisation of softwood finishes plus field monitoring in an occupied childcare facility and an unoccupied university classroom — before remodelling, 24 h after, and at four weeks — covering formaldehyde, anthropogenic and biogenic VOCs, particulate matter and total airborne bacteria, with a life-cycle global warming potential assessment. α -Pinene dominated the biogenic emission profile. In the occupied, ventilated building PM and bacterial counts fell or stabilised and most VOCs declined within four weeks; in the unoccupied, poorly ventilated classroom **α -pinene rose from 28.7 to 1713.6 $\mu\text{g}/\text{m}^3$ over four weeks** — a 60-fold accumulation. Net GWP in that building fell to 37.7 kgCO₂eq, 9–11 times below non-wood alternatives, on biogenic carbon storage. With $n = 2$ buildings differing simultaneously in occupancy, ventilation and use, those factors cannot be separated, and the PM measurements are reported only qualitatively. The transferable point is the ventilation dependence: a material acceptable in one air-exchange regime is not in another, which argues for continuous indoor monitoring rather than commissioning-time spot checks.

doi: [10.1016/j.scitotenv.2026.182065](https://doi.org/10.1016/j.scitotenv.2026.182065)

Qi et al. 2026 Ann Work Expo Health

Emission rates of respirable crystalline silica and dust from engineered stone fabrication

Extracts emission rates (mass per time) from four published experimental simulation studies of cutting and grinding engineered stone, tests control technology and task as determinants by regression, then drives a two-zone mathematical model and validates predictions against task-duration field measurements. RCS emission rate rose with slab crystalline-silica content and fell with control technology; **near-field predictions tended to underestimate measured exposures** though the distributions overlapped. Four source studies is a thin evidence base and the underestimation direction is exactly the wrong way for a protective standard. Methodologically the most transferable record in this cluster — an explicit emission-rate-to-exposure model rather than another exposure survey.

doi: [10.1093/annweh/wxag060](https://doi.org/10.1093/annweh/wxag060)

Ramsden 2026 J Xenobiot

Xenobiotic hazards in aircraft cabin air

Review of bleed-air contamination in the ~95% of airliners drawing cabin air from turbine compressors: engine-oil traces, tricresyl phosphate, and ultrafine particles abraded from turbine blades. Argues no safe lower TCP limit given inter-individual sensitivity, and presents a life-quality-index affordability calculation. The only ultrafine-focused record in today's corpus.

doi: [10.3390/jox16040119](https://doi.org/10.3390/jox16040119)

Subway PSD study 2026 J Environ Sci (China)

PM and CO₂ in Chinese subway stations with varying platform screen door forms

Seven stations instrumented across platform, hall and outdoor locations, contrasting full-height platform screen doors (PSD) with half-height platform bailout doors (HPBD). Platform > hall > outdoor for PM; CO₂ tracks passenger count linearly. The piston effect on train arrival raises **PM_{2.5} by 20% and PM₁₀ by 36% under HPBD**, appreciably more than under PSD, while CO₂ falls 1.8% (PSD) and 4.6% (HPBD). Fine particles carry 38.22–62.57% of total mass, highest behind PSD, and Fe dominates the elemental profile across six apportioned sources. The design is a snapshot without seasonal or diurnal repetition, and the door form is confounded with station age, depth and ventilation design, so the PSD-versus-HPBD contrast is associational. Still the cleanest demonstration in today's corpus that a metric-and-geometry-specific enclosed microenvironment needs its own sensor siting logic — a platform-edge node and a concourse node are measuring different populations of particles.

doi: [10.1016/j.jes.2025.12.027](https://doi.org/10.1016/j.jes.2025.12.027)

RECOVERED BY THE MONTH-WIDE HARVEST — EXTRACTIVE PRECIS, NOT AN ASSESSED SUMMARY (6 RECORDS)

Bolden et al. 2026 J Air Waste Manag Assoc

Wildfire smoke indoors: a review of infiltration risk factors and exposure mitigation strategies

Exposure to the particulate matter (PM_{2.5}) component of wildfire smoke is associated with cardiovascular and respiratory morbidity and mortality.

doi: [10.1080/10962247.2026.2692674](https://doi.org/10.1080/10962247.2026.2692674)

Gonzalez-Colom et al. 2026 BMJ Open Respir Res

Should household air quality monitoring be considered in selected patients with asthma and COPD?

Tobacco smoking, either active or passive, was significantly associated with PM_{2.5} levels ($p < 0.001$).

doi: [10.1136/bmjresp-2025-003899](https://doi.org/10.1136/bmjresp-2025-003899)

Lai 2026 J Surg Res

Time-resolved Impact of Smoke Evacuation on Particulate Exposure During Laparoscopic Cholecystectomy

Continuous Vac use was associated with lower intraoperative PM concentrations with greater reductions in complex LC procedures.

doi: [10.1016/j.jss.2026.05.070](https://doi.org/10.1016/j.jss.2026.05.070)

Li 2026b J Hazard Mater

Mechanically driven aerosol emissions from electric toothbrushing: An overlooked indoor source with exposure and infection risk implications

Results showed that emissions were dominated by respirable fine particles, with over 85% below 2.5 μm , while ultrafine particle levels showed no clear increase relative to the pre-brushing baseline, indicating a distinct emission regime compared to respiratory processes.

doi: [10.1016/j.jhazmat.2026.142993](https://doi.org/10.1016/j.jhazmat.2026.142993)

Rogula-Kopiec et al. 2026 Environ Geochem Health

Underground versus outdoor air in the Wieliczka Salt Mine health resort: particulate matter and selected gaseous pollutants

The underground chamber was characterised by markedly lower concentrations of anthropogenic pollutants than the outdoor environment.

doi: [10.1007/s10653-026-03338-x](https://doi.org/10.1007/s10653-026-03338-x)

Szczap 2026 J Hazard Mater

Size-dependent characterization of toxic aerosol emissions from ignition of cryogenically processed black mass from fully charged lithium-ion batteries

Higher battery state of charge (SOC) produced substantially greater particulate emissions, while metallic contamination intensified ignition and further increased aerosol concentrations.

doi: [10.1016/j.jhazmat.2026.143028](https://doi.org/10.1016/j.jhazmat.2026.143028)

Burden, policy & mitigation

29 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (12 OF 29 RECORDS IN THIS SUBTOPIC)

Abi Nader et al. 2026 Disaster Med Public Health Prep

Short-term respiratory and skin disease risks from air pollution during the 2024 Beirut conflict

Satellite-derived PM_{2.5} over Beirut's southern suburbs compared pre-war (1 Aug–22 Sep 2024) against war (23 Sep–29 Nov 2024), plus ground PM_{2.5} at five sites over ten days. Satellite median rose 15.11 → 20.21 µg/m³; ground median in the direct war zone 19.33 → 22.16 µg/m³. Applying BAPHE exposure–response functions gives RR 1.011 (AR 1.13%) respiratory and RR 1.032 (AR 3.10%) skin admissions, corroborated by WHO AirQ+. The estimates are transferred, not observed — no admissions data were analysed — so these are projections conditional on a borrowed concentration–response function.

doi: 10.1017/dmp.2026.10412

Hu et al. 2026 Front Public Health

Older-adult life expectancy in China: Bayesian mortality estimation and spatially heterogeneous determinants

Two-stage design on the 2015 China 1% Population Sample Survey: Bayesian random-effects smoothing of age-specific mortality for 60+ across 31 provinces, abridged life tables, then a bake-off of MLR, spatial error, spatial lag, GWR and MGWR for socio-environmental determinants. **PM₁₀ was not significant in any global model**; only GWR suggested a more negative association in south-western China, and GWR outperformed MGWR (R^2 0.789 vs 0.664), which the authors correctly attribute to bandwidth instability at $n = 31$. Read this as a negative result about method, not about PM: 31 aggregated provincial units, one cross-section, provincial mean PM₁₀ as the exposure, and no lag structure cannot recover a pollution–longevity gradient that individual-level cohorts detect routinely. The paper is a clean illustration of how ecological aggregation destroys an exposure contrast that a denser monitoring grid would preserve.

doi: 10.3389/fpubh.2026.1851714

Karamzad et al. 2026 Health Sci Rep

Ischaemic heart disease burden in MENA, 1990–2021 (GBD 2021)

21 countries. 2021 age-standardised IHD prevalence 6,404.8/100,000, mortality 202.8, DALY rate 4,023.2. **Ambient PM pollution accounts for 28.2% of IHD DALYs**, third after systolic blood pressure (49.6%) and LDL cholesterol (39.6%). Age-standardised rates declined 1990–2021 but burden remains high and inversely non-linearly related to SDI.

doi: 10.1002/hsr2.72886

Kraus et al. 2026 Int J Hyg Environ Health

A risk-based air quality index for Germany

Rebuilds the German AQI on risk equivalence rather than convention: pollutant–outcome pairs with causal or likely-causal evidence were identified, effect estimates extracted from systematic reviews, daily-mean effects transformed to hourly using German monitoring data, and equivalence coefficients derived by Monte Carlo simulation standardised to PM_{2.5} as reference. PM_{2.5} thresholds anchor to WHO 2021 and the EU information threshold; PM₁₀, NO₂, O₃ and SO₂ thresholds follow by risk equivalence, with an added multi-pollutant module. Evaluated on 2019 and 2022 hourly station data: a marked shift from “very good” to “good”/“moderate” and modestly more “poor” days. The daily-to-hourly effect transformation is the load-bearing and least verifiable assumption, and the index inherits every bias of the source meta-analyses. The most directly policy-relevant record today, and a clean template for making an index defensible rather than traditional.

doi: 10.1016/j.ijheh.2026.114867

Li et al. 2026b Ann Acad Med Singap

Burden trends of Alzheimer disease and other dementias in China, 1990–2023

GBD 2023 secondary analysis, GATHER-compliant. Deaths rose 122,411 → 586,002; DALYs and prevalence more than tripled; age-standardised rates rose only modestly, so the burden growth is demographic. Women carry higher ASRs in 2023 but male rates are rising faster; burden concentrates at ≥75 years. Among the modelled risk factors, **ambient particulate matter accounts for the largest attributable share in 2023**, while household solid-fuel air pollution declined. That attributable fraction is a GBD model output, not an epidemiological estimate from Chinese cohorts, and it inherits the comparative risk assessment's exposure–response assumptions wholesale.

doi: 10.47102/annals-acadmedsg.2026146

Makkar et al. 2026 Environ Sci Pollut Res

Fungal biotechnology for air-pollution mitigation: mechanisms and engineered systems

Proposes mycofiltration as a complement to conventional filtration on the argument that HEPA-class media remove particulate mass efficiently but interact weakly with the chemically reactive fraction (aldehydes, PAHs, VOCs, transition metals). Mechanisms offered: extracellular oxidoreductases for aromatic VOC/PAH oxidation, chitin–glucan–melanin wall polymers for metal and soot immobilisation, porous mycelium for interception. Laboratory and pilot-scale evidence only — no field deployment and no bioaerosol-emission safety data. Speculative.

doi: 10.1007/s11356-026-38029-8

Qian et al. 2026 J Hazard Mater

Stronger actions are needed to advance PM_{2.5} mitigation and its health benefits under population aging in China

A new daily 1 km PM_{2.5} product for mainland China 2015–2024, used to track concentration, WHO AQG exceedance and population-weighted exposure, with attributable cardiovascular mortality and a decomposition isolating the ageing offset. National PM_{2.5} fell 35.57 → 20.62 µg/m³ (population-weighted –37.35%), and the area above the WHO interim target of 35 µg/m³ collapsed from 51.07% to 1.49% — yet **over 99% of the population still lives above the 5 µg/m³ guideline**. Attributable CVD deaths fell from 2.16 (1.64–2.58) million in 2015 to 1.55 (1.14–1.91) million in 2022, then rebounded to 1.76 (1.30–2.15) million in 2024. The share of health benefit cancelled by population ageing rose from around 40% before 2020 to 56.54% in 2024, exceeding 70% in some subregions. The burden estimates inherit exposure–response functions from external cohorts, so the uncertainty intervals understate structural uncertainty, and the 1 km product's own error is not propagated into the mortality intervals. The demographic-offset framing is the contribution and it generalises beyond China.

doi: 10.1016/j.jhazmat.2026.143101

Rees et al. 2026 PLoS One

AFRI-c HEPA filtration in care homes: mixed-methods process evaluation

Process evaluation of a cluster RCT across 22 care homes (25 staff, 20 resident and 12 relative interviews; 1,158 resident questionnaires), using normalisation process theory. Self-reported fidelity was high with low contamination — **the trial's null infection result is therefore unlikely to be an adherence artefact**. HEPA units did not change infection-control practice or environmental satisfaction; some residents disliked the draught. An important negative-result anchor for indoor filtration claims.

doi: 10.1371/journal.pone.0347989

Shan et al. 2026 J Environ Sci (China)

Policy-driven emission reductions and post-pandemic rebound in China

High-resolution integrative inventories for 2018, 2020 and 2023 covering CO₂ and nine pollutants. In 2020 vs. 2018, SO₂ (–18.1%), VOCs

(−32.6%) and CO (−8.8%) fell while CO₂ (+11.6%), NO_x (+16.8%) and PM (+21.9%) rose on expanded power generation and metal production. 2023 saw 3.2–15.1% rebound across pollutants. The Fenwei Plain shows persistent PM growth. Useful for anyone building policy-counterfactual exposure scenarios.

doi: [10.1016/j.jes.2026.01.009](https://doi.org/10.1016/j.jes.2026.01.009)

Sun et al. 2026 Br J Cancer

Clarifying causal and translational inference in studies of ambient PM_{2.5} and lung cancer

Correspondence, not primary research, arguing that the PM_{2.5}–lung-cancer literature conflates aetiological causal claims with translational or prognostic ones, and that the distinction changes what an estimate licenses. No new data. Worth keeping in the register because the same conflation runs through exposure-assignment practice: a satellite- or LUR-derived annual mean supports population-attributable-fraction arithmetic but not the individual-level counterfactual that translational framing implies, and adding a denser sensor network narrows the measurement error without touching the identification problem.

doi: [10.1038/s41416-026-03555-2](https://doi.org/10.1038/s41416-026-03555-2)

Vellaiyan 2026 J Environ Health Sci Eng

Diesel NO_x emissions and public health: a source–pathway–receptor synthesis

Review spanning combustion science, atmospheric chemistry, exposure assessment and epidemiology — disciplines that normally publish past each other. Useful on near-road gradients, street-canyon trapping, and the gap between certification cycles and real-world driving emissions. A synthesis, not new evidence; the epidemiological half is thinner than the combustion half.

doi: [10.1007/s40201-026-00995-z](https://doi.org/10.1007/s40201-026-00995-z)

Zhou et al. 2026b Toxics

Sources of ultrafine particles during severe haze periods: a spatial Durbin analysis

Spatial Durbin model over the 28 “2+26” Beijing-Tianjin-Hebei channel cities, relating pollution-intensive industrial activity to haze, and attributing ultrafine particle generation to desulfurisation, denitrification and dust-removal operations — specifically lax wet-FGD management, ratcheting NO_x standards, and electrostatic precipitators that are ineffective in the ultrafine size range. Direct effect 0.028, total spillover-inclusive effect 0.151. **No ultrafine particles were measured anywhere in this study.** The UFP attribution is inferred from city-level industrial statistics and a haze proxy, so the central claim — that abatement equipment is itself a UFP source — is an untested hypothesis dressed as a finding, however physically plausible the ESP argument is. Logged tier C for the hypothesis, not the evidence; it is exactly the claim a size-resolved mobility-sizer campaign near a controlled stack could settle.

doi: [10.3390/toxics14070588](https://doi.org/10.3390/toxics14070588)

RECOVERED BY THE MONTH-WIDE HARVEST — EXTRACTIVE PRECIS, NOT AN ASSESSED SUMMARY (17 RECORDS)

Akhter 2026 Front Epidemiol

Global information-seeking behavior of air pollution and cardiovascular disease: insights from google trends analysis

Technical AP indicators showed weak or negative zero-order associations with most CVD terms; however, sensitivity analysis controlling for pandemic-related confounding revealed complete directional reversal for PM_{2.5}, with partial correlations becoming significantly positive, indicating that original negative associations were pandemic-driven artefacts.

doi: [10.3389/fepid.2026.1874654](https://doi.org/10.3389/fepid.2026.1874654)

Alwadi 2026 J Epidemiol Glob Health

A Policy Support Analysis of Air Pollution Epidemiology and Hospital Admissions in Kuwait

A 10 µg/m³ reduction could prevent up to 403 (95%CI: 252–551) admissions/year; a 20 µg/m³ reduction up to 810 (95%CI: 505–1,109) admissions/year.

doi: [10.1007/s44197-026-00609-9](https://doi.org/10.1007/s44197-026-00609-9)

Aydin 2026 Sci Rep

Associations between ambient air pollution and mortality during a 7-year ecological time-series study

A 10 µg/m² increase in PM₁₀ was associated with a 1.7% increase in daily mortality (RR: 1.017; 95% CI: 1.009–1.025), while a similar increase in PM_{2.5} was associated with a 3.3% increase (RR: 1.033; 95% CI: 1.021–1.046).

doi: [10.1038/s41598-026-54782-9](https://doi.org/10.1038/s41598-026-54782-9)

Cui et al. 2026 Environ Sci Technol

Future chronic and acute air pollution deaths in China under climate change and emission reduction

Future climate warming and population aging may lead to increased health risks from air pollution, while emission reductions can help alleviate the associated risks.

doi: [10.1021/acs.est.6c03570](https://doi.org/10.1021/acs.est.6c03570)

Guo 2026 Front Public Health

The impact of the zero-waste city pilot policy on the synergistic reduction of CO₂ and air pollutant emissions: evidence from China

Against the backdrop of China’s carbon peaking and carbon neutrality agenda and intensified pollution control efforts, advancing the synergistic control of CO₂ and air pollutant emissions has become an important pathway toward green transformation and the reduction of environmental health risks.

doi: [10.3389/fpubh.2026.1853126](https://doi.org/10.3389/fpubh.2026.1853126)

Ji 2026 Front Public Health

Association between ambient air pollution and daily non-accidental mortality: evidence from the coastal city of Shantou, China

Non-accidental mortality in Shantou was significantly associated with exposure to PM_{2.5}, PM₁₀, SO₂, NO₂ and O₃, but not with CO.

doi: [10.3389/fpubh.2026.1821846](https://doi.org/10.3389/fpubh.2026.1821846)

Ji 2026b Front Public Health

Attributable risk of hospital admissions for ischemic stroke due to ambient air pollution: a time-series study in Zhangzhou, China

Ischemic stroke admissions were significantly associated with exposure to PM_{2.5}, PM₁₀, NO₂ and O₃, but not with SO₂ or CO.

doi: [10.3389/fpubh.2026.1850457](https://doi.org/10.3389/fpubh.2026.1850457)

Jiang et al. 2026 Health Data Sci

Global disease burden attributable to ambient air pollution: disparities, determinants, and implications

Despite the decline in age-standardized rates, the absolute number of deaths attributable to ambient PM_{2.5} and O₃ both increased, driven primarily by population aging, population growth, and increasing exposure levels, although reductions in baseline mortality rates partly offset these increases.

doi: [10.34133/hds.0447](https://doi.org/10.34133/hds.0447)

McDuffie et al. 2026 Environ Sci Technol

Meteorologically driven changes in future global air quality: physical and monetized impacts

We also highlight striking regional differences in the sign of net pollutant impacts by 2100, with net pollution decreases in the Northern Hemisphere driven by reductions in nitrate aerosol, while increases in both ozone and organic aerosol at higher temperatures lead to net increases in pollutant impacts in the Southern Hemisphere.

doi: [10.1021/acs.est.5c14713](https://doi.org/10.1021/acs.est.5c14713)

Pei 2026 Environ Res

Unraveling health disparities across cause-specific outcomes associated with emerging air pollutants: Evidence on black carbon and ultrafine particles in Buffalo, New York

For BC exposure (per IQR increase of 0.7 $\mu\text{g}/\text{m}^3$), relatively larger associations were observed for eye (OR: 1.08; 95% CI: 1.01, 1.17), endocrine (OR: 1.05; 95% CI: 1.04, 1.07), and circulatory diseases (OR: 1.04; 95% CI: 1.03, 1.06).

doi: [10.1016/j.envres.2026.125167](https://doi.org/10.1016/j.envres.2026.125167)

Rajagopal 2026 Environ Monit Assess

Air quality management during the G20 summit: strategies for urban pollution reduction

The study showed that stringent measures could reduce PM₁₀ concentrations to around 30 $\mu\text{g}/\text{m}^3$, which was ~5 times less than the concentrations observed during normal scenarios.

doi: [10.1007/s10661-026-15624-3](https://doi.org/10.1007/s10661-026-15624-3)

Sangkham 2026 J Prev Med Public Health

Fine Particulate Matter (PM_{2.5}), the Exposome, and Cumulative Health Risk: A Life-course Perspective

Fine particulate matter (PM_{2.5}; aerodynamic diameter $\leq 2.5 \mu\text{m}$) is a leading contributor to the global burden of disease and is associated with substantial morbidity and mortality.

doi: [10.3961/jpmph.26.219](https://doi.org/10.3961/jpmph.26.219)

Shkembi et al. 2026 Environ Sci Technol Lett

Workplace exposures may mask wildfire smoke-related exposure inequities and mortality

Despite experiencing less ambient exposure to wildfire PM_{2.5}, counties with higher portions of non-Hispanic Black and Hispanic Americans experienced higher work-related exposures.

doi: [10.1021/acs.estlett.6c00273](https://doi.org/10.1021/acs.estlett.6c00273)

Spira-Cohen et al. 2026 Environ Epidemiol

Changes in respiratory and cardiovascular morbidity air pollution risk and attributable burden in New York City

NO₂-attributable burden was higher than PM_{2.5} burden for asthma and CVD outcomes.

doi: [10.1097/EE9.0000000000000497](https://doi.org/10.1097/EE9.0000000000000497)

Stewart 2026 Geohealth

Blazing a Trail for Fire Management and Smoke Health Collaboration: Takeaways From the 2024 Rocky Mountain Wildfire Smoke Symposium

Public health and forest management are often viewed as having conflicting goals—reducing smoke exposure versus restoring fire to ecosystems—but opportunities for collaboration exist.

doi: [10.1029/2025GH001668](https://doi.org/10.1029/2025GH001668)

Wang 2026b Bull Environ Contam Toxicol

Regional Disparities in Air Pollution Control: A Comparative Analysis of Beijing-Tianjin-Hebei, Yangtze River Delta, Pearl River Delta, and Fenwei Plain, China

However, rising VOC emissions threaten further O₃ increases.

doi: [10.1007/s00128-026-04293-0](https://doi.org/10.1007/s00128-026-04293-0)

Xu 2026b iScience

Evaluating air pollution and economic impacts in Chinese megacity clusters under clean air policies

To address worsening air pollution issues, the Chinese government has incurred substantial economic costs by implementing emission reductions.

doi: [10.1016/j.isci.2026.116492](https://doi.org/10.1016/j.isci.2026.116492)

Other clinical endpoints

28 records

CARRIED FROM THE DAILY ISSUES — FULL SUMMARIES (10 OF 28 RECORDS IN THIS SUBTOPIC)

Doğan Karataş et al. 2026 J Clin Med

Air pollution and sudden sensorineural hearing loss: nine-year retrospective study

223 SSHL patients (mean 52.7 ± 7.4 y) at a single ENT clinic in Sivas, Türkiye, 2016–2025, with monthly mean PM₁₀, PM_{2.5}, SO₂, NO_x, NO, CO and NO₂ from ministry monitoring stations and a “cumulative risk” computed by a public-health specialist. Post-treatment audiometric change was significant at all frequencies (greatest at 2 kHz) and no significant seasonal variation in case occurrence was found. The design cannot answer the question it poses: a case series with no control period and no non-case comparison cannot estimate an exposure–occurrence association, and pairing 223 events with monthly city-wide pollutant means guarantees exposure misclassification far exceeding any plausible effect. The audiometric outcomes are a treatment-response analysis with no pollution content. Included for completeness of the register.

doi: [10.3390/jcm15145687](https://doi.org/10.3390/jcm15145687)

He et al. 2026 Toxics

PM_{2.5}-bound chlorinated paraffins and sleep disorders in youth, with body weight as mediator

Largest study in today's corpus: 122,965 valid questionnaires from school-aged children across the Pearl River Delta, with PM_{2.5}-bound chlorinated paraffin exposure analysed by generalised linear mixed models, restricted cubic splines, WQS regression and quantile g-computation, plus mediation analysis. Per-IQR Σ CP exposure raised sleep-disorder odds with ORs reported as a **1.08–3.20 range**; short-chain CPs were the dominant contributor; body weight mediated 3.40–35.67%; associations were stronger in girls and in children under 12. Two serious problems. The exposure is almost certainly modelled or area-assigned rather than personally measured — CP speciation on PM_{2.5} is a specialist analysis not deployable at $n > 10^5$ — so the exposure contrast is spatial and confounded with everything else that varies across the PRD. And an OR range spanning 1.08–3.20 without a stated model set is uninterpretable: it could be lag structure, subgroup, or CP congener. Large n does not fix either.

doi: [10.3390/toxics14070607](https://doi.org/10.3390/toxics14070607)

Ion-Andrei et al. 2026 Biomedicines

Chronic air pollution exposure and the immunomolecular profile of lung cancer

Regional retrospective series in Constanța County, Romania, linking ~15-year historical monitor-based pollution exposure to tumour histology, PD-L1 expression and EGFR status, with ALK/KRAS described only. **Chronic NO₂ was associated with higher PD-L1 expression (OR 1.035, $p = 0.017$)**, with PM₁₀, NO and NO_x trending in the same direction and an exploratory CO–EGFR signal. The idea — that ambient exposure shapes tumour immunophenotype and therefore immunotherapy eligibility — is genuinely interesting and underexplored. The execution does not support it: no CI or sample size in the abstract, no adjustment for smoking described, an exposure assigned from county monitors over 15 years with no residential-history weighting, and multiple correlated pollutants tested without correction. Hypothesis-generating at best, and the authors concede causality cannot be established.

doi: [10.3390/biomedicines14071615](https://doi.org/10.3390/biomedicines14071615)

Knapova et al. 2026 Appl Psychol Health Well Being

Daily relationship between air pollution and physical activity in an industrial region

693 adults in the industrial Ostrava region of the Czech Republic, 9,702 person-days, device-measured step counts against daily PM₁₀ over 14 days, multilevel Bayesian models with within-person centring and adjustment for temperature, sunshine and precipitation. On days when PM₁₀ exceeded a participant's own 14-day mean, step counts fell ($b = -0.027$, 95% CI -0.047 to -0.006); **a 1-SD PM₁₀ increase corresponded to 361 fewer steps**. Self-reported tendency to monitor air quality did not moderate the effect, which argues the response is not deliberate avoidance behaviour. Exposure is ambient monitor data assigned to residence, not personal, so the estimate confounds pollution with unmeasured weather and visibility cues; 14 days is short for a within-person design; and non-probabilistic quota sampling limits transport. The finding matters for exposure modelling regardless: **behavioural response makes time-activity patterns endogenous to concentration**, which biases any exposure model that treats mobility as exogenous.

doi: [10.1111/aphw.70186](https://doi.org/10.1111/aphw.70186)

Lupita et al. 2026 Dent J

PM_{2.5}, preventive-care capacity and oral disease burden across the EU

Ecological panel across 27 EU member states, 2020–2023, GBD outcomes vs. national mean PM_{2.5} with unmet dental care need as an effect modifier. PM_{2.5} was *not* independently associated with caries after adjustment, but the PM_{2.5} × unmet-need interaction was significant; PM_{2.5} was independently associated with edentulism. Ecological fallacy risk is high and the authors say so.

doi: [10.3390/dj14070440](https://doi.org/10.3390/dj14070440)

Ouyang et al. 2026 J Hazard Mater

Independent and joint effects of short-term air pollutant exposure on eczema in two Chinese cities

879,356 daily eczema outpatient records from two Guangdong dermatology hospitals matched to daily PM_{2.5}, PM₁₀, NO₂, SO₂ and O₃; distributed lag non-linear models for single pollutants and quantile g-computation for two four-pollutant mixtures. Per 10 µg/m³, **SO₂ gave the largest single-pollutant effect at +22.63%**. Each quartile increase in the PM₁₀-based mixture raised visits 4.14% (49.5% of the weight on PM₁₀); the PM_{2.5}-based mixture raised them 4.21%, but with the weight on SO₂ (38.1%) and NO₂ (37.5%). Both mixtures correspond to ≈18 extra visits per day. Outpatient visits confound severity with care-seeking behaviour, and no confidence intervals are reported for the mixture estimates.

doi: [10.1016/j.jhazmat.2026.143124](https://doi.org/10.1016/j.jhazmat.2026.143124)

Park et al. 2026 JAAD Int

Long-term particulate matter exposure and risk of androgenetic alopecia

Nationwide Korean cohort research letter linking long-term PM_{2.5}/PM₁₀ to androgenetic alopecia incidence, with hair-follicle inflammation and oxidative stress as the proposed mechanism. Abstract not indexed; full text required for effect sizes.

doi: [10.1016/j.jdin.2026.06.008](https://doi.org/10.1016/j.jdin.2026.06.008)

Sun et al. 2026 J Neurosurg Pediatr

Environmental exposure and area deprivation vs. paediatric hydrocephalus outcomes

881 patients, 2008–2023 single-institution registry, geocoded to Area Deprivation Index and CDC/ATSDR environmental measures plus satellite-derived ground-level PM_{2.5}. Multivariable Cox with cluster-robust SEs: high ozone HR 1.38 (1.05–1.82) and high ADI HR 1.46 (1.15–1.86) for shunt failure; high coal exposure HR 1.92 (1.28–2.88) for shunt infection; **high PM_{2.5} HR 3.16 (1.11–8.99) for all-cause mortality**. The mortality CI is very wide — treat as signal, not estimate.

doi: [10.3171/2026.3.PEDS2615](https://doi.org/10.3171/2026.3.PEDS2615)

Zhou et al. 2026 BMJ Open

Environmental exposome study for prostate cancer in elderly men from Western China

580 histologically confirmed localised prostate cancers after radical prostatectomy, matched 1:2 to 1,160 cancer-free controls from the WeEndPd cohort, with urinary bisphenols and phthalate metabolites by LC-MS/MS and geocoded five-year SO₂ and PM_{2.5}. Highest-quartile five-year SO₂ (9.37–28.79 µg/m³) gave **OR 1.65 (95% CI 1.08–2.72)** against the lowest; MECPP, MEHHP, MEHP, BPA and BPZ were all higher in cases; the WQS mixture of endocrine-disrupting chemicals gave **OR 1.26 (1.20–1.33)**. The PM_{2.5} term is reported but is not the headline. Two structural problems: urinary EDC metabolites have half-lives of hours to days and were measured at or after diagnosis, so they cannot index the aetiologically relevant window — this is reverse-causation-prone by construction; and cases are surgical patients from one tertiary centre, which selects on operability and on health-seeking behaviour that also tracks residential air quality.

doi: [10.1136/bmjopen-2025-110957](https://doi.org/10.1136/bmjopen-2025-110957)

Zurita et al. 2026 Pediatr Blood Cancer

Childhood leukaemia and proximity to benzene emission sources in the Mexico City area

Ecological study of acute lymphoblastic leukaemia cases from one paediatric hospital, 2010–2021, against counts of benzene-emitting establishments (petroleum and coal product manufacture, fuel and lubricant retail, printing, chemical industry) within distance buffers, using multivariate spatial models by municipality, sex and age group. Significant positive association with source counts for ages 2–17. Single-hospital case ascertainment plus municipality-level exposure is a textbook ecological design: no individual exposure, no residential history, and referral patterns to one hospital are almost certainly spatially structured. Hypothesis-generating at best — the authors say as much.

doi: [10.1002/1545-5017.70482](https://doi.org/10.1002/1545-5017.70482)

RECOVERED BY THE MONTH-WIDE HARVEST — EXTRACTIVE PRECIS, NOT AN ASSESSED SUMMARY (18 RECORDS)

Al-Kindi 2026 Am J Prev Cardiol

Impact of air pollution exposure on chronic kidney disease and type 2 diabetes: A FIDELITY analysis

Finerenone reduced composite CV (hazard ratio [HR], 0.86; 95% CI, 0.78–0.95), kidney (HR, 0.76; 95% CI, 0.66–0.88) and combined CV and kidney (HR, 0.83; 95% CI, 0.76–0.90) outcomes versus placebo across PM_{2.5} exposures (P interaction=0.37, 0.14, and 0.74, respectively).

doi: [10.1016/j.ajpc.2026.101670](https://doi.org/10.1016/j.ajpc.2026.101670)

Baek 2026 Cent Eur J Public Health

Association between air pollution and paediatric intestinal intussusception in Seoul, Korea: a time-series regression approach

DLNM analysis with a 3-day cumulative lag demonstrated a gradual increase in intussusception incidence with rising NO₂ levels, with significant rate ratio changes at lag 0 and lag 1 but not at lag 2 or lag 3.

doi: [10.21101/cejph.a8806](https://doi.org/10.21101/cejph.a8806)

Dai 2026 BMC Nephrol

Short-term exposure to ambient air pollutants increases the mortality risk of chronic kidney disease: a time-stratified case-crossover study in Zhejiang Province

Using multi-day moving average lags (calculated as the mean exposure from the current day up to the specified preceding day), the most prominent ERs were observed for PM_{2.5} at lag02 (ER: 15.16%, 95% CI: 12.69–17.68), SO₂ at lag03 (ER: 8.80%, 95% CI: 5.95–11.72), O₃ at lag01 (ER: 5.67%, 95% CI: 2.07–9.39), and ...

doi: [10.1186/s12882-026-05103-9](https://doi.org/10.1186/s12882-026-05103-9)

Gryko 2026 J Clin Med*Mid-Term Exposure to Air Pollution and Acute Kidney Injury Incidence: A 10-Year Study in Eastern Poland*

In mid-term exposure analyses (lag 0-30), each 10 $\mu\text{g}/\text{m}^3$ increase in PM_{2.5}, PM₁₀, NO₂ and CO, and each 1 $\mu\text{g}/\text{m}^3$ increase in BaP, was associated with higher AKI risk, with the strongest effect observed for NO₂ (RR 1.066, 95% CI 1.033-1.099).

doi: 10.3390/jcm15134929

Hevia-Ramos et al. 2026 J Expo Sci Environ Epidemiol*Long-term exposure to air pollution and chronic kidney disease incidence: the Danish Nurse Cohort*

We found positive associations of modest magnitude between long-term exposure to air pollutants and CKD, with hazard ratios (95% confidence intervals) per interquartile range: 1.18 (0.93-1.50) per 2.86 $\mu\text{g}/\text{m}^3$ for PM_{2.5}, 1.14 (0.93-1.40) per 3.33 $\mu\text{g}/\text{m}^3$ for PM₁₀, 1.13 (0.99-1.28) per 8.09 $\mu\text{g}/\text{m}^3$ for NO₂, and 1.09 (1.00-1.20) per 0.34 $\mu\text{g}/\text{m}^3$ for BC.

doi: 10.1038/s41370-026-00943-x

Huang 2026 Environ Res*Long-term exposure to air pollution, proteomic aging clocks, and liver-related outcomes in Chinese adults*

A 1-unit increment in the weighted pollutant score was associated with an 81% higher risk of LRE (hazard ratio [HR] = 1.81 [1.65-1.98]) and a 17% higher risk of LRM (HR = 1.17 [1.05-1.31]).

doi: 10.1016/j.envres.2026.125154

Ihenacho 2026 Environ Res*Association between ambient air pollution and colorectal cancer risk: The Multiethnic Cohort Study*

CRC incidence increased with PM_{2.5} exposure (per 10 $\mu\text{g}/\text{m}^3$; hazard ratio [HR] = 1.13, 95% confidence interval [CI] = 0.96-1.33), mainly among female (HR = 1.29, 95% CI = 1.03-1.62) but not among male participants (Pheterogeneity = 0.08).

doi: 10.1016/j.envres.2026.125250

Li 2026c Int J Biometeorol*Short and long-term exposure to fine particulate matter, nitrogen dioxide and meteorological factor and the risk of glaucoma: evidence from the China health and retirement longitudinal study, distributed lag non-linear and Mendelian randomization models*

To assess causality, Mendelian randomization analyses provided genetic evidence that genetically predicted exposure to PM_{2.5} (IVW OR = 1.367) and NO₂ (IVW OR = 1.463) were causally linked to an increased risk of glaucoma.

doi: 10.1007/s00484-026-03267-5

Li 2026d Clin Exp Pharmacol Physiol*Relationship Between Long-Term Exposure to Airborne Particulate Matter and the Intrinsic Capacity of Middle-Aged and Older Adults in China: A Retrospective Cohort Study Based on CHARLS*

Long-term exposure to PM₁, PM_{2.5} and PM₁₀ was significantly associated with a higher risk of IC decline and lower composite IC scores.

doi: 10.1111/1440-1681.70144

Liu 2026c J Dtsch Dermatol Ges*Association between exposure to air pollution and the risk of vitiligo: a nationwide cohort study*

Each standard deviation increase in mean daily exposure levels of SO₂, CO, PM₁₀, PM_{2.5}, NO_x, NO, and NO₂ was associated with an increased risk of vitiligo of 27%, 99%, 59%, 98%, 74%, 64%, and 67%, respectively.

doi: 10.1111/ddg.70349

Liu 2026d J Endocrinol Invest*Association between long-term exposure to environmental coarse particulate matter and the prevalence of thyroid disorders: a cross-sectional study in China*

Each 10 $\mu\text{g}/\text{m}^3$ increase in the PM₁₀ concentration was associated with higher odds of overt hypothyroidism (OR 1.16 [1.07-1.25]) and subclinical hypothyroidism (1.07 [1.04-1.10]).

doi: 10.1007/s40618-026-02982-5

Qin 2026 Environ Pollut*Ambient air pollution and glaucoma among middle-aged and older adults in China: Evidence from the China health and retirement longitudinal study*

Ambient PM₁₀ and PM_{2.5} concentrations in the Q2 and Q3 quartiles were associated with a higher prevalence of glaucoma in the study population.

doi: 10.1016/j.envpol.2026.128802

Salinas-Rodríguez 2026 Am J Epidemiol*Air pollution and healthy aging: a longitudinal inquire on the association of PM2.5 with intrinsic capacity and its domains*

No abstract was available from PubMed at entry; this record is listed on title, journal and metadata alone.

doi: 10.1093/aje/kwag160

Varol 2026 Iran J Public Health*The Relation of Air Pollution on Morbidity and Mortality of SARS-CoV-2 Infection in Tekirdag, Türkiye*

Correlation analyses between air pollutants and SARS-CoV-2 cases and deaths revealed significant positive associations between SARS-CoV-2 cases ($r=0.220$ for PM₁₀, $P<0.001$; $r=0.290$ for PM_{2.5}, $P<0.001$; $r=0.218$ for SO₂, $P<0.001$) and deaths ($r=0.203$ for PM₁₀, $P<0.001$; $r=0.289$ for PM_{2.5}, $P<0.001$; $r=0.278$ for SO₂, $P<0.001$).

doi: 10.18502/ijph.v54i11.20390

Viau et al. 2026 Cancer Epidemiol Biomarkers Prev*Traffic-related air pollution and the risk of invasive and borderline epithelial ovarian tumours*

For borderline tumours, the adjusted OR (95% CIs) per IQR of NO₂ was 1.36 (0.99-1.87), while for invasive cancer was 0.98 (0.79-1.21).

doi: 10.1158/1055-9965.EPI-26-0435

Wu 2026 Chem Biol Interact*Ambient PM2.5 exposure is associated with clear cell renal cell carcinoma aggressiveness and promotes tumor progression through ROS-mediated AKT/HIF-2 α signaling*

Each 10 $\mu\text{g}/\text{m}^3$ increase in PM_{2.5} exposure was independently associated with advanced AJCC stage (adjusted OR, 1.42; 95% CI, 1.15-1.74) and higher WHO/ISUP grade (adjusted OR, 1.36; 95% CI, 1.12-1.65).

doi: 10.1016/j.cbi.2026.112262

Yuan et al. 2026 Ecotoxicol Environ Saf*Proteomic signatures of ambient air pollution and risk of type 2 diabetes*

Higher proteomic signature scores of PM_{2.5}, NO₂, SO₂, and benzene were also associated with an elevated risk of T2D, with HRs (95% CIs) of 1.05 (1.00, 1.09), 1.17 (1.12, 1.23), 1.11 (1.06, 1.16), and 1.11 (1.06, 1.16) for a per-standard-deviation increase, respectively.

doi: 10.1016/j.ecoenv.2026.120474

Zeng et al. 2026c Int J Epidemiol*Ambient air pollution and risk of active tuberculosis: a population-based cohort study in Taiwan*

After adjustment for age, sex, body mass index, cigarette smoking, alcohol use, education level, income, and tuberculosis history, the adjusted hazard ratio was 1.16 (95% confidence interval: 1.00-1.34) for every 10- $\mu\text{g}/\text{m}^3$ increase in the PM_{2.5} level.

doi: 10.1093/ije/dyag100

Complete corpus register — July 2026

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Calderon-Garcidueñas	Abnormal eye movements reflect early cortical and volumetric brain changes in PM _{2.5} -exposed urban youth	<i>J Alzheimers Dis</i>	Cross-sectional imaging	PM _{2.5}	Mexico
Cao	Short-term effects of ambient air pollutants on neurological hospital admissions: a time-series analysis in a northern Chinese city	<i>BMC Public Health</i>	Time-series	PM _{2.5} + PM ₁₀ jointly	China
Cattarinussi	Independent and joint effects of outdoor air pollution and genetic risk on adolescent mental health trajectories	<i>medRxiv</i>	Prospective cohort	PM _{2.5}	USA
Dales	Wildfire-sourced PM _{2.5} and ED visits for migraine and other primary headache	<i>JAMA Netw Open</i>	Case-crossover	PM _{2.5} (wild-fire)	Canada
Ghassabian et al.	Prenatal air pollution exposure and autism spectrum disorder in the ECHO Consortium	<i>Environ Health Perspect</i>	Prospective cohort	PM _{2.5} + NO ₂ + O ₃	USA
Glover	Umbrella meta-review of urban environmental exposures and cognitive health	<i>Age Ageing</i>	Umbrella review / meta-analysis	PM _{2.5}	Global
Goossens	In utero air pollution exposure and pre-/postnatal brain development	<i>Neurosci Appl</i>	Narrative review	PM _{2.5} / PM ₁₀ / PM _{coarse}	Global
Kabir	Neighbourhood socioeconomic inequalities in mental health: moderation by environmental attributes	<i>Health Place</i>	Scoping review	PM (unspecified) / emissions	Global
Kwon	Untargeted serum metabolomics and air pollution in Parkinson's disease	<i>NPJ Parkinsons Dis</i>	Observational exposure study	PM _{2.5}	USA
Lichtiger	Air pollution and autism-like traits: Sensitive periods and joint effects of exposure to sources and constituents of fine particulate matter	<i>Environ Res</i>	Prospective cohort	PM _{2.5}	USA
Lu	Subchronic PM _{2.5} exposure exacerbates chronic stress-induced depression-like pathology in mice	<i>Environ Res</i>	Murine inhalation exposure	PM _{2.5}	China
Ma	PM _{2.5} -induced neurotoxicity and depression: exposure risks and mechanisms	<i>iScience</i>	Narrative review	PM _{2.5} + components	Global
Ma et al.	Impacts of contemporaneous air pollution exposure on cognitive performance in Kenya	<i>J Health Econ</i>	Panel study	PM _{2.5}	Kenya
Pan	Maternal PM exposure and offspring neurodevelopment: effects and mechanisms	<i>Zhonghua Yu Fang Yi Xue Za Zhi</i>	Narrative review	PM (bulk)	China
Qi et al.	The role of gut microbiota in the association between air pollution and cognitive function in older adults	<i>Environ Health Perspect</i>	Repeated-measures biomarker	PM _{2.5} + O ₃	China
Qiao	Beyond air pollution: Green space, water minerals, and genetic predisposition in migraine onset	<i>iScience</i>	Prospective cohort	PM _{2.5} + PM ₁₀ jointly	China
Sarkar	Particulate-matter-induced neurotoxicity: neuroinflammation and organelle dysfunction	<i>J Appl Toxicol</i>	Narrative review	PM _{2.5} + PM ₁₀ jointly	Global
Tien-Smith et al.	Long-term outdoor air pollution and incidence of Parkinson's disease, multiple sclerosis and motor neuron disease: a systematic review and meta-analysis	<i>Environ Int</i>	Umbrella review / meta-analysis	PM _{2.5} + PM ₁₀ + NO ₂	Global
Wang et al.	The evolving clinical landscape of PM _{2.5} -attributable hemorrhagic stroke in China, 1990-2050	<i>Neurol Sci</i>	GBD secondary analysis	PM _{2.5}	China
Cardiovascular & metabolic					
Belay	Long-term bushfire-related PM _{2.5} and coronary plaque burden in asymptomatic adults	<i>Eur J Prev Cardiol</i>	Prospective cohort	PM _{2.5} (bush-fire)	Australia
Chen	Long-term particulate matter exposure and coronary inflammation by pericoronary fat attenuation index	<i>Environ Pollut</i>	Retrospective cohort	PM _{2.5} + PM ₁₀ jointly	China
Chrysoshoou	Preventing cardiovascular disease within the built and living environment: an expert opinion	<i>Eur J Prev Cardiol</i>	Narrative review	PM (unspecified)	EU-27
Huang	Association between air pollution and incident cardiovascular diseases among a population with cardiovascular-kidney-metabolic syndrome stages 0-3: the first evidence from the China Health and Retirement Longitudinal Study	<i>Front Endocrinol (Lausanne)</i>	Prospective cohort	PM _{2.5} + PM ₁₀ jointly	China

First author	Focus	Journal	Design	Metric	Region
Huang	PM _{2.5} and dyslipidaemia in rural Chinese children; dietary effect modification	<i>Zhonghua Liu Xing Bing Xue Za Zhi</i>	Cross-sectional (multistage)	PM _{2.5}	China
Kazmierska	Air pollution and cardiovascular disease: review of current data	<i>Przegl Epidemiol</i>	Narrative review	PM _{2.5} / PM ₁₀	Global
Kim	Associations of short-, mid-, and long-term particulate matter exposure with elevated LDL-C levels in Korean adults: analysis of KNHANES 2016-2022	<i>Int Arch Occup Environ Health</i>	Cross-sectional (multistage)	PM _{2.5} + PM ₁₀ jointly	South Korea
Kuntic	[Effects of air pollution on cardiovascular health]	<i>Herz</i>	Narrative review	PM _{2.5}	Global
Lelieveld et al.	Air quality and health implications of exposure to ultrafine particle pollution	<i>Cardiovasc Res</i>	Umbrella review / meta-analysis	Ultrafine (particle number)	Global
Li	Associations of green space and air pollution with insulin resistance: Analysis of repeated-measurements health examination data	<i>Environ Pollut</i>	Observational exposure study	PM _{2.5} + PM ₁₀ jointly	China
Li et al.	Sequential mediating roles of MMP-12 and DNA methylation in air pollution-induced atherosclerotic cardiovascular disease	<i>Ecotoxicol Environ Saf</i>	Repeated-measures biomarker	PM _{2.5} + PM ₁₀ + PM1 components	China
Liang	Synergistic effects of daytime, nighttime, and compound extreme temperature events and ambient air pollution on cardiovascular mortality	<i>Environ Pollut</i>	Case-crossover	PM _{2.5} + PM ₁₀ jointly	China
Liu	Association of air pollutant levels and blood pressure variability in patients with essential hypertension: a distributed lag analysis study	<i>Blood Press</i>	Observational exposure study	PM _{2.5} + PM ₁₀ jointly	China
Luo	Beyond PM _{2.5} Mass: Constituent-Specific Cardiovascular Risks and Implications for Air Pollution Control	<i>J Am Coll Cardiol</i>	Observational exposure study	PM _{2.5}	China
Luo	Short-term ambient air pollution and cardiovascular mortality: an 11-year time-series in Nanning	<i>Front Public Health</i>	Time-series	PM _{2.5} + PM ₁₀	China
Paolisso et al.	Micro- and nano-plastics in the coronary circulation and air pollution exposure in ischaemic heart disease presentation	<i>Eur Heart J</i>	Retrospective clinical series	PM _{2.5} (+ micro/nanoplastics)	Italy
Rajzer	Ultrafine particles - rethinking air pollution risk	<i>Cardiovasc Res</i>	Observational exposure study	Ultrafine / nanoscale	Poland
Rusconi	Long-term ultrafine particle exposure 2010-2019 in high-income countries and cardiovascular burden	<i>GeoHealth</i>	Ecological panel	Ultrafine particles	Global
Wang	Size-specific health effects of particulate matter components on cardiovascular mortality: a weekly time-series study in Tianjin, China	<i>Int J Environ Health Res</i>	Time-series	PM _{2.5}	China
Xi	Air pollution, built environment and temperature vs. prediabetes phenotypes	<i>Diabetologia</i>	Prospective cohort	PM _{2.5}	Germany
Xu	Independent and interactive associations of green space and air pollution with blood lipid biomarkers: A longitudinal study	<i>Ecotoxicol Environ Saf</i>	Prospective cohort	PM (unspecified) / emissions	China
Yang	Short-term PM _{2.5} and hospitalisation/ED use in atrial fibrillation	<i>J Am Heart Assoc</i>	Case-crossover	PM _{2.5} (wildfire-influenced)	USA
Yang et al.	Joint effects of heat and fine particulate matter on out-of-hospital cardiac arrest in Japan	<i>J Am Heart Assoc</i>	Case-crossover	PM _{2.5}	Japan
Zhang et al.c	Association of joint exposure to PM _{2.5} and extreme heat with cardiovascular disease: a population-based cohort study	<i>iScience</i>	Prospective cohort	PM _{2.5}	China
Zhao	Compound exposure to extreme heat and substantial landscape fire-sourced air pollution and cardiovascular mortality	<i>J Am Coll Cardiol</i>	Case-crossover	PM _{2.5} (fire-sourced)	Global
Zhou	Air pollution and hypertension: umbrella review of systematic reviews and meta-analyses	<i>iScience</i>	Umbrella review / meta-analysis	PM _{2.5} / PM ₁₀	Global
Reproductive & developmental					
Chen	Prenatal air pollution, fetal growth trajectory phenotypes and child neurodevelopment	<i>J Environ Sci (China)</i>	Prospective cohort	PM _{2.5} / PM ₁₀	China
Gao	Association of atmospheric oxidants and fine particulates with polycystic ovary syndrome and the reproductive trajectory: Protective role of residential greenness	<i>J Hazard Mater</i>	Prospective cohort	PM _{2.5}	China

First author	Focus	Journal	Design	Metric	Region
Gomez-Herrera et al.	Prenatal air pollution exposure and placental biomarkers of oxidative stress and ageing: the BiSC cohort	<i>Int J Hyg Environ Health</i>	Repeated-measures biomarker	PM _{2.5} + BC + NO ₂	Spain
Hu et al.	Prenatal exposure to ambient air pollution and fetal growth trajectories in Wuhan, China	<i>Environ Health Perspect</i>	Prospective cohort	PM _{2.5} + gases	China
Huang	Joint prenatal bisphenol + air-pollutant mixture exposure and preschool IQ	<i>Ecotoxical Environ Saf</i>	Prospective cohort (mixtures)	PM _{2.5} / PM ₁₀	China
Jiang	Association of surrounding greenness with preterm birth and the potential mediating role of ambient air pollution	<i>Sci Rep</i>	Prospective cohort	PM _{2.5}	China
Kahroba et al.	Cord-blood black carbon particle burden and a C19MC extracellular-vesicle miRNA signature enriched for neurodevelopmental pathways	<i>Environ Res</i>	Repeated-measures biomarker	Black carbon (cord blood)	UK
Leec	Prenatal exposure to particulate matter and risk of autism spectrum disorder in children: a population-based cohort study in South Korea	<i>Environ Res</i>	Prospective cohort	PM _{2.5} + PM ₁₀ jointly	South Korea
Li	Ambient air pollutant exposure and semen quality: a refined multi-window panel study	<i>Ecotoxical Environ Saf</i>	Panel study	PM _{2.5} + PM ₁₀	China
Li et al.c	Maternal ambient air pollution exposure and risk of stillbirth in Georgia, USA	<i>Environ Epidemiol</i>	Case-control	PM _{2.5} + PM ₁₀ + 8 pollutants	USA
Lin	Maternal particulate matter exposure during pregnancy and offspring liver function	<i>Environ Pollut</i>	Prospective cohort	PM _{2.5} + PM ₁₀ jointly	China
Liu	Deoxynivalenol, cytokines and particulate matter in human trophoblast cell models	<i>Int J Mol Sci</i>	In vitro (cell lines)	PM _{2.5}	Netherlands
Liue	Impacts of extreme air pollution events on pregnancy outcomes: A systematic review	<i>Environ Res</i>	Umbrella review / meta-analysis	PM _{2.5} + PM ₁₀ jointly	UK
McCracken	Prenatal Air Pollution Exposure and Child Lungs: Innovations in Measurement and Causal Inference	<i>Respirology</i>	Observational exposure study	PM _{2.5}	USA
Peterson	Ambient air pollution, PAHs and pubertal development: critical review	<i>Curr Environ Health Rep</i>	Critical review	PM _{2.5} components	Global
Sezer et al.	Preconception and early pregnancy exposure to ambient air pollution and risk of cleft lip and/or palate	<i>Birth Defects Res</i>	Case-control	PM _{2.5} + PM ₁₀ + O ₃	Turkiye
Sieber et al.	The impact of improved air quality on low birth weight and health inequalities in Spain, 2010-2018	<i>Int J Epidemiol</i>	Registry cohort + geospatial	PM _{2.5} + NO ₂	Spain
Smith	Air Pollution Mixture and Folate Status Indicators among Pregnant Women in Canada: Maternal-Infant Research on Environmental Chemicals (MIREC) Study, 2008 to 2011	<i>Curr Dev Nutr</i>	Prospective cohort	PM _{2.5}	USA
Tang	Window-specific effects of ultraviolet radiation and air pollution on ovarian reserve in infertile women: A retrospective cohort study	<i>Environ Pollut</i>	Prospective cohort	PM _{2.5}	China
Xie	Long-term PM _{2.5} and timing of pubertal development in boys and girls	<i>Environ Res</i>	Prospective cohort	PM _{2.5}	China
Yin et al.	PM _{2.5} -dominant disruption of male reproductive hormone homeostasis: fine versus coarse particulate matter	<i>Ecotoxical Environ Saf</i>	Prospective cohort	PM _{2.5} + PM _{2.5-10} + PM ₁₀	China
Zhang	Maternal exposure to fine particulate matter and autism spectrum disorder in children: population based case-control study	<i>Front Public Health</i>	Case-control	PM _{2.5}	China
Respiratory & allergic					
Bayar Muluk	Climate change and otolaryngology: airway disease impacts and emission reduction	<i>Int J Biometeorol</i>	Narrative review	PM _{2.5}	Global
Bbuye	Ambient air pollution and risk of respiratory tract infections in low- and middle-income countries: a systematic review and meta-analysis	<i>BMC Public Health</i>	Umbrella review / meta-analysis	PM _{2.5}	Uganda
Chaiwong	PM _{2.5} and systemic inflammatory biomarkers in elderly COPD patients versus healthy elderly	<i>Int J Mol Sci</i>	Panel study	PM _{2.5}	Thailand
Duan et al.	Long-term exposure to PM _{2.5} and ozone and the risk of acute respiratory infections: community-based prospective cohort	<i>JMIR Public Health Surveill</i>	Prospective cohort	PM _{2.5} + O ₃	China
Ghana AIRCLARE	Short-term ambient PM _{2.5} /PM ₁₀ and acute respiratory morbidity in Ghana: time-stratified case-crossover pilot	<i>Research Square (preprint)</i>	Case-crossover	PM _{2.5} / PM ₁₀	Ghana

First author	Focus	Journal	Design	Metric	Region
He	Air pollution-induced airway epithelial barrier dysfunction in COPD: a narrative review of mechanisms and therapeutic implications	<i>Eur Respir Rev</i>	Narrative review	PM (unspecified) / emissions	China
Jackson	Respiratory health outcomes following co-exposure to subchronic high temperature, episodic wildfire smoke exposure, and high-cholesterol diet	<i>Inhal Toxicol</i>	Murine inhalation exposure	PM (unspecified) / emissions	USA
Kaewsri	Does diet modify the effects of air pollution on lung function? A large cross-sectional study	<i>Environ Res</i>	Prospective cohort	PM _{2.5} + PM ₁₀ jointly	Italy
Krecl	Wildfire Smoke and Human Health: A Global Perspective on Respiratory Risks and Emerging Interventions	<i>Environ Health (Wash)</i>	Observational exposure study	PM (unspecified) / emissions	Brazil
Lopez Domínguez	Short-term exposure to urban air pollution and asthma-related morbidity in children under five: a PRISMA-guided systematic review with narrative synthesis	<i>Expert Rev Respir Med</i>	Umbrella review / meta-analysis	PM _{2.5}	Mexico
Li	Human genetic variation, airway microbiome, environmental exposure and respiratory health	<i>mSystems</i>	Prospective cohort	PM _{2.5} only	China
Liao	Short-term air pollution, respiratory symptoms and lung function in asthma	<i>Med Gas Res</i>	Panel study	PM _{2.5} / PM ₁₀	China
Matheny	Residential Air Pollution, Inhalation Injury, and Burn Recovery: A Burn Model System Study	<i>J Burn Care Res</i>	Prospective cohort	PM _{2.5}	USA
Meng	Genetic susceptibility to respiratory health effects from outdoor air pollution: a structured narrative review	<i>Eur Respir Rev</i>	Narrative review	PM (unspecified) / emissions	Canada
Milibari	The impact of air pollution on paediatric hospital admissions for acute lower respiratory infections (ALRI) in Makkah, Saudi Arabia, during Hajj events	<i>J Air Waste Manag Assoc</i>	Time-series	PM ₁₀	Australia
Mokoena	Socio-economic conditions, air pollution sources and eczema symptoms among teenagers in Soshanguve	<i>Public Health Chall</i>	Cross-sectional (multistage)	PM (unspecified) / emissions	South Africa
Odimba et al.	Sex-specific associations between ambient air pollution exposure thresholds and lung function in middle-aged and older Canadians	<i>Inhal Toxicol</i>	Cross-sectional (multistage)	PM _{2.5} + NO ₂ + O ₃	Canada
Sanchez de Prada	Ambient PM and asthma-coded admissions in children 0-3 y, Spain	<i>Epidemiologia</i>	Ecological panel	PM _{2.5} / PM ₁₀	Spain
Sinha	Incorporating ambient air pollution as a risk factor for tuberculosis mortality in Delhi, India	<i>Clin Infect Dis</i>	Observational exposure study	PM _{2.5}	USA
Sun et al.	Fine particulate matter exposure and long-term lung-function trajectory in adults with cystic fibrosis	<i>Ann Am Thorac Soc</i>	Retrospective cohort	PM _{2.5}	USA
Tangsangworntamma	Ambient PM _{2.5} and respiratory tract disorder-related outpatient visits among healthcare workers: an ecological time-series study	<i>BMC Public Health</i>	Time-series	PM _{2.5}	Thailand
Wang	Environmental and socioeconomic influences on childhood asthma and pneumonia burden	<i>Front Med</i>	Retrospective cohort	PM _{2.5}	China
Yang	Allergenic vs. toxic risk of airborne seed hairs compared with ambient PM _{2.5}	<i>J Hazard Mater</i>	Field measurement	PM _{2.5} + bioaerosol	China
Yue et al.	Short-term traffic-related gaseous pollutant exposure and childhood bronchopneumonia hospitalization in Guangzhou	<i>Front Public Health</i>	Time-series	Traffic gases (NO ₂ , SO ₂)	China
Zhang	Air pollution exposure in infancy and incidence of atopic dermatitis in preschool-aged children	<i>Epidemiology</i>	Prospective cohort	PM _{2.5} only	USA
Zhu et al.	Air pollution and pneumonia hospitalization in Qingyang, China: a time-series analysis	<i>Medicine (Baltimore)</i>	Time-series	PM _{2.5} + PM ₁₀ + 4 gases	China
Mechanistic toxicology					
Balu et al.	SRM 1650b administration to isolated rat heart aggravates ischemia-reperfusion injury via mitochondrial dysfunction	<i>J Biochem Mol Toxicol</i>	Ex vivo perfused organ	Diesel PM (SRM 1650b)	India
Cai	Network toxicology and multi-omics identify potential interactions between air pollutants and interferon-related signaling in tuberculosis	<i>Mol Biol Rep</i>	Prospective cohort	PM (unspecified) / emissions	China

First author	Focus	Journal	Design	Metric	Region
Chen	Asiatic acid mitigates PM _{2.5} -elicited cardiomyocyte pyroptosis via suppression of mtDNA-driven cGAS-STING-NLRP3 signalling	<i>J Cardiovasc Pharmacol</i>	Murine inhalation exposure	PM _{2.5}	China
Ding	PM _{2.5} promotes vascular calcification via miR-27b-5p/HES1 signalling	<i>J Environ Sci (China)</i>	In vitro + organoid + murine	PM _{2.5}	China
Easton et al.	Ultrafine particulate matter emitted from ships drives inflammation and susceptibility to viral infection	<i>Environ Int</i>	In vitro + organoid + murine	Ultrafine (ship-source)	Europe
Ediagbonya	Dosimetry of Inhaled Aerosols: From Exposure to Target Tissue Dose in Risk Assessment and Drug Delivery	<i>J Appl Toxicol</i>	Narrative review	Ultrafine / nanoscale	Global
El Mais	In vitro bioanalytical assessment of aryl hydrocarbon and steroid receptor-mediated activities of primary and aged particulate emissions from residential pellet and logwood stoves	<i>Environ Int</i>	In vitro (cell lines)	PM _{2.5}	France
Fakfum	Seasonal PM _{2.5} exposure and plasma metabolome changes related to metabolic syndrome	<i>Toxics</i>	Repeated-measures biomarker	PM _{2.5}	Thailand
Gao	Epigenetically Regulated NOX4/NRF2 Axis Mediates PM _{2.5} -Induced Ferroptosis and Inflammatory Response in Membranous Nephropathy	<i>J Appl Toxicol</i>	Murine inhalation exposure	PM _{2.5}	China
Giorgione	Exploring oxidative stress and epigenetic biomarkers in bees: insights into health effects of air pollution and diet - a pilot study	<i>Environ Sci Pollut Res Int</i>	Observational exposure study	PM (unspecified) / emissions	Italy
Karuna	Nanoparticle-Encapsulated 5-Azacytidine Attenuates Air Pollution-Induced Human Cardiac Fibroblast Activation	<i>Cardiovasc Toxicol</i>	In vitro (cell lines)	Ultrafine / nanoscale	Thailand
Kern-Allely	Dairy calves exposed to ambient fine particulate matter have increased bovine respiratory disease risk	<i>Am J Vet Res</i>	Observational exposure study	PM _{2.5}	Global
Killur	Air pollution risks to wheat and mustard in Delhi agricultural airshed during the winter cropping season	<i>Environ Monit Assess</i>	Observational exposure study	PM _{2.5} + PM ₁₀ jointly	India
Kovács	Combined Effects of Air Pollution and Drought Stress in Tomato Landraces	<i>Bull Environ Contam Toxicol</i>	Observational exposure study	PM (unspecified) / emissions	Hungary
Kwon	Ambient fine particulate matter (PM _{2.5}) induces AhR-dependent proinflammatory responses and reactive oxygen species production in human conjunctival epithelial cells	<i>Cutan Ocul Toxicol</i>	Observational exposure study	PM _{2.5}	South Korea
Lee	Heat-Killed Lactocaseibacillus paracasei ATG-E1 Improves Particulate Matter 10 Plus Diesel Exhaust Particles (PM _{10D})-Induced Airway Inflammation	<i>Int J Mol Sci</i>	Murine inhalation exposure	PM ₁₀	South Korea
Liu	Resveratrol ameliorates PM _{2.5} -aggravated IBS-D via gut bacteria and hypoxanthine/linoleic acid metabolism	<i>Environ Toxicol Pharmacol</i>	Murine inhalation exposure	PM _{2.5} only	China
Lu	Toxicity of PM _{2.5} from incomplete solid-fuel burning in <i>Caenorhabditis elegans</i>	<i>Toxics</i>	Controlled exposure (ecotox)	PM _{2.5} (biomass)	China
Maraschi	Osmoregulatory disruption and metal bioaccumulation from settleable atmospheric PM	<i>J Comp Physiol B</i>	Controlled exposure (ecotox)	Settleable APM	Brazil
Massarsky	Developmental exposure to cigarette smoke particulate matter induces cross-generational DNA methylation changes in zebrafish	<i>Comp Biochem Physiol C Toxicol Pharmacol</i>	Observational exposure study	Ultrafine / nanoscale	USA
Noor	Wildfire smoke exposure impacts wing development of two pest outbreaking moths in boreal forests	<i>Environ Sci Pollut Res Int</i>	Observational exposure study	PM _{2.5}	Canada
Pan et al.	Effects of air pollution exposure on nasal mucosal immune-inflammatory markers in animal models of allergic rhinitis	<i>Front Pharmacol</i>	Umbrella review / meta-analysis	PM _{2.5} / PM ₁₀ / DEP (animal)	Global
Peng	From Size-Resolved PAC Emissions to Age-Specific Respiratory Deposition: Toxicity Potential of Particulate PACs from Residential Coal and Biomass Combustion	<i>Environ Health (Wash)</i>	Observational exposure study	Ultrafine / nanoscale	China
Ribble et al.	Exposure timing is a determinant of fine particulate matter pulmonary, vascular, and metabolic toxicity in male mice	<i>Toxicol Sci</i>	Murine inhalation exposure	PM _{2.5}	USA

First author	Focus	Journal	Design	Metric	Region
Shi	Mometasone Furoate Alleviates PM _{2.5} -Induced Pyroptosis of Nasal Mucosa Cells via Blocking JAK2/STAT3/AIM2 Inflammasome	<i>Immun Inflamm Dis</i>	Observational exposure study	PM _{2.5}	China
Strader	Near-road particulate matter from Washington D.C. has variable levels of 6PPD and 6PPD-quinone: acute and developmental impacts in zebrafish	<i>Environ Sci Pollut Res Int</i>	Observational exposure study	PM (unspecified) / emissions	USA
Wang	PM exposure, mitochondrial stress and extracellular matrix remodelling in scleral fibroblasts	<i>Exp Eye Res</i>	In vitro + organoid + murine	PM (unspecified) / emissions	China
Wangd	CC16 alleviates PM _{2.5} -induced airway inflammation in asthmatic mice by targeting the E-cadherin-mediated ferroptosis regulatory axis	<i>Curr Res Toxicol</i>	Murine inhalation exposure	PM _{2.5}	China
Williams et al.	Liver-derived, circulating xanthine oxidoreductase drives vascular impairment associated with inhalation of ultrafine particulates	<i>Free Radic Biol Med</i>	Murine inhalation exposure	Ultrafine	USA
Wu	Macrophage NLRP3 inflammasome in real-ambient PM-induced hepatic glucose dysregulation	<i>J Environ Sci (China)</i>	Real-ambient murine + co-culture	Real-ambient PM	China
Zeng et al.	Circular RNA circDNAJC5 regulates ultrafine particle-disrupted microglial lipid metabolism via sphingolipid signalling	<i>Environ Health (Wash)</i>	In vitro (cell lines)	Ultrafine (PM0.1)	China
Occupational & indoor					
Bolden et al.	Wildfire smoke indoors: a review of infiltration risk factors and exposure mitigation strategies	<i>J Air Waste Manag Assoc</i>	Scoping review	Wildfire PM _{2.5} (indoor)	USA
Chen	Nested suction instrument to cut intraoperative surgical-smoke PM _{2.5}	<i>Front Med Technol</i>	Bench / device experiment	PM _{2.5} (surgical smoke)	China
Cole	Determinants of respirable crystalline silica exposure in Australian tunnelling	<i>Ann Work Expo Health</i>	Observational exposure study	Respirable dust / RCS	Australia
Ding	Stage-resolved hair Zn-isotope dynamics as a firefighting combustion-exposure tracer	<i>J Hazard Mater</i>	Repeated-measures biomarker	Combustion-derived PM	China
Ghosh	Respiratory quality of life and pulmonary function among traffic police in eastern India	<i>Indian J Occup Environ Med</i>	Cross-sectional (multistage)	PM (unspecified) / emissions	India
Gonzalez-Colom et al.	Should household air quality monitoring be considered in selected patients with asthma and COPD?	<i>BMJ Open Respir Res</i>	Prospective cohort	PM _{2.5} (indoor)	Spain
Gwak	Size-resolved modelling of airborne viral RNA copies in indoor environments	<i>Risk Anal</i>	Physical exposure model	Bioaerosol / coarse	Global
Komonmithiphong	PM _{2.5} -bound PAHs and carcinogenic risk in charcoal barbecue (Moo Kratha) restaurants	<i>Toxics</i>	Field measurement	PM _{2.5} components	Thailand
Lai	Time-resolved Impact of Smoke Evacuation on Particulate Exposure During Laparoscopic Cholecystectomy	<i>J Surg Res</i>	Observational exposure study	PM _{2.5} + PM ₁₀ jointly	Taiwan
Li	Mechanically driven aerosol emissions from electric toothbrushing: An overlooked indoor source with exposure and infection risk implications	<i>J Hazard Mater</i>	Observational exposure study	Ultrafine / nanoscale	China
Park et al.	Environmental evaluation of wood-based remodeling in aging buildings: indoor air quality and life-cycle carbon	<i>Sci Total Environ</i>	Field measurement	PM (indoor, unspecified)	South Korea
Qi	Emission rates of respirable crystalline silica and dust from engineered stone fabrication	<i>Ann Work Expo Health</i>	Bench / device experiment	Settleable / respirable dust	Global
Ramsden	Xenobiotic hazards in aircraft cabin air	<i>J Xenobiot</i>	Review + risk economics	Ultrafine particles	Global
Rogula-Kopiec et al.	Underground versus outdoor air in the Wieliczka Salt Mine health resort: particulate matter and selected gaseous pollutants	<i>Environ Geochem Health</i>	Field measurement	Size-resolved PM (PM1-PM _{tot})	Poland
Subway PSD	Characterisation and source apportionment of PM and CO ₂ in subway stations with differing platform door forms	<i>J Environ Sci (China)</i>	Field measurement	PM _{2.5} / PM ₁₀	China
Szczap	Size-dependent characterization of toxic aerosol emissions from ignition of cryogenically processed black mass from fully charged lithium-ion batteries	<i>J Hazard Mater</i>	Observational exposure study	Ultrafine / nanoscale	USA
Sensing, forecasting & instrumentation					
Acker et al.	Comparison of policy-relevant air quality metrics calculated with sparse in situ monitoring and contiguous satellite-derived data	<i>Geohealth</i>	Satellite retrieval evaluation	PM _{2.5}	USA

First author	Focus	Journal	Design	Metric	Region
Aher	Comment on advancing air pollution forecasting: operational challenges beyond predictive accuracy	<i>Environ Sci Pollut Res</i>	Commentary / methods letter	PM (unspecified)	India
Ainembabazi	Trends in atmospheric pollution research in Uganda, 1990-2025: a scoping review	<i>Toxics</i>	Scoping review	PM _{2.5} / PM ₁₀	Uganda
Ali et al.	Evaluation of low-cost sensors for size-resolved indoor particle monitoring	<i>Sci Rep</i>	Chamber sensor evaluation	PM _{2.5} + PM ₁₀ (size-resolved)	India
Bang et al.	Application of stationary IoT sensor networks to reduce exposure to particulate matter during local hotspot events in a megacity	<i>Sensors (Basel)</i>	Real-time sensor campaign	PM _{2.5} + PM ₁₀	South Korea
Campbell et al.	Comparison of black carbon measurements using filter-specific reference transmittance to lab blanks or averaged unloaded filters	<i>J Air Waste Manag Assoc</i>	Sensor co-location	Black carbon (filter eBC)	Global
Chang	Development of a predictive model for PM _{2.5} over the greater Athens metropolitan area, Greece, at a 1 km by 1 km grid using satellite measurements and machine learning methods	<i>PLoS One</i>	Machine-learning model	PM _{2.5}	USA
De Jesus	Spike-Driven Neuromorphic Sensing for Energy-Proportional Indoor Air Quality Monitoring in Multi-Zone IoT-Enabled Smart Building Environments	<i>Sensors (Basel)</i>	Machine-learning model	PM _{2.5} + PM ₁₀ jointly	Global
Feng	Calibration strategies and measurement frameworks for low-cost particulate matter sensors: a review	<i>Meas Sci Technol</i>	Critical review	PM _{2.5} only	Global
Górka-Kostrubiec	Real-time record of particulate matter along traffic roads and health risk assessment	<i>Environ Sci Pollut Res Int</i>	Observational exposure study	PM _{2.5} + PM ₁₀ jointly	Poland
Gharibzadeh et al.	Long-term performance evaluation and seasonal field calibration of a low-cost particulate matter sensor	<i>J Environ Health Sci Eng</i>	Sensor co-location	PM _{2.5} + PM ₁₀	Iran
Ginsburg	Global and regional calibration of low-cost PM _{2.5} sensors (PurpleAir): transferability of US-based corrections	<i>Aerosol Sci Technol</i>	Field measurement	PM _{2.5} only	Global
Holloway et al.	Applying satellite-derived PM _{2.5} data to policy-relevant air quality metrics	<i>Geohealth</i>	Satellite retrieval evaluation	PM _{2.5}	USA
Huda	Low-cost sensor network for wildfire monitoring and air quality at remote community locations	<i>J Air Waste Manag Assoc</i>	Field measurement	PM _{2.5} only	Canada
Ismail	Application of machine learning algorithms in building health diagnostics: predictive analytics evaluating indoor air quality and sick building syndrome in educational settings	<i>PeerJ</i>	Machine-learning model	PM (unspecified) / emissions	Italy
Jung et al.	Development of a high-resolution PM _{2.5} model in Taiwan using integrated low-cost air quality sensors and satellite observations	<i>J Expo Sci Environ Epidemiol</i>	Machine-learning model	PM _{2.5}	Taiwan
Kos Sinhala et al.	Determinants of in-home personal exposure to black carbon: the MobilSense sensor-based study	<i>Int J Environ Health Res</i>	Real-time sensor campaign	Black carbon	France
Majeed	Air quality index prediction using machine learning regression models: A comparative analysis	<i>PLoS One</i>	Machine-learning model	PM _{2.5} + PM ₁₀ jointly	Pakistan
Mamani-Suclla	PM _{2.5} /PM ₁₀ Forecasting System with Benchmarking of 44 Machine Learning Algorithms and Ensemble Learning Approaches	<i>Sensors (Basel)</i>	Machine-learning model	PM _{2.5} + PM ₁₀ jointly	Global
Msigwa	AIoT real-time sensing and calibration of low-cost PM sensors using the TCM network	<i>IEEE Internet Things J</i>	Deep-learning model	PM _{2.5} only	Global
Olivares Lopez et al.	Sensitivity of ultrafine particle and black carbon simulations to model resolution and emission data in urban environments	<i>Environ Pollut</i>	Chemical transport model	Ultrafine (PN) + black carbon	Spain
Patel	Scalable air-quality sensor placement via gradient-based mutual information maximization	<i>Proc AAAI</i>	Deep-learning model	PM _{2.5} only	Global
Patton	Human health risk of particle exposure in outdoor environments in southwestern Pennsylvania	<i>Environ Sci Pollut Res</i>	Real-time sensor campaign	PM _{2.5} only	USA
Rocha	Non-destructive characterization of atmospheric particulate matter using SEM-EDS: Elemental mapping as a tool for chemical composition assessment	<i>An Acad Bras Cienc</i>	Field measurement	Bioaerosol / coarse	Italy
Ryu et al.	Spatial variability and pollutant-specific mitigation effects of an urban vegetated park observed with a high-density air quality sensor network	<i>Environ Pollut</i>	Real-time sensor campaign	PM _{2.5} + gases	South Korea

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Salari	Hybrid multilayer perceptron models optimized by evolutionary algorithms for urban air quality forecasting: a case study of Shiraz, Iran	<i>Sci Rep</i>	Machine-learning model	PM (unspecified) / emissions	Iran
Shi	Adaptive spatiotemporal graph transformer for multi-site multi-step PM _{2.5} forecasting	<i>J Environ Manage</i>	Deep-learning model	PM _{2.5}	China / India
Shishegaran	Ensemble model based on time series and regression to predict daily air quality in Tehran	<i>Environ Sci Pollut Res Int</i>	Time-series	PM (unspecified) / emissions	Iran
Singh et al.	A spatiotemporal machine learning framework for high-resolution PM _{2.5} estimation using reconstructed satellite aerosol observations	<i>Environ Sci Pollut Res Int</i>	Machine-learning model	PM _{2.5}	India
Sribudda	Development of a Hospital Air Quality Monitoring Application Utilizing IoT Technology and Google Workspace: A Case Study of a Hospital in Maharakham, Thailand	<i>Stud Health Technol Inform</i>	Observational exposure study	PM _{2.5} + PM ₁₀ jointly	Italy
Tsuchiya	Seasonal and indoor-outdoor characteristics of fluorescent aerosol particles in offices	<i>Biosensors</i>	Real-time sensor campaign	Bioaerosol / coarse	Japan
Wang	Joint monitoring and early warning of SARS-CoV-2 in outdoor PM _{2.5} and wastewater	<i>Viruses</i>	Environmental surveillance campaign	PM _{2.5} (bioaerosol)	China
Yang	A Chemical and Biological Portable Mass Spectrometer (CB-pMS) with Automated Sampling and Hybrid Ionization for Online Detection of Bioaerosols and VOCs	<i>Anal Chem</i>	Observational exposure study	Bioaerosol / coarse	China
Yu et al.	Integrating audiovisual data for short-term particulate matter concentration prediction on urban sidewalks	<i>Environ Sci Technol</i>	Machine-learning model	PM _{2.5} + PM ₁	China
Exposure assessment & modelling					
Abinash	Modulation of fine aerosol ionic profiles and chemical characteristics by crop residue burning over the North-Western Indo-Gangetic Plain	<i>Environ Monit Assess</i>	Observational exposure study	PM _{2.5}	India
Al-Saleh	Long-term trends and exposure implications of ambient air pollution in Riyadh, Saudi Arabia: A nine-year multi-site assessment	<i>Chemosphere</i>	Observational exposure study	PM _{2.5} + PM ₁₀ jointly	USA
Chen	Dual-isotope evidence for shipping contributions to atmospheric nitrate in coastal Chinese cities	<i>J Hazard Mater</i>	Source apportionment	PM _{2.5}	China
Chen et al.	Geospatial-based neighborhood hazards and asthma in adults: interaction with social vulnerability	<i>J Hazard Mater</i>	Exposome-wide association	PM _{2.5} constituents + VOCs	USA
Chengdu NO ₃	Gas-phase reactions accelerate nitrate formation in spring and early summer in a Sichuan Basin megacity	<i>J Environ Sci (China)</i>	Supersite observation	PM _{2.5} components	China
Christoforou et al.	Ascertaining household-level exposure to air pollution in people with schizophrenia and other psychotic disorders	<i>Int J Popul Data Sci</i>	Registry cohort + geospatial	PM _{2.5} + PM ₁₀ + NO _x	UK
Collarin	Semi-parametric methods for long-term trend detection in air quality: The North-Eastern Italian Alps case	<i>Environ Pollut</i>	Observational exposure study	PM _{2.5} + PM ₁₀ jointly	Italy
Devan	Integrated weighted geospatial analysis and spatiotemporal assessment of key criteria air pollutants for hotspot identification and air quality risk assessment	<i>Environ Toxicol Chem</i>	Observational exposure study	PM ₁₀	India
Ding	Urban-rural balanced estimation framework for PM _{2.5} exposure assessment across China	<i>Geo-spat Inf Sci</i>	Deep-learning model	PM _{2.5} only	China
Firooz et al.	Characterization of traffic-related particulate matter in an urban tunnel: source apportionment of exhaust and non-exhaust emissions	<i>Environ Pollut</i>	Source apportionment	PM _{2.5} + PM ₁₀	Iran
Goel	Ambient black carbon and its dependence on in situ stubble burning	<i>Environ Monit Assess</i>	Source apportionment	Black carbon	India
Hoppa	Roadside trees along school routes enhance foliar particulate-matter retention but increase children's pedestrian-level exposure	<i>Sci Rep</i>	Observational exposure study	PM _{2.5} + PM ₁₀ jointly	Poland
Hu	Fusion of ground monitoring, satellite observation and chemical transport modelling for seamless hourly 1-km PM _{2.5}	<i>Geo-spat Inf Sci</i>	Deep-learning model	PM _{2.5} only	China

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Lee	Spatiotemporal response of urban bike-sharing ridership to weather, air quality, and future climate change in major U.S. cities	<i>J Environ Manage</i>	Observational exposure study	PM (unspecified) / emissions	USA
Liu	Synergistic control of greenhouse gases and air pollutants in the Greater Bay Area, China: Regional, sectoral and supply chain pathways	<i>J Environ Sci (China)</i>	Observational exposure study	PM (unspecified) / emissions	China
Lu	Combustion Sources Dominated Ammonia Concentrations During Severe Air Pollution Episodes During "Two Sessions" 2025 in Beijing	<i>Bull Environ Contam Toxicol</i>	Source apportionment	PM _{2.5}	China
Mămureanu	Spatial clustering of air pollutant trends and exposure burden across Romania	<i>Environ Pollut</i>	Observational exposure study	PM _{2.5} + PM ₁₀ jointly	Romania
Nam	Assessment of Air Pollution in Hai Phong City (Vietnam) Using Lichens as Biomonitors	<i>Bull Environ Contam Toxicol</i>	Ecological panel	PM (unspecified) / emissions	Italy
Nguyen	Satellite-driven prediction of PM _{2.5} : machine learning and explainable AI	<i>Eng Res Express</i>	Critical review	PM _{2.5} only	Global
Nyoni et al.	Performance of road-traffic-based exposure proxies against personal PM _{2.5} measurements in three Sub-Saharan African countries	<i>Air Qual Atmos Health</i>	Proxy validation vs personal exposure	PM _{2.5} (personal, filtered)	The Gambia / Kenya / Mozambique
Pajak	Toxicological legacy of PAHs from an urban tire fire: soil contamination and cancer risk	<i>Toxics</i>	Dispersion model + soil sampling	PM ₁₀ / PAH	Poland
Patwary	Spatio-temporal assessment of PM ₁₀ and meteorological correlates in Cumilla, Bangladesh	<i>Res Sq (preprint)</i>	Field measurement	PM ₁₀	Bangladesh
Raj	Spatiotemporal ambient air quality across urban, tourist and industrial Himachal Pradesh	<i>Res Sq (preprint)</i>	Field measurement	PM _{2.5} + PM ₁₀	India
Sablan	Wildfire Smoke and Cardiorespiratory Emergency Visits in New Mexico 2022: Sensitivity to Exposure Estimates and Referent Periods	<i>Geohealth</i>	Case-crossover	PM _{2.5}	USA
Sanchez	Morbidity related to level-two eruption of Popocatepetl in communities near the crater	<i>Indian J Occup Environ Med</i>	Ecological panel	PM _{2.5} + PM ₁₀ jointly	Mexico
Solorzano-Araque	Multiscale exposure modelling: LOTOS-EUROS + mobility surveys + particle toxicity	<i>GeoHealth</i>	CTM + mobility integration	PM _{2.5} / PM ₁₀	Colombia
Su et al.	Industrial condensable particulate matter: a missing source in PM _{2.5} source apportionment driving winter pollution in China	<i>Environ Sci Technol</i>	Source apportionment	PM _{2.5}	China
Tang	Global and local impacts of multi-dimensional urban form on PM _{2.5} concentrations: A case study of the downstream urban agglomeration in the Yangtze River Delta	<i>PLoS One</i>	Machine-learning model	PM _{2.5}	China
Teng	Greenland Emissions Contribute to Black Carbon Aerosols in the Arctic Marine Boundary Layer	<i>Environ Sci Technol</i>	Observational exposure study	Black carbon	China
Wangc	Nighttime Organic Nitrates Drive New Particle Formation and Aerosol Growth in Urban Beijing	<i>Environ Sci Technol</i>	Observational exposure study	PM (unspecified) / emissions	China
Xiang	Identifying Tracing Species from Source Regions of PM _{2.5} on the Tibetan Plateau by Integrating Measurement and Modeling	<i>Environ Sci Technol</i>	Emission inventory	PM _{2.5}	China
Xie	Matrix effects reshape organic aerosol volatility and atmospheric persistence	<i>Proc Natl Acad Sci U S A</i>	Observational exposure study	PM (unspecified) / emissions	Global
Yan	Evaluating the impact of different green infrastructure strategies on particulate pollutant concentrations in street canyons: a numerical simulation study in Qingdao	<i>Sci Rep</i>	Field measurement	PM ₁₀	China
Zerin	Integrated spatial analysis and Gaussian dispersion modelling of airborne particulate matter from rice processing mill hubs in rural Bangladesh and associated public health risks	<i>Environ Sci Pollut Res Int</i>	Observational exposure study	PM (unspecified) / emissions	Bangladesh
Zhang	High-resolution aerosol liquid water content across the contiguous US using machine learning	<i>npj Clim Atmos Sci</i>	Machine-learning model	PM _{2.5}	USA
Zhang	Investigating Ultrafine Aerosol Turbulent Fluxes during Atmospheric New Particle Formation Events	<i>Environ Sci Technol</i>	Observational exposure study	Ultrafine / nanoscale	USA

First author	Focus	Journal	Design	Metric	Region
Zhang et al.	Extrapolating personal exposure measurements to air pollution from outdoor sources into annual mean estimates	<i>Environ Int</i>	Physical exposure model	PM _{2.5} + BC + NO ₂ + O ₃	UK
Zhou	Identifying sources of black carbon associated with high mortality risk in Beijing, China based on long-term hourly continuous measurements	<i>Environ Pollut</i>	Source apportionment	PM _{2.5}	China
Zhu	Significant contributions of sea spray aerosol to vertical CINO ₂ formation over coastal South China	<i>J Environ Sci (China)</i>	Chemical transport model	PM (unspecified) / emissions	China
Burden, policy & mitigation					
Abi Nader	Short-term respiratory and skin disease risks from air pollution during the 2024 Beirut conflict	<i>Disaster Med Public Health Prep</i>	Environmental surveillance campaign	PM _{2.5}	Lebanon
Akhter	Global information-seeking behavior of air pollution and cardiovascular disease: insights from google trends analysis	<i>Front Epidemiol</i>	Time-series	PM (unspecified) / emissions	Pakistan
Alwadi	A Policy Support Analysis of Air Pollution Epidemiology and Hospital Admissions in Kuwait	<i>J Epidemiol Glob Health</i>	Observational exposure study	PM _{2.5}	USA
Aydin et al.	Associations between ambient air pollution and mortality during a 7-year ecological time-series study	<i>Sci Rep</i>	Time-series	PM _{2.5} + PM ₁₀ jointly	Türkiye
Cui et al.	Future chronic and acute air pollution deaths in China under climate change and emission reduction	<i>Environ Sci Technol</i>	Chemical transport model	PM _{2.5} + O ₃	China
Guo	The impact of the zero-waste city pilot policy on the synergistic reduction of CO ₂ and air pollutant emissions: evidence from China	<i>Front Public Health</i>	Panel study	PM (unspecified) / emissions	China
Hu	Older-adult life expectancy in China: Bayesian mortality estimation and spatially heterogeneous determinants	<i>Front Public Health</i>	Bayesian + geospatial regression	PM ₁₀	China
Ji	Association between ambient air pollution and daily non-accidental mortality: evidence from the coastal city of Shantou, China	<i>Front Public Health</i>	Time-series	PM _{2.5} + PM ₁₀ jointly	China
Ji	Attributable risk of hospital admissions for ischemic stroke due to ambient air pollution: a time-series study in Zhangzhou, China	<i>Front Public Health</i>	Time-series	PM _{2.5} + PM ₁₀ jointly	China
Jiang et al.	Global disease burden attributable to ambient air pollution: disparities, determinants, and implications	<i>Health Data Sci</i>	GBD secondary analysis	PM _{2.5} + O ₃	Global
Karamzad	Ischaemic heart disease burden in MENA 1990-2021; PM attributable fraction 28.2%	<i>Health Sci Rep</i>	GBD secondary analysis	Ambient PM	MENA
Kraus	Development and evaluation of a risk-based air quality index for Germany	<i>Int J Hyg Environ Health</i>	Bayesian + geospatial regression	PM _{2.5} + PM ₁₀ jointly	Germany
Li	Burden trends of Alzheimer disease and other dementias in China, 1990-2023	<i>Ann Acad Med Singap</i>	GBD secondary analysis	PM (unspecified)	China
Makkar	Fungal biotechnology for air-pollution mitigation: mechanisms and engineered systems	<i>Environ Sci Pollut Res</i>	Critical review	PM _{2.5}	India
McDuffie et al.	Meteorologically driven changes in future global air quality: physical and monetized impacts	<i>Environ Sci Technol</i>	Chemical transport model	PM _{2.5} + O ₃	Global
Pei	Unraveling health disparities across cause-specific outcomes associated with emerging air pollutants: Evidence on black carbon and ultrafine particles in Buffalo, New York	<i>Environ Res</i>	Case-crossover	Ultrafine / nanoscale	USA
Qian et al.	Stronger actions are needed to advance PM _{2.5} mitigation and its health benefits under population aging in China	<i>J Hazard Mater</i>	Machine-learning model	PM _{2.5}	China
Rajagopal	Air quality management during the G20 summit: strategies for urban pollution reduction	<i>Environ Monit Assess</i>	Observational exposure study	PM _{2.5} + PM ₁₀ jointly	India
Rees	HEPA filtration to reduce respiratory infection in care homes	<i>PLoS One</i>	RCT process evaluation	Filtered PM	UK
Sangkham	Fine Particulate Matter (PM _{2.5}), the Exposome, and Cumulative Health Risk: A Life-course Perspective	<i>J Prev Med Public Health</i>	GBD secondary analysis	PM _{2.5}	Thailand
Shan	Policy-driven emission reductions and post-pandemic rebound, high-resolution inventories	<i>J Environ Sci (China)</i>	Emission inventory	PM emissions	China

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Shkembi et al.	Workplace exposures may mask wildfire smoke-related exposure inequities and mortality	<i>Environ Sci Technol Lett</i>	Ecological panel	Wildfire PM _{2.5}	USA
Spira-Cohen et al.	Changes in respiratory and cardiovascular morbidity air pollution risk and attributable burden in New York City	<i>Environ Epidemiol</i>	Time-series	PM _{2.5} + NO ₂ + O ₃	USA
Stewart	Blazing a Trail for Fire Management and Smoke Health Collaboration: Takeaways From the 2024 Rocky Mountain Wildfire Smoke Symposium	<i>Geohealth</i>	Observational exposure study	PM _{2.5}	USA
Sun	Clarifying causal and translational inference in studies of ambient PM _{2.5} and lung cancer	<i>Br J Cancer</i>	Commentary / methods letter	PM _{2.5}	Global
Vellaiyan	Diesel NOx emissions and public health: source-pathway-receptor synthesis	<i>J Environ Health Sci Eng</i>	Narrative review	PM (unspecified)	Global
Wang	Regional Disparities in Air Pollution Control: A Comparative Analysis of Beijing-Tianjin-Hebei, Yangtze River Delta, Pearl River Delta, and Fenwei Plain, China	<i>Bull Environ Contam Toxicol</i>	Observational exposure study	PM _{2.5} + PM ₁₀ jointly	China
Xu	Evaluating air pollution and economic impacts in Chinese megacity clusters under clean air policies	<i>iScience</i>	Observational exposure study	PM _{2.5} + PM ₁₀ jointly	China
Zhou	Sources of ultrafine particles during severe haze periods: a spatial Durbin analysis	<i>Toxics</i>	Spatial econometric model	Ultrafine particles	China
Other clinical endpoints					
Al-Kindi	Impact of air pollution exposure on chronic kidney disease and type 2 diabetes: A FIDELITY analysis	<i>Am J Prev Cardiol</i>	Observational exposure study	PM _{2.5}	USA
Baek	Association between air pollution and paediatric intestinal intussusception in Seoul, Korea: a time-series regression approach	<i>Cent Eur J Public Health</i>	Narrative review	PM ₁₀	South Korea
Dai	Short-term exposure to ambient air pollutants increases the mortality risk of chronic kidney disease: a time-stratified case-crossover study in Zhejiang Province	<i>BMC Nephrol</i>	Case-crossover	PM _{2.5}	China
Dogan Karatas	Air pollution and sudden sensorineural hearing loss: nine-year retrospective study	<i>J Clin Med</i>	Retrospective clinical series	PM ₁₀ / PM _{2.5}	Turkiye
Gryko	Mid-Term Exposure to Air Pollution and Acute Kidney Injury Incidence: A 10-Year Study in Eastern Poland	<i>J Clin Med</i>	Prospective cohort	PM _{2.5} + PM ₁₀ jointly	Poland
He	PM _{2.5} -bound chlorinated paraffins and sleep disorders in youth, with body weight as mediator	<i>Toxics</i>	Cross-sectional (multistage)	PM _{2.5} components	China
Hevia-Ramos et al.	Long-term exposure to air pollution and chronic kidney disease incidence: the Danish Nurse Cohort	<i>J Expo Sci Environ Epidemiol</i>	Prospective cohort	PM _{2.5} + PM ₁₀ + BC + NO ₂	Denmark
Huang	Long-term exposure to air pollution, proteomic aging clocks, and liver-related outcomes in Chinese adults	<i>Environ Res</i>	Observational exposure study	PM _{2.5} + PM ₁₀ jointly	China
Ihenacho	Association between ambient air pollution and colorectal cancer risk: The Multiethnic Cohort Study	<i>Environ Res</i>	Prospective cohort	PM _{2.5} + PM ₁₀ jointly	USA
Ion-Andrei	Chronic air pollution exposure and the immunomolecular profile of lung cancer	<i>Biomedicines</i>	Retrospective cohort	PM ₁₀ (with NO ₂ /NO _x)	Romania
Knapova et al.	Daily relationship between air pollution and physical activity in an industrial region	<i>Appl Psychol Health Well Being</i>	Panel study	PM ₁₀	Czech Republic
Lic	Short and long-term exposure to fine particulate matter, nitrogen dioxide and meteorological factor and the risk of glaucoma: evidence from the China health and retirement longitudinal study, distributed lag non-linear and Mendelian randomization models	<i>Int J Biometeorol</i>	Time-series	PM _{2.5} + PM ₁₀ jointly	China
Lid	Relationship Between Long-Term Exposure to Airborne Particulate Matter and the Intrinsic Capacity of Middle-Aged and Older Adults in China: A Retrospective Cohort Study Based on CHARLS	<i>Clin Exp Pharmacol Physiol</i>	Prospective cohort	PM _{2.5} + PM ₁₀ jointly	China
Liuc	Association between exposure to air pollution and the risk of vitiligo: a nationwide cohort study	<i>J Dtsch Dermatol Ges</i>	Prospective cohort	PM _{2.5} + PM ₁₀ jointly	Taiwan

First author	Focus	Journal	Design	Metric	Region
Liud	Association between long-term exposure to environmental coarse particulate matter and the prevalence of thyroid disorders: a cross-sectional study in China	<i>J Endocrinol Invest</i>	Cross-sectional (multistage)	PM ₁₀	China
Lupita	PM _{2.5} , preventive-care capacity and caries/edentulism burden across the EU	<i>Dent J</i>	Ecological panel	PM _{2.5}	EU-27
Ouyang	Independent and joint effects of short-term air pollutant exposure on eczema in two Chinese cities	<i>J Hazard Mater</i>	Time-series	PM _{2.5} + PM ₁₀	China
Park	Long-term PM exposure and risk of androgenetic alopecia	<i>JAAD Int</i>	Nationwide cohort	PM _{2.5} / PM ₁₀	South Korea
Qin	Ambient air pollution and glaucoma among middle-aged and older adults in China: Evidence from the China health and retirement longitudinal study	<i>Environ Pollut</i>	Observational exposure study	PM _{2.5} + PM ₁₀ jointly	China
Salinas-Rodríguez	Air pollution and healthy aging: a longitudinal inquire on the association of PM _{2.5} with intrinsic capacity and its domains	<i>Am J Epidemiol</i>	Observational exposure study	PM _{2.5}	Mexico
Sun	Environmental exposure and area deprivation vs. paediatric hydrocephalus outcomes	<i>J Neurosurg Pediatr</i>	Registry cohort + geospatial	PM _{2.5} (satellite-derived)	USA
Varol	The Relation of Air Pollution on Morbidity and Mortality of SARS-CoV-2 Infection in Tekirdag, Turkey	<i>Iran J Public Health</i>	Ecological panel	PM _{2.5} + PM ₁₀ jointly	Global
Viau et al.	Traffic-related air pollution and the risk of invasive and borderline epithelial ovarian tumours	<i>Cancer Epidemiol Biomarkers Prev</i>	Case-control	NO ₂ (land-use regression)	Canada
Wu	Ambient PM _{2.5} exposure is associated with clear cell renal cell carcinoma aggressiveness and promotes tumor progression through ROS-mediated AKT/HIF-2α signaling	<i>Chem Biol Interact</i>	In vitro (cell lines)	PM _{2.5}	China
Yuan et al.	Proteomic signatures of ambient air pollution and risk of type 2 diabetes	<i>Ecotoxicol Environ Saf</i>	Repeated-measures biomarker	PM _{2.5} + NO ₂ + SO ₂ + benzene	UK
Zeng et al.	Ambient air pollution and risk of active tuberculosis: a population-based cohort study in Taiwan	<i>Int J Epidemiol</i>	Prospective cohort	PM _{2.5}	Taiwan
Zhou et al.	Environmental exposome study for prostate cancer in elderly men from Western China	<i>BMJ Open</i>	Case-control	PM _{2.5} + SO ₂	China
Zurita	Childhood leukaemia and proximity to benzene emission sources in the Mexico City area	<i>Pediatr Blood Cancer</i>	Ecological panel	PM (unspecified)	Mexico

Provenance and method — read this before citing any count above

This issue breaks the standing rollup contract, deliberately and on request. Rollups in this series normally aggregate previously published daily issues and never re-query an API. Daily coverage of July exists only for 27–31 July, so the 1–26 July portion was **harvested independently on 1 August 2026** and had never appeared in any daily issue. Sources: PubMed E-utilities on the [Date - Entry] field, 2026/07/01–2026/07/26, along two independent axes — health (352 records) and instrumentation (53) — giving **368 unique records** after merge. Of those, 22 were already in `state/seen.json` from the daily issues and were dropped; **166 passed a PM-relevance title screen**; 2 were rejected on reading as water-phase chemistry rather than airborne particulate (stream-sediment geochemistry; sorption of antibiotics to dissolved black carbon), logged with reasons to `state/rejected.json`; and **all 164 remaining records were classified and are listed in this issue**.

What this issue is, and what it is not. It is a complete listing of every record in the July entry window that passed the PM-relevance title screen — 270 records, none omitted for space or interest. It is *not* a systematic review: the title screen is itself a filter, and the 202 records that did not pass it were excluded on title alone, so a paper about particulate matter whose title never says so would have been missed. Recall was not measured against any external gold standard.

Two classification tiers, and the weaker one is the larger. 157 records were classified by hand — the 106 from daily issues plus 51 selected on 1 August. The remaining 113 were classified by rule from MeSH terms, author keywords, title and abstract, then reviewed record by record; **43 of the 113 auto-assignments were corrected** on that review, which is a 38% correction rate and the honest measure of how far a keyword classifier gets you here. Auto-classified records carry `auto: true` in `state/corpus/2026-07-26.json` and are all tier C. Their subtopic and design labels should be treated as indicative; their titles, journals and DOIs are exact.

The daily-trend panel is labelled a harvest diagnostic for the same reason: the 26 July bar is one batch, not one day. Records were deduplicated against `state/seen.json` on PMID, lowercased DOI and title signature before any summarisation, so no record appears twice across the daily issues and the backfill. Effect estimates were transcribed verbatim from abstracts and only where an interval was reported; 62 qualified. Subtopic, design, particle-metric and geography labels are single-label judgements from title, abstract, keywords and MeSH terms, not a controlled vocabulary. Two records fetched without abstracts (*Am J Epidemiol* PMID 42429684; *J Am Coll Cardiol* PMID 42455106) were **excluded rather than summarised from their titles**, despite the JACC piece being directly on this issue's theme. DOIs carry through from PubMed's own record. Content derived from PubMed, Europe PMC and ClinicalTrials.gov metadata; abstracts remain the copyright of their respective publishers.